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Doctor of Philosophy

**Development of the Specialized
Question-Answering System using a
Bridge Maintenance Knowledge Base
and Large Language Model**

February 2024

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Seoul National University

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Development of the Specialized Question-Answering System using a Bridge Maintenance Knowledge Base and Large Language Model

A dissertation submitted to the Graduate School of
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in partial fulfillment of the requirements for the degree of
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By
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Development of the Specialized Question-Answering System using a Bridge Maintenance Knowledge Base and Large Language Model

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To my family, with love and hope

ABSTRACT

Development of the Specialized Question-Answering System using a Bridge Maintenance Knowledge Base and Large Language Model

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The importance of public facilities maintenance cannot be overstated. Numerous countries have institutionalized regulations and manualized standards for managing facilities. A Bridge Management System (BMS) was developed as a platform to collect and manage bridge specifications and

maintenance records for supporting the data-based decision-making of facility managers. However, compared to the vast amount of unstructured and heterogeneous data and bridges that need to be managed, there is a shortage of human resources, such as experts who can analyze data and make decisions. Despite numerous research developing databases and machine learning-based predictive models for data-driven decision-making in the bridge maintenance field, the practical application for end-users to access necessary information remains arduous.

To address this issue, this research aims to develop a question-answering (QA) system to assist individuals responsible for bridge maintenance in searching for necessary data and analyzing them for data-driven decisions. Firstly, the author integrated the discrete, heterogeneous bridge maintenance-related data from the BMS and several related platforms to form a comprehensive knowledge base. Secondly, the system to chain the large language model (LLM) was generated to answer bridge maintenance-related questions, including simple retrieval, composite queries, and predictive reasoning functions. With the assistance of an LLM, the proposed system can transform a user's natural language inquiry into a structured query, retrieve information from the knowledge base, and produce a corresponding answer. Finally, user testing was conducted to whether the developed system returns answers aligned with the user's intent.

To verify the generated knowledge base and developed QA system, a competency question was composed and tested to ascertain if it could generate answers based properly. The competency questions confirmed that the knowledge base provides an environment suitable for generating necessary answers. Furthermore, the user testing with ten participants showed that the system revealed an above-average satisfaction level (4.57 in 1 to 7 Likert-scale) and an average response generation time of 14.97 seconds. An accuracy review using the generated question-answer sets showed an accuracy rate of 85.7%.

In conclusion, this dissertation presents the generation of a knowledge base that integrates heterogeneous data for the bridge maintenance sector and the development of a QA system using an LLM by expanding accessibility to data search and analysis for bridge maintenance personnel. This system can be utilized to support data-driven decision-making in bridge maintenance practices, ultimately contributing to the safe usage of infrastructural facilities.

Keywords: Bridge Maintenance, Knowledge Base, Large Language Model, Question and Answering

Student Number: 2020-39040

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Chapter 1. Introduction

1.1 Research Background

The efficient management of public facilities has always been a significant concern for civil engineers around the world (Wu et al. 2021). Especially, for over a decade, maintenance of them has become more important due to the increase in the number of facilities requiring maintenance coupled with their deterioration. In the United States, 621,510 bridges are recorded in U.S bridge inventory in 2023, and approximately 36 percent of bridges are in a situation requiring major repairs or replacement (ARTBA 2023). Similarly, in South Korea, there is a rising demand for efficient bridge maintenance due to the increase in bridge numbers and the onset of their aging (Park and Lee 2021).

To ensure the efficient management of a vast number of bridges, a combination of physical agencies and institutional systems has been systematically established. Such as the regulatory frameworks that provide the foundational guidelines and standards for bridge maintenance, uniformity in management practices. And localized bridge management agencies are established for the direct oversight and upkeep of bridge infrastructure within their respective jurisdictions. Also, as a pivotal role in bridge management,

BMS was developed to store extensive data regarding bridge specifications and maintenance records. The BMS is particularly instrumental in facilitating data-driven decision-making for managers. By leveraging the stored information, managers can make informed choices about maintenance priorities, resource allocation, and long-term planning for bridge infrastructure, thereby enhancing the overall efficiency and safety of bridge management. (Miyamoto et al., 2000; Wu et al., 2021).

However, it takes a lot of time and effort for bridge maintenance managers to find the necessary information from the numerous data buried in the database. In order to find the data that is divided into several tables or stored in an unstructured form in the BMS and obtain the desired information, knowledge to control the data is required, but only a very small number of people are capable of doing so. In other words, poor data integration and retrieval techniques make it difficult to retrieve and analyse the necessary data that can immediately help bridge maintenance managers make decisions (Yang et al. 2022).

Three types of stakeholders involved in bridge maintenance are experiencing data-related challenges. First, if personnel directly inspecting bridges in the field want to find past data, they can access the BMS through a mobile phone application, but they have to go through several steps to reach the desired information, so its usability is very low. Second, in the case of

management personnel at local agencies, not only are there a large number of bridges to manage, but they are often unable to master all the manuals they need to know due to rotating work. This is an anxiety factor that can become a major obstacle to efficient bridge maintenance. Third, for high-level decision makers who must make decisions such as budget execution for bridges across the country, comprehensive data analysis is necessary to devise data-based strategies, but it is difficult to draw comprehensive conclusions due to scattered data. To address these challenges, a database structure that not only efficiently stores and manages collected data but also improves usability is required.

In general, BMS have designed as a relational database (RDB), which is a highly structured database in a table form. It has the advantage of consistent data management with its structured schema, but it also limits the flexibility to modify the data structure and perform complex joins between tables (Nayak et al., 2013). For example, the basic specifications of bridges and annual inspection and repair records are often stored in separate tables in BMS. In these situations, relationships between latent knowledge and variables are missed and data analysis is limited.

Therefore, it is necessary to design a system that integrates the knowledge required for bridge maintenance managers into a single knowledge base and helps them find and analyse the desired information

based on this. In addition, there is a need to develop an intelligent system that can easily find and analyze data based on a knowledge base by expanding natural language processing capabilities by utilizing large language models that have been actively developed recently.

1.2 Problem Statement

There are many challenges inherent in the current practices of bridge maintenance data applications. This dissertation summarizes several points that require improvement in the current situation as follows.

First, bridge maintenance managers who need to make data-based decisions cannot find the information they need. Currently, BMS and its field applications have the function of showing data for users, but it is difficult to know where to find the necessary data, and the data is distributed across multiple tables, making it difficult to use integrated analysis functions for decision-making.

Although a knowledge database in the bridge maintenance field is being created through various research, the results using it for specific tasks such as Question-answering have not been delivered in practice. It is because their practical applicability in real-world scenarios was limited, and also, the natural language processing capabilities of the language model were limited.

Recently, even with the recent launch of LLM, it was still in its early stages, and a suitable system was not yet in place.

This highlights that, large amounts of new data such as bridge inspection history and repair history are generated and accumulated in the system every year, but it is difficult to analyze the data in an integrated manner due to the fragmented data format and collection unit. Only a few people in charge who can analyze data take the time to analyze bridge maintenance data and share the results with the managers, and it is difficult for field staff who lack data analysis skills to analyze the data themselves. These gaps indicate the need for more sophisticated and adaptive systems to compensate for the problem. These gaps indicate the need for more sophisticated and adaptive systems to compensate for the problem.

1.3 Research Objectives

This dissertation aims to provide the framework to support bridge maintenance workers in answering their questions by generating the knowledge base and integrating a large language model for improving usability. Three specific objectives were established to address the challenges presented below. The overall research framework is illustrated in Figure 1.1.

Objective 1: Compare the database schema and generate a database suitable for bridge maintenance knowledge data to integrate the heterogeneous bridge maintenance-related data easily for retrieval and query.

Objective 2: Design the custom agents to execute the task of answering the bridge maintenance related question by retrieving and querying the data and reasoning the predicted results with machine learning model.

Objective 3: Perform user testing and evaluate the results on a system that can generate answers based on a bridge maintenance knowledge base using a large language model.

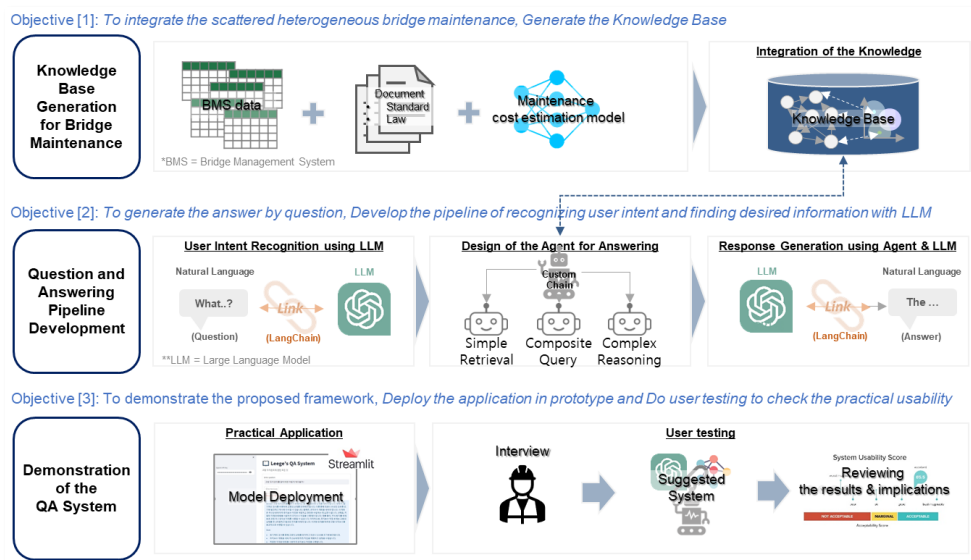


Figure 1.1 Research framework

The entire research framework was verified with the competency question and validated with the user testing. Additionally, the comparative advantage of the selected methodologies in the framework was validated through additional experiments at each stage.

1.4 Research Scope

In this dissertation, the scope of data correction and system development was set as follows. First, the scope of data was limited to the bridges on national roadways in South Korea. Bridges in this range are managed under the Ministry of Land, Infrastructure, and Transport (MOLIT) of government and in accordance with the “Special Act on Safety and Maintenance of Facilities” and its subsequent enforcement degrees and implementation rules. Therefore, the knowledge base of bridge maintenance has contained the information within the relevant scope.

Second, the development scope of the system is the generation of the knowledge base and an LLM-based system framework that demonstrates it. LLM, which is used for natural language processing, including querying the data and analyzing it, utilized the GPT-4 model using the API. The application for user testing was prototyped with the Streamlit application.

1.5 Dissertation Outline

This dissertation consists of six chapters. The content of each chapter is described below. Chapters 3, 4, and 5 include the development process of the components that make up the framework in Figure 1.2.

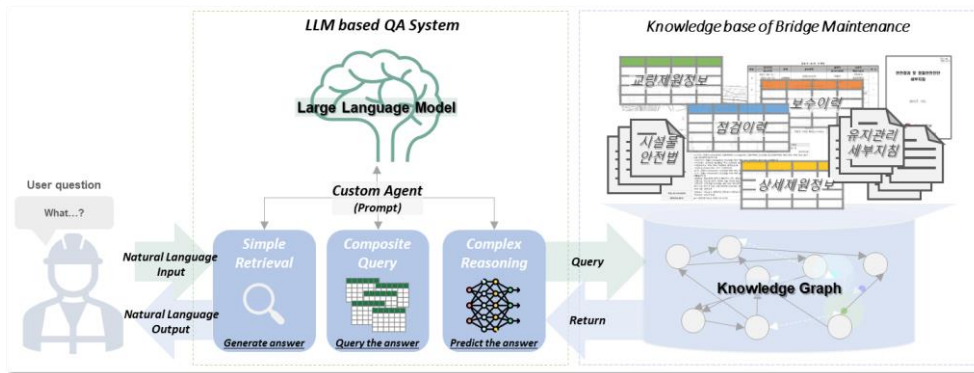


Figure 1.2 Concept of the proposed framework

Chapter 1 Introduction: This chapter introduced the research background and motivations. The importance of the bridge maintenance knowledge base, the necessity of the question-answering system, and the limitations of the previous approaches are described. The objectives, scope, framework, and process of the research are also presented.

Chapter 2 Literature Review: This chapter introduces the research to date in the bridge maintenance knowledge base, from generation of ontology

to graph databases. In addition, the previous attempts are reviewed, of which objectives were to generate a question and answering system to assist the worker in the construction field, focusing on the technical approaches and applications.

Chapter 3 Generation of Bridge Maintenance Knowledge Base: This chapter covers the first step in integrating the heterogeneous and unlinked data in bridge maintenance field. The collection and data processing methods of both structured and unstructured data related to bridge maintenance, modeling of the relationships between entities, and finally verification of the database is explained.

Chapter 4 Implementation of the Question-Answering System using a Large Language Model: This chapter provides in-depth discussion on topics ranging from establishing the QA scope through reviewing user requirements, designing data retrieval and analysis agents for generating the response for questions by leveraging large-scale language models in natural language processing and inference.

Chapter 5 Demonstration of the QA system through User Testing: This chapter provides the interface design of the proposed framework, usage

scenarios for each function for users, and user testing results to demonstrate the usability of the system proposed in this dissertation.

Chapter 6 Conclusions: This chapter summarizes the research findings and contributions. Opportunities for further improvement and future research works are also provided.

Chapter 2. Literature Review

This chapter covers the historical progression of research related to the knowledge base for bridge maintenance, extending the recent studies on developing question-answering systems based on large language models. Through this review, the necessity and contribution of the research presented in this dissertation can be clearly elucidated.

2.1 Knowledge Base in Context of Data Engineering

The concept of knowledge base emerged from the challenges posed by vast, imprecise, and incomplete data sets (Debenham 2012). It is an assembly of facts, processes, and meanings that surpasses conventional databases due to its reasoning capabilities that aid in addressing complex queries. When knowledge-based technologies intersect with bridge maintenance, they present novel solutions to long-standing challenges faced by civil engineers. To create a tailored knowledge base for bridge maintenance, the author initiates the formalization of the bridge maintenance ontology, which serves as the foundation of the knowledge base, and then delves into the exploration of existing research on the development of a graph database. For this purpose,

the following bridge maintenance ontology and knowledge databases in related fields were reviewed.

2.1.1 Ontological Frameworks in Bridge Maintenance

The term “ontology,” originally from philosophy signifying a structured understanding of existence, has recently become significant in the Knowledge Engineering field (Corcho et al. 2003). Moreover, within the construction sector, numerous researchers have endeavoured to undertake ontology-based tasks (Turk 2006). This section provides an overview of several notable examples of ontologies tailored for bridge maintenance tasks and delineates the distinctions between each study.

Of late, Liu and El-Gohary have consistently focused their research on ontology and knowledges utilizing bridge maintenance data. They conceived an ontology referred to as “BridgeOnto” (Liu and Nora El-Gohary 2016), which aimed to semantically represent knowledge on bridge deterioration. This ontology made use of data from National Bridge Inventory (NBI) and National Bridge Elements (NBE) data sets, as well as from bridge inspection reports. The modeling of relations was executed manually, and system verification was also conducted manually. This research subsequently expanded into a project leveraged the CRF-based information extraction

method to extract information from inspection and maintenance reports (Liu and El-Gohary 2017). In their latest paper published in 2022 (Liu and El-Gohary 2022a), they detailed the hierarchy of an ontology referred to as “BridgeOnto” and conducted expert interviews as part of evaluation of it.

Additionally, various research teams are conducting related studies. An ontology called Bridge maintenance Ontology (“BrMontology”) has been developed to help engineers make decisions in bridge maintenance (Ren et al. 2019). They set the SWRL rules to generate the potential reasons for damage, required materials for fixing, and maintenance methods of the damage automatically.

By building an “BIontology”, Zhang et al. (2022) proposed an enhanced information retrieval method based on ontology to enhance the efficiency of bridge inspection. They are modeling the spatial relationships of the bridge to help to find the target elements.

Through this endeavour, they introduced an efficient mechanism to unearth data embedded within the inspection and maintenance documents. They used ontology to help bridge terminology gaps across different data sources and support the analysis of heterogeneous text characteristics and patterns and the recognition of complex terms based on content and domain-specific meaning.

Although these ontologies cover essential concepts for document

analysis and serve as a basis for knowledge expansion, but they are still limited in their scope of use and usability.

2.1.2 Advancements in Knowledge Graph Databases

Knowledge graph is an advanced knowledge management technology. It often incorporates an ontology to provide a systematic organization and representation of knowledge (Ehrlinger and Wöß 2016). This means that while the ontology provides the structured framework of the domain's knowledge, the knowledge graph uses this framework to create an interconnected structure of information. By adopting an ontology as its foundational structure and using a graph database for storing this information, a knowledge graph effectively manages heterogeneous and discrete data. However, the complexity of the process cannot be underestimated. Building a knowledge base is a costly activity as it requires domain experts to understand the ontology schema and then manually curate the knowledge. Nevertheless, related research is continuously being conducted, which is consistent with the development of the direction of the question and answering field. Examples are as follows:

Wang and El-Gohary (2023) suggested the deep-learning-based automatic relation extraction from the Occupational Safety and Health Administration (OSHA) data and formed the knowledge graph-based

representation. They draw the knowledge graph-based relation representation by applying the named entity extraction, tokenization, and structuring the triple with a deep learning model. However, there is still an important limitation in generalizability because the identified relations are not necessarily complete or exhaustive.

Pandithawatta et al. (2023) developed a knowledge graph for Job Hazard Analysis (JHA) in the construction field. They systematically developed the ontology using “METHONTOLOGY” and subsequently evaluated the JHA knowledge graph with competency questions and a case study centered on hazard-related queries and responses. Drawing conclusions from their work, they recognize the imminent need to incorporate chatbot technology with the JHA knowledge graph. Moreover, they foresaw the potential for querying the knowledge base using natural language.

Zhang et al. (2023) built a knowledge base of bridge maintenance using the knowledge graph, called the bridge maintenance knowledge graph (BMKG). They coded the domain ontology with Protégé and stored it in the Neo4j graph database through data format conversion. The mentioned research demonstrated methodologies for creating a knowledge base using ontology and knowledge graphs and showed the scalability of graph databases to various types of data.

2.2 Innovations in Question-Answering Techniques

Question and Answering (QA) is a specific sub-task within Natural Language Processing (NLP) (Kolomiyets and Moens 2011). The objective of QA is to provide concise, relevant answers to user queries by the sort of advanced information retrieval (Bouziane et al. 2015). The queries can be posed in natural language, and the answers can be extracted from a set of documents, a database, or generated based on knowledge. The entire pipeline of QA systems has evolved significantly over the past few decades, driven by advances in natural language processing, machine learning, and computational linguistics (Abbasiantaeb and Momtazi 2021). In this part, the author introduced the knowledge graph-based QA and the latest research derived from the emergence of pre-trained large language models.

2.2.1 Knowledge Graph-based QA

Knowledge Graph-based QA (KGQA) is primarily driven by the necessity to translate natural language inquiries into queries suitable for a graph database. The fundamental step for KGQA is to compose a pipeline that facilitates this translation. The author briefly introduced research on QA based on the knowledge graph.

Agarwal et al. (2022) introduced a knowledge graph-guided deep learning based QA system designed to efficiently access the diverse repositories, addressing the aviation industry’s need, including maintenance, compliance, and safety. In their experiment, “KGQA+GPT3-QA” model outperformed others across several metrics. However, their rule-based approach for converting natural language into SPARQL queries exhibited certain vulnerabilities, leading to errors.

Luo et al. (2022) constructed a knowledge graph and complex question answering corpus for bridge inspection. Previously, knowledge graph-based QA could only answer simple questions, but this study tried a structure capable of five types of multi-hop reasoning.

2.2.2 Integration of Large Language Models in QA

Recently, Large Language Models (LLMs) have revolutionized the field of natural language processing and have a great impact on improving QA task performance (Zhao et al. 2023). Traditional QA systems often operated within the confines of predefined rules, structured databases, and a specific set of training data. However, with the advent of LLMs, which have impressive reasoning ability in various downstream tasks, the paradigm has shifted (Liu et al. 2023b).

Especially, since the launch of ChatGPT in November 2022, numerous articles and preprints have been published about its potential impact on various fields. These models, equipped with billions of parameters, can understand context, semantics, and nuances in questions, providing more accurate and contextually relevant answers (Ray 2023). Unlike previous systems that relied heavily on exact keyword matching or specific phrasing, LLMs can interpret a broader range query, even those phrased ambiguously or indirectly. Furthermore, they have reduced the need for extensive manual annotations and fine-tuning, as they are pretrained on vast corpora, encompassing diverse knowledge and linguistic structures. As a result, LLMs have expanded the horizons of what's achievable in QA, delivering more human-like responses and significantly enhancing user experience.

Typically, answers are generated based on probability, which often causes a hallucination problem. There may be issues with biased information and explainability by AI models (Ray 2023; Sohail et al. 2023). There is also real-time responsiveness, safety concerns, privacy concerns, etc. Nevertheless, there is no doubt that this technology will continue to be developed and utilized in the future, and it is essential to modify it to suit each task to achieve a method to minimize limitations.

Yasunaga et al. (2021) did the research to prove the importance of the knowledge graph for QA. They suggested the two challenges in combining

large language model and knowledge graphs: given a QA context, methods need to identify informative knowledge from a large KG; and capture the nuance of the QA context and the structure of the KGs to perform joint reasoning over these two sources of information.

2.3 Identification of Research Gaps

In this chapter, the author introduced the current state of research on knowledge bases in the construction field and the remarkable advancements in large language models in natural language processing. Through a review of existing research, the author was able to identify the following limitations and found research gaps for future development.

First, although research related to ontology has been conducted for a long time, it is often used as a somewhat unclear concept in research. In addition, the tasks sought to be achieved through ontology construction can serve as the basis for future research, but are often lacking in completeness in themselves (see Table 2.1)

Table 2.1 Limitations of the previous research

Research	Sources	Task to Achieve	Limitations
BrOntology	(Ren et al. 2019)	Rule-based Answering	Limited on specific bridge damage “crack”, limited in the Protégé context.
BridgeOnto	(Liu and El-Gohary 2022b)	Extraction of the relations buried in documents	Cannot cover all bridge deterioration concepts, only validated for extracting five target information types from bridge inspection reports
BIontology	(Zhang et al. 2022)	Retrieval of the information	Limited to modeling inspection data for spatial relationships between elements

Thus, the meaning of the word “ontology” tends to remain a bit vague, the author set in this dissertation as a basis of the knowledge bases which is formal structures at the semantic level. Also, the author able to see why the creation of ontology and knowledge base is important that the author referred in detail to previously developed bridge maintenance-related ontology. In addition, by analysing in detail the previously developed bridge maintenance-related ontology, the author was able to confirm that for practical QA, the relative DB information collected in the BMS must also be turned into a knowledge base.

Secondly, it was observed that knowledge graph-based QA operates within a constrained scope, leading to limited usability. Much of the QA was restricted to rule-based methods and probabilistic answer generation through deep learning. While LLM-based QA systems, such as ChatGPT, have considerably broadened the QA horizon by generating answers from vast pre-trained datasets, there is the "hallucination" problem that generate the plausible fake answers. Consequently, the author ascertained that it is essential to confine the answer-generation scope using the LLM to the established knowledge base. This precaution helps prevent the production of spurious answers, and an architecture that leverages the efficient functionalities of LLM within this constrained scope was developed.

Chapter 3. Generation of Bridge Maintenance Knowledge Base

This chapter delves into the construction of a bridge maintenance knowledge base, encompassing a vast array of information from various sources such as the Bridge Management System (BMS) and related documentation. The process involves the collection, formatting, and integration of data into a query-friendly knowledge base. Herein, the author elaborates on the types of data gathered, their unique attributes, and the methodologies employed to transform this data into a structured format suitable for answering the questions and efficient storage in the knowledge base.

3.1 Data Transformation of BMS

3.1.1 Collection of BMS Data

Since 2012, the Ministry of Land, Infrastructure, and Transport (MOLIT) of South Korea has been utilizing the Bridge Management System (BMS) to effectively manage the country's bridge infrastructure (Sun and Park 2012). Korean BMS accumulates various data during the bridge maintenance

<Bridge Maintenance process>

```

graph TD
    A[Completion of the Facility] --> B[Maintenance strategy]
    B --> C[Regular inspection]
    B --> D[Detailed inspection]
    B --> E[Emergent inspection]
    C --> F{Defect}
    D --> F
    E --> F
    F --> G[Decision of repair]
    G --> H[Repair/Rehabilitation]
    G --> I[Reconstruction]
    H --> J[Repair Report]
    I --> J
    
```

<Interface of BMS>

Inspection Data Table:

Span	부위명	종상상태	종상점수	부수상태	부수점수	부수잔여 수명(잔여)	부수부수(잔여)	비고
000	하수관측 호통	양호	1.25	하수관측 호통보수	2	250	600	120 호통보수
000	하수관측 호통	양호	1	하수관측 호통보수	2	250	600	120 호통보수
000	하수관측 호통	양호	0.5	하수관측 호통	1	250	200	20 호통부수(양호보수)
000	하수관측 호통	하수관측	0.5	하수관측	2	250	200	100 호통보수
000	하수관측 호통	하수관측	2	하수관측	2	250	200	40 호통보수
000	배수시설	제거/수입	4	배수시설 보수	4	250	50	200 철거
000	배수시설	배수시설	7	배수시설 관리	7	150	150	150 배수시설 관리
000	배수시설	배수시설	5	배수시설	15	150	200	300 배수시설(양호보수)
000	배수시설	배수시설	14.5	배수시설	22	150	50	1,100 배수시설
000	배수시설	배수시설	8	배수시설	14	300	50	700
000	배수시설	배수시설	6	배수시설	1	250	200	200
000	배수시설	배수시설	6	배수시설	4	300	50	400
000	배수시설	배수시설	6.38	배수시설	1	300	50	400
000	PS10-75C1	부수/수리	1.8	배수시설	3	300	50	150
000	PS10-75C1	양호	1.3	배수시설	2	300	100	100
000	PS10-75C1	양호	0.8	배수시설	1	150	50	50
000	양호	양호	32	배수시설	32	300	300	300
000	양호	양호	9.1	배수시설	14	300	200	200 배수시설
000	양호	양호	0.5	배수시설	1	150	50	50 배수시설
000	양호	양호	6.38	배수시설	1	300	50	50

Figure 3.1. Data generation process during bridge maintenance

Table 3.1 Columns of the bridge specification data

Column name	Data description	Data type
Bridge number	Unique number used for bridge management	S
Bridge name	Name of the bridge (Korean)	S
Facility class	Type classified according to bridge size and management importance (e.g., Type1, Type2, Type3, other)	C
Completion date	Date the bridge was completed	D
Competent authority	Classification between direct management or delegated management	C
Management agency (region)	Regional agency responsible the bridge management	C
Management agency (sub-region)	Sub-regional agency responsible for the bridge management	C
Road lane	Road classification number on which the bridge is located	C
Total length	Length of the bridge	N
Total width	Width of the bridge	N
Total height	Height of the bridge	N
Water depth	Depth of the bridge under water	N
Maximum span length	Maximum length of span in bridge	N
Super-structure type	Type of superstructure (e.g., PSC, RC)	N
Sub-structure type	Type of substructure	C
Design live load	Structural design criteria	C
Seismic design application	Seismic design applicability code	C
Average daily traffic	Average daily traffic volume passing over the bridge	N
Average daily truck traffic	Average daily truck traffic volume passing over the bridge	N

Note: PSC= pre-stressed concrete; RC= reinforced concrete,
N= Numerical variable, C= Categorical variable, S= String, D= Datetime

Table 3.2 Columns of the bridge inspection record

Column name	Data description	Data type
Inspection project number	Unique number of inspection project	S
Bridge number	Unique number used for bridge management	S
Element code	Codes representing the element	C
Element detail	Detailed category of element	C
Inspection date	Date of inspection was finished	D
Condition grade of bridge	Comprehensive grade of the defect status of the element (A: Excellent, B: Good, C: Usual, D: Bad E: Poor)	C
Condition grade of element	Comprehensive grade of the defect status of the bridge	C
Defect name	Type of defect	C
Defect quantity	Quantity of defect	N
Unit of defect	Unit of defect	C
Priority of repair	Order of defects that must be repaired first	C
Expected repair methods	Methods expected to be used when performing repairs	C
Expected repair cost	Estimated costs of carrying out repairs (low confidence)	N

Note: N= Numerical variable, C= Categorical variable, S= String, D= Datetime

Table 3.3 Columns of the bridge repair record

Column name	Data description	Data type
Repair project number	Unique number indicating bridge on which repairs were performed together	S
Bridge number	Unique number used for bridge management	S
Span number	Number of span where repairs were performed	C
Element code	Codes representing the element	C
Element detail	Detailed category of element	C
Repair date	Date of repair work was finished	D
Repair method	Methods of repair work	C
Quantity of repair method	Amount of repair methods applied	N
Unit of repair method	Unit of repair method	C
Unit cost of repair method	Unit cost for application of repair method	N

Note: N= Numerical variable, C= Categorical variable, S= String, D= Datetime

3.1.2 Current Limitations of BMS Data

The author draws the entity relationship diagram of the BMS as depicted in Figure 3.2. It can be seen that a one-to-many relationship is being formulated based on the “Bridge specifications” of the bridge. Herein, a one-to-many relationship means that multiple pieces of information are connected to information in one bridge. For example, one bridge can have multiple spans, multiple elements, multiple inspection records, and multiple repair records.

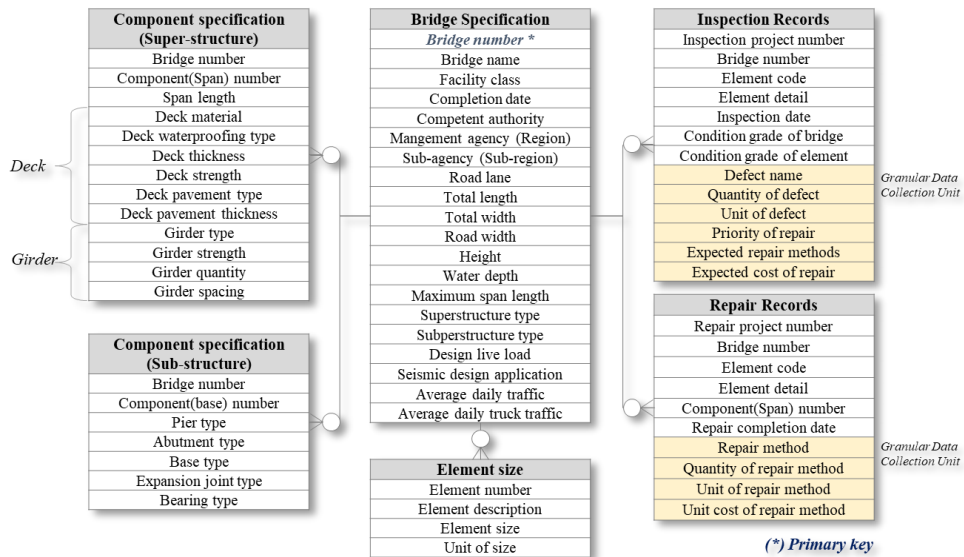


Figure 3.2 Entity relationship diagram (ERD) of BMS

In the existing relational database structure of the BMS, the author identifies several issues based on the ERD of BMS, as follows:

(1) Absence of Primary Keys in Tables:

In database management, the 'primary key' is an essential variable that uniquely identifies each record in the database (Storey 1991). It plays a crucial role in ensuring data integrity and in facilitating efficient data retrieval and relationship establishment. However, most tables in the BMS, except for the 'Bridge specification', lack a unique primary key for each row, potentially causing confusion in data integration. This issue is evident in 'Inspection records' (Tables 3.2) and 'Repair records' (Table 3.3), where the lack of a unique identifier complicates data analysis and necessitates the creation of primary keys for clarity.

(2) Table Denormalization:

As seen in Figure 3.3, the inspection records collect data at the defect unit level, with unnecessary duplication of information such as, 'Bridge number,' 'Element code,' 'Inspection code,' 'Inspection date,' and 'Condition grade of element'. To improve database integrity and query speed, it is imperative to eliminate redundant data.

(3) Unnecessarily Merged Rows:

Illustrated in Figure 3.2, multiple elements of a single span (e.g., deck and girder) are combined in a single row. This structure prevents each row

from representing an independent entity. Since the data processing unit vary for each bridge element, it is more analytically practical to create separate tables for each element.

(4) Inconsistency in Units:

A distinctive characteristic of bridge maintenance data is the discrepancy in units between the inspection records, which enters defect data at element level, and the repair record, which enters repair methods information at the span level. This inconsistency poses limitations in analyzing which repair methods have been applied to specific defects.

Bridge number	Element code	Inspection code	Inspection date	Condition grade of element	Defect name	Quantity of defect
010042	000	2015110155	20150720	D	재료분리	375.8
010042	000	2016120098	20160919	D	기타	5
010042	000	2016120098	20160919	D	망상균열	17.25
010042	000	2016120098	20160919	D	균열	10
010042	000	2016120098	20160919	D	누수및백태	1.5
010042	000	2016120098	20160919	D	누수및백태	49.3
010042	000	2016120098	20160919	D	박락	10.2
010042	000	2016120098	20160919	D	재료분리	2.5
010042	000	2016120098	20160919	D	파손	0.4
010042	000	2016120098	20160919	D	파손	9.3
010042	000	2016120098	20160919	D	재료분리	168
010042	000	2020110082	20200902	C	균열	0.16
010042	000	2020110082	20200902	C	철근노출/부식	1.2
010042	000	2020110082	20200902	C	박락	0.4
010042	000	2020110082	20200902	C	망상균열	18.75
010042	000	2020110082	20200902	C	누수및백태	1.5
010042	000	2020110082	20200902	C	박리	55
010042	000	2021100027	20210615	C	망상균열	24
010042	000	2021100027	20210615	C	철근노출/부식	0.03
010042	000	2021100027	20210615	C	박락	1.77
010042	000	2021100027	20210615	C	누수및백태	48
010042	000	2021100027	20210615	C	망상균열	400
010042	000	2021100027	20210615	C	박리	0.1

Figure 3.3 Current data duplication of inspection records

Bridge number	Element code	Span code	Repair code	Repair date	Repair method name	Repair cost
010042	000	1	2016120145	20170120	표면보수	12528
010042	000	1	2016120145	20170120	단면보수	4229
010042	000	1	2016120145	20170120	단면보수	2987
010042	000	2	2016120145	20170120	표면보수	1675
010042	000	2	2016120145	20170120	단면보수	296
010042	000	2	2016120145	20170120	단면보수	2308
010042	000	1	2017110040	20171222	도장	7496
010042	000	1	2017110040	20171222	단면보수	353
010042	000	1	2017110040	20171222	단면보수	3874
010042	000	2	2017110040	20171222	도장	6358
010042	000	2	2017110040	20171222	표면보수	17432
010042	000	1	2019020047	20181123	표면보수	46815

Figure 3.4 Current data duplication of repair records

To address these issues, the author has clearly defined the relationship between bridge elements and spans. A unique primary key at the element level was established, and defects identified in each ten of element were organized as shown in Table 3.4. For repair methods, a unique primary key at the span level was created, and the application of these methods at the element level was compiled as in Table 3.5. This reorganization of BMS data in the relational database structure facilitates a more effective and coherent analysis of bridge maintenance activities.

Table 3.4 Check of defects by bridge element

NO	Defects from inspection records	Deck (000)	Girder (100)	Crossbeam (200)	Pier/Abutment(300)	Base (400)	Bearing (500)	Expansion (600)	Pavement (700)	Drainage (800)	Curb/Railing (900)
		11,120	8,629	4,161	12,519	793	9,289	10,958	7,830	9,859	11,004
1	Key(Brdg_elmnt_pjt)	O	O	O	O	O	O	O	O	O	O
2	점검일자	O	O	O	O	O	O	O	O	O	O
3	부재등급	O	O	O	O	O	O	O	O	O	O
4	강재 변형	-	O	O	-	-	-	-	-	-	O
5	강재 파손	-	O	-	-	-	-	O	-	-	-
6	고무재 마모	-	-	-	-	-	-	O	-	-	-
7	고무재 변형	-	-	-	-	-	O	-	-	-	-
8	고무재 손상	-	-	-	-	-	O	-	-	-	-
9	고무재 파손	-	-	-	-	-	-	O	-	-	-
10	균열	O	O	O	O	O	O	-	O	-	O
11	기울음/전도	-	-	-	O	O	-	-	-	-	-
12	기능 저하	-	-	-	-	-	O	-	-	-	-
13	기타	O	O	O	O	O	O	O	O	O	O
14	길이부족	-	-	-	-	-	-	-	-	O	-
15	누수 및 백태	O	O	O	O	O(백태)	-	O(누수)	-	O(누수)	O(백태)
16	단차	-	-	-	-	-	-	-	O	-	-
17	도장 손상	O	-	-	-	-	-	-	-	-	-
18	도장 탈락	O	O	O	-	-	-	-	-	-	O
19	마모	-	-	-	-	-	-	-	O	-	-
20	망상균열	O	O	O	O	-	-	-	O	-	O
21	모르타르 손상	-	-	-	-	-	O	-	-	-	-

22	물고임	-	-	-	O	-	-	-	O	-	-
23	밀착(유간부족)	-	-	-	-	-	O	-	-	-	-
24	박락	O	O	O	O	O	-	-	-	-	O
25	박리	O	O	O	-	-	-	-	O	-	-
26	방수재(씰링재)파손	-	-	-	-	-	-	O	-	-	-
27	방수층 손상	-	-	-	-	-	-	-	O	-	-
28	배수관 파손	-	-	-	-	-	-	-	-	O	-
29	변형	O	-	-	-	-	-	O	-	-	O
30	볼트 체결불량	-	O	O	-	-	-	-	-	-	-
31	볼트 탈락불량	-	O	O	-	-	-	-	-	-	-
32	부속물 파손	-	-	-	-	-	O	-	-	-	-
33	부식	O	-	-	-	-	O	O	-	-	O
34	설계도면 상이	-	O	-	-	-	-	-	-	-	-
35	설치 누락/불량	-	-	-	-	-	-	-	-	O	-
36	세굴	-	-	-	-	O	-	-	-	-	-
37	소성변형	-	-	-	-	-	-	-	O	-	-
38	수면접촉부 침식	-	-	-	O	-	-	-	-	-	-
39	쉬스간노출/부식	-	O	-	-	-	-	-	-	-	-
40	연결부 불량	-	-	-	-	-	-	-	-	-	O
41	연결부 변형	-	O	-	-	-	-	-	-	-	-
42	연결부 부식	-	O	-	-	-	-	-	-	-	-
43	연결부 이완파손	-	-	-	-	-	-	O	-	-	-
44	연석 손상	-	-	-	-	-	-	-	-	-	O
45	오염	O	O	O	O	-	-	-	-	-	-
46	용접 불량	-	O	O	-	-	-	-	-	-	-

47	유간부족/과다	-	-	-	-	-	-	O	-	-	-
48	이물질(오물/토사)퇴적	-	O	-	-	-	-	O	O	O	-
49	재료분리(공동,공극)	O	O	O	O	-	-	-	-	-	-
50	정착구 부식	O	O	-	-	-	-	-	-	-	-
51	철근노출/부식	O	O	O	O	O	-	-	-	-	O
52	채수	-	O	-	-	-	-	-	-	O	-
53	측방유동	-	-	-	-	O	-	-	-	-	-
54	침식	-	-	-	-	O	-	-	-	-	-
55	침하	-	-	-	-	O	-	-	-	-	-
56	콘크리트 백태	O	-	-	-	-	-	-	-	-	-
57	파손	O	O	O	O	-	-	-	O	O	O
58	편기	-	-	-	-	O	-	-	-	-	-
59	표면 부식	-	O	O	-	-	-	-	-	-	-
60	표면 오염	-	O	-	-	-	-	-	-	-	-
61	피복두께 부족	O	-	-	-	-	-	-	-	-	-
62	함몰(포트홀)	-	-	-	-	-	-	-	O	-	-
63	후타재 균열	-	-	-	-	-	-	O	-	-	-
64	후타재 단차	-	-	-	-	-	-	O	-	-	-
65	후타재 파손	-	-	-	-	-	-	O	-	-	-

* O = Include, - = Not include

Table 3.5 Check of the repair methods by bridge element

Repair methods from repair records	Deck (000)	Girder (100)	Crossbeam (200)	Pier/Abutme nt(300)	Base (400)	Bearing (500)	Expansion (600)	Pavement (700)	Drainage (800)	Curb/Railing (900)
	8,189	4,175	977	9,846	116	5,038	6,537	12,890	2,938	14,717
Key(Brdg elmnt span_pjt]	O	O	O	O	O	O	O	O	O	O
보수일자	O	O	O	O	O	O	O	O	O	O
주입(및충전)공법	O	O	O	O	-	-	-	-	-	-
표면보수	O	O	O	O	-	-	-	-	-	-
아스팔트 표면보수	-	-	-	-	-	-	-	O	-	-
콘크리트 표면보수	-	-	-	-	-	-	-	O	-	-
(콘크리트)단면보수	O	O	O	O	O	-	-	-	-	-
도장	O	O	O			O	-	-	-	-
균열방지공법	O	-	-	-	-	-	-	-	-	-
외부긴장	-	O	-	O	-	-	-	-	-	-
중형증설	-	-	O	-	-	-	-	-	-	-
단면증설	-	-	-	O	-	-	-	-	-	-
볼트교체	-	O	O	-	-	-	-	-	-	-
보강(보강판)	-	O	-	-	-	-	-	-	-	-
보강재 덧댐공법	O	-	-	-	-	-	-	-	-	-
충전공법	-	-	O	O	-	-	-	-	-	-
접착공법	-	-	-	O	-	-	-	-	-	-
새굴방지공	-	-	-	-	O	-	-	-	-	-
받침본체 보수	-	-	-	-	-	O	-	-	-	-
받침대 보강	-	-	-	-	-	O	-	-	-	-
교량받침 교체	-	-	-	-	-	O	-	-	-	-
고무재 교체	-	-	-	-	-	O	-	-	-	-
고무재 기타	-	-	-	-	-	O	-	-	-	-

신축이음 보수	-	-	-	-	-	-	O	-	-	-
후타재 보수	-	-	-	-	-	-	O	-	-	-
신축이음 교체	-	-	-	-	-	-	O	-	-	-
아스팔트 팻칭	-	-	-	-	-	-	-	O	-	-
아스팔트 절삭오버레이	-	-	-	-	-	-	-	O	-	-
콘크리트 팻칭	-	-	-	-	-	-	-	O	-	-
표면청소	-	-	-	-	-	-	-	O	-	-
콘크리트 포장교체	-	-	-	-	-	-	-	O	-	-
아스팔트 포장교체	-	-	-	-	-	-	-	O	-	-
아스팔트 기타	-	-	-	-	-	-	-	O	-	-
배수시설 보수	-	-	-	-	-	-	-	-	O	-
배수시설 교체	-	-	-	-	-	-	-	-	O	-
유입구 보수	-	-	-	-	-	-	-	-	O	-
유입구 교체	-	-	-	-	-	-	-	-	O	-
방호울타리 보수	-	-	-	-	-	-	-	-	-	O
연석 보수	-	-	-	-	-	-	-	-	-	O
방호울타리 교체	-	-	-	-	-	-	-	-	-	O
연석 교체	-	-	-	-	-	-	-	-	-	O
RC기타	O	-	-	-	-	-	-	O	-	O
STEEL기타	-	O	O	-	-	O	-	-	-	O
PC기타	-	O	-	-	-	-	-	-	-	-
기타	O	O	O	O	O	O	O	O	O	O

* O = Include, - = Not include

3.1.3 Results of BMS Data Transformation

The primary key must be created by combining information that can indicate the characteristics of each data (e.g., 'bridge number_span number_project number').

- Inspection record: Bridge number_Element number_Project number (e.g., “10042_000_2015110155”)
- Repair record: Bridge number_Element number_Span number_Project number (e.g., “10045_000_1_2022050037”)

Based on its primary key, each row in the database represents a distinct record. In the case of inspection, the quantity of defects is aggregated and entered under each respective defect type column. Similarly, for repair quantity, the repair cost is cumulatively entered according to each repair method column. This approach allows for a meaningful interpretation when a single row is queried, revealing the extent of defect sustained and the level of repairs carried out. The transformed data can be viewed in Figure 3.5 (for inspection record) and 3.6 (for repair record). These revised data structures will serve as the fundamental entity units in graph database modeling. This reorganization not only enhances data clarity but also significantly improve the efficiency of data retrieval and analysis in bridge management.

BRDG_ELMT_PJT	INSPCT_DATE	ELMT_GRADE	균열_QTY	기타_QTY	누수및배터도장손상	(도장탈락)	(양상균열)	(박락_QTY)	박리_QTY	변형_QTY	부식_QTY	오염_QTY	재료분리(정착구)	부-함근노출/콘크리트박손_QTY	피복두께누
10042_0_2015110155	2015-07-20	D	0	0	0	0	0	0	0	0	0	0	375.8	0	0
10042_0_2016120098	2016-09-19	D	10	5	50.8	0	0	1725	102	0	0	0	170.5	0	9.7
10042_0_2020110082	2020-09-02	C	0.16	0	1.5	0	0	1875	0.4	55	0	0	0	1.2	0
10042_0_2021100027	2021-06-15	C	0	0	48	0	0	424	1.77	0.1	0	0	0	0.03	0
10044_0_2016120091	2016-09-22	C	169	0	14	0	0	0	69.5	0.75	0	0	0	0.04	0
10044_0_2021010079	2020-12-07	C	12.44	0	0	0	0	0	0	0	0	0	15.93	1.32	0
10045_0_2016120091	2016-09-22	C	108	0	91.54	0	0	0	34.9	0.5	0	0	0	5.3	0
10045_0_2021010079	2020-12-07	C	207.62	0	0	0	0	0	0	0	0	0	0	0.3	6.09
10046_0_2015103181	2015-02-01	C	7.5	0	22.5	0	0	0	0	0	0	0	0	3.22	0
10046_0_2021010079	2020-12-07	C	229.5	0	44.63	0	0	0	9.74	0	0	0	0	0.21	0
1047_0_2017090001	2017-05-08	B	26.8	2	0.64	0	0	0	8.86	0	0	0	0	1.08	0
1047_0_2019100056	2019-12-26	C	275.97	2	3.76	0	0	0	0.4	0	0	0	0	2.5	0
1055_0_2020110087	2020-06-11	B	44.6	0	6.5	0	0	0	0	0	0	0	0	0	0
106_0_2016120113	2016-12-16	C	92.14	57.79	2.1	0	0	1.5	0	11.33	0	0	1	0.59	0
106_0_2021010192	2020-06-10	C	1593.2	85	0	0	0	0	0	0	0	0	14.07	2.55	0

Figure 3.5 Modified data of inspection records

BRDG_ELMT_GG_PJT	CMPLT_DT	COST_주입공발	COST_표면보수	COST_단면보수	COST_균열방지	COST_도장	COST_RC	COST_보강	COST_기타	COST_합계
10042_000_1_2016120145	2017-01-20	0	12528	7216	0	0	0	0	0	19744
10042_000_1_2017110040	2017-12-22	0	0	4227	0	7496	0	0	0	11723
10042_000_1_2019020047	2018-11-23	0	46815	0	0	0	0	0	0	46815
10042_000_2_2016120145	2017-01-20	0	16751	2604	0	0	0	0	0	19355
10042_000_2_2017110040	2017-12-22	0	17432	0	0	6358	0	0	0	23790
10044_000_1_2017110040	2017-12-22	97	218	13613	0	0	0	0	0	13928
10044_000_1_2022050043	2021-12-28	0	0	1586	0	0	0	0	0	1586
10044_000_2_2017110040	2017-12-22	0	5350	17843	0	0	0	0	0	23193
10044_000_2_2022050043	2021-12-28	0	242	6548	0	0	0	0	0	6790
10044_000_3_2017110040	2017-12-22	0	146	17361	0	0	0	0	0	17507
10044_000_3_2022050043	2021-12-28	0	0	2936	0	0	0	0	0	2936
10045_000_1_2017110055	2017-12-19	0	4756	9935	0	0	0	0	0	14691
10045_000_1_2022050037	2021-10-13	0	2493	1202	0	0	0	0	0	3695
10045_000_2_2017110055	2017-12-19	3596	3298	11587	0	0	0	0	0	18481
10045_000_2_2022050037	2021-10-13	0	6581	317	0	0	0	0	0	6898
10045_000_3_2017110055	2017-12-19	0	0	18608	0	0	0	0	0	18608
10045_000_3_2022050037	2021-10-13	0	6427	306	0	0	0	0	0	6733
10046_000_1_2016020134	2015-12-19	1108	0	13960	0	0	0	0	0	15068
10046_000_2_2016020134	2015-12-19	595	0	669	0	0	0	0	0	1264
1047_000_1_2022040260	2021-11-05	0	3794	51586	0	0	0	0	0	55380
1047_000_2_2022040260	2021-11-05	0	2234	54767	0	0	0	0	0	57001

Figure 3.6 Modified data of repair records

3.2 Data Processing of Unstructured Data

This chapter summarizes the process of structuring key knowledge by preprocessing unstructured data related to bridge maintenance.

3.2.1 Collection of the Unstructured Data

The data encompassing regulations and procedures necessary for the maintenance of bridges, often composed by complex document structures and extensive content. Providing essential information to manage bridges effectively is crucial for bridge management professionals. Such information is indispensable for bridge management professionals, aiding them in effective maintenance strategies. To streamline the acquisition and identification of this essential knowledge, the author has compiled a list of key unstructured data sources in Table 3.6. This table was derived from South Korea's Facility Management System (KALIS 2023a), encompasses fundamental laws and guidelines integral to the work of bridge maintenance personnel. The listed documents, accessible via the FMS platform (KALIS 2023a) and the Korean Law Information Center (Legislation 2023), were initially collated in PDF format. Utilizing the PyPDF2 library, the size of the data was quantified by the word count, highlighting the voluminous nature of these documents. As Table 3.6 illustrates, the extensive word count across various documents poses a significant challenge for facility maintenance

managers in efficiently locating specific information, underscoring the importance and necessity of effectively organizing and accessing unstructured data in bridge maintenance.

Table 3.6 List of the collected unstructured data

Category	File title	Number of words	Number of tables
/Law (from Legislation)	시설물의 안전 및 유지관리에 관한 특별법 (시설물안전법)	7,695	-
	The special act on safety and maintenance of facilities		
	시설물의 안전 및 유지관리에 관한 특별법 시행령 The enforcement ordinance on safety and maintenance of facilities	5,076	16
	시설물의 안전 및 유지관리에 관한 특별법 시행규칙 The enforcement regulation on safety and maintenance of facilities	3,007	6
/Manuals(from Guideline)	시설물의 안전 및 유지관리 실시 등에 관한 지침 Guidelines on the implementation of safety and maintenance for facilities	20,308	28
	시설물의 안전 및 유지관리 실시 세부지침(안전점검·진단편)_교량편 Detailed guidelines for the implementation of safety and maintenance of facilities (safety inspection and diagnosis edition) – Bridge section	47,083	
	시설물의 안전 및 유지관리 실시 세부지침(성능평가편)_교량편 Detailed guidelines for the implementation of safety and maintenance of facilities (performance evaluation edition) – bridge section	105,148	
	시설물 통합정보관리체계 운영규정 Operating regulations for the integrated information management system of facilities	20,894	

3.2.2 Pattern Extraction from Unstructured Data

To structuring the unstructured documents, the process of manually checking the documents is necessary. As shown in Table 3.7, the document of legislation or laws were composed of text with semantic hierarchy. In the existing law, the semantic hierarchy was structured from higher-level concept consists of “Chapter,” “Section,” “Article,” “Paragraph,” “Subparagraph,” and “Item” that encompass lower-level concepts. Through this, the author gained a clue that the structure can be modeled based on the upper and lower-level structures of the sentences.

Table 3.7 Hierarchy of the sentence in ‘Special act’

Category	Expressions	Description example
장 (Chapter)	제1장, 2장, 제3장, ...	제1장 총칙
절 (Section)	제1절, 2절, 제3절, ...	제1조(목적)
조 (Article)	제1조, 2조, 제3조, ...	제1항 이 법은 시설물의 안전점검과 적절한 유지관리를 통하여 재해와 재난을 예방하고 시설물의 효용을 증진시킴으로써 공중의 안전을 확보하고 나아가 국민의 복리증진에 기여함을 목적으로 한다.
항 (Paragraph)	①, ②, ③, ...	제1항 시설물의 종류는 다음 각 호와 같다.
호 (Subparagraph)	1, 2, 3, ...	제1호 제1종시설물: 공중의 이용편의와 안전을 도모하기 위하여 특별히 관리할 필요가 있거나 구조상 안전 및 유지관리에 고도의 기술이 필요한 대규모 시설물로서 다음 각 목의 어느 하나에 해당하는 시설물 등 대통령령으로 정하는 시설물
목 (Item)	가, 나, 다, ...	가)목 고속철도 교량, 연장 500미터 이상의 도로 및 철도 교량

As when processing the structured data of BMS, a unique key was created for each sentence-level text in unstructured data. A unique key was created by combining the hierarchy names representing each sentence (See Figure 3.7)

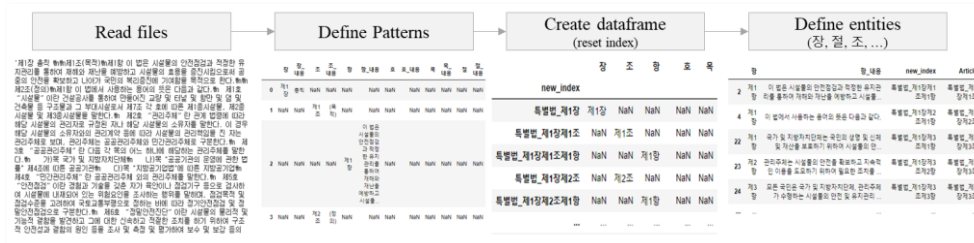


Figure 3.7 Data extraction process from unstructured document

The documents such as instructions and user manuals also contain images and tables, which are used to support actual maintenance tasks. Especially, the information buried in unstructured data, important standards are contained in tables. Thus, the author considered several techniques that extract the tables in more accurate form. “Tabula-py,” “pdfminer,” and “camelot-py” package were tested and finally selected the “camelot-py” package, which gave the best results. In dataframe format tables extracted from PDF, the columns and rows do not match and there are cases where unnecessary characters are included (see Figure 3.8). Accordingly, the author manually checked the number of extracted tables and the results, established rules, and selected key information in a short amount of time.

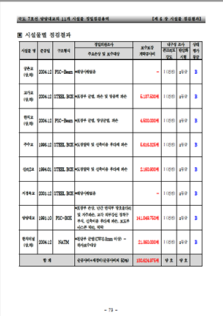
Read the tables in PDF	Extract the tables	Standardize the table using the Regex
		

Figure 3.8. Table extraction process from unstructured document

A package that can extract the images from PDF files is also supported in Python. But it was excluded in the scope of this dissertation. The detailed versions of Python, each package, and library used in Chapter 3, were summarized in Table 3.8.

Table 3.8 System environment of the knowledge base development

Category	Name	Description with version information
Language	Python	3.9.12
Software	Neo4j	Community-4.4.21
Python packages	Regex	2022.3.15
	PyPDF2	3.0.1 (3.12.2)
	camelot-py	0.11.0
	Py2neo	Version: 2021.2.3
	neo4j	Python client library and toolkit for Neo4j Version: 5.9.0 Neo4j Bolt driver for Python

3.2.3 Structuring of the Unstructured Data

Data cleaning is necessary as the final step in formalizing unstructured data. Regular expressions, often abbreviated as “regex”, is a powerful tool in the Python programming language used for pattern matching and text manipulation (Chapman and Stolee 2016). When processing unstructured data, regular expressions were used in a more complex but important way than general special character processing. In many cases, regular expressions are mainly used when processing special characters in a sentence or extracting a desired word. Basically, it was used to delete Chinese characters or symbols included in sentences in the special laws and detailed guidelines on facility maintenance. However, more importantly, the document structure was stratified and cut using regex to create entities and relations. The detailed description is as shown in the pseudo-code in Figure 3.9.

Each index has the following format:

- Index of legislation and law: "document name"_"chapter number"_"section number"_"article number"_"paragraph number"_"subparagraph number"_"item number"

1. Open the file '02_시설물의안전및유지관리에관한특별법_시행령(대통령령).txt' for reading with UTF-8 encoding.
2. Read the contents of the file into a variable.
3. Define regular expression patterns for the hierarchical structure:
 - Chapter: Regular expression pattern
 - Section: Regular expression pattern
 - Article: Regular expression pattern
 - Paragraph: Regular expression pattern
 - Subparagraph: Regular expression pattern
 - Item: Regular expression pattern
4. Initialize an empty list to store data.
5. Split the file content into individual lines.
6. For each line in the split content:
 - a. Trim any leading or trailing whitespace.
 - b. For each pattern (Chapter, Section, Article, Paragraph, Subparagraph, Item):
 - i. Check if the current line matches the pattern.
 - ii. If a match is found, extract and store the relevant data, then stop checking further patterns.
7. If extracted data is not empty, append it to the data list.
8. Define the order of columns for data presentation.
9. Create a new index for each row:
 - a. For each row in the dataset:
 - i. Initialize an empty list for indices.
 - ii. For each column in the row, in reverse order:
 - If the cell is not empty, prepend its value to the indices list.
 - iii. Join the collected indices into a single string with '시행령_' as a prefix.
 - iv. Add the new index string to the list of indices.
10. Set the new index as the primary index of the dataset.
11. Reset the index of the dataset to include the new index.
12. Create sub-datasets for each hierarchical structure (Chapter, Section, Article, Paragraph, Subparagraph, Item) including the new index.
13. Add columns to the sub-datasets to indicate their respective hierarchical levels based on the new index.
14. Close the file.

Figure 3.9. Pseudo-code for processing of unstructured data

Table 3.9 Example of the flattened unstructured data

Higher Entity	Index	Article number	Article content
시행규칙_제1장	시행규칙_제1장_제1조	제1조	(목적)
시행규칙_제1장	시행규칙_제1장_제2조	제2조	(안전점검의 종류)
시행규칙_제2장	시행규칙_제2장_제3조	제3조	(안전 및 유지관리계획의 제출)
시행규칙_제2장	시행규칙_제2장_제4조	제4조	(제3종시설물의 지정 및 해제 통보등)
시행규칙_제2장	시행규칙_제2장_제5조	제5조	(제3종시설물의 지정 · 해제 요청)
시행규칙_제2장	시행규칙_제2장_제6조	제6조	(설계도서 등의 제출)

In this way, a data frame was created for each class to prepare for modeling to be performed in the next chapter. An entity was created based on the pattern of this document, and each entity has a description as a property. It is modeled so that each sub-syntax is included under the upper-level syntax (see Table 3.9). For more information, please refer to the code in Appendix A.

3.3 Graph Database Modeling

Database modeling is the most important part of this dissertation. The form in which the knowledge base is created greatly affects the final performance of a specific task (in this case, question answering).

The differences between the existing relational database and the relationship-based graph database can be seen in the data structure, search method, and processing method. First, the data structure of relational database is individual records in the form of a table composed of rows and columns, a structure that is limited in expressing complex relationships, but graph database modeling based on nodes, edges, and properties expresses the connectivity between complex data based on relationships. As an example, to find records that meet specific conditions in a csv file, you must search the entire file or most of it, but graph queries enable efficient search based on the connectivity of nodes. There are also differences in data processing methods. Compared to relational database, which can have significant memory and performance problems when filtering data in a specific column or finding rows that meet conditions, graph database has a clear difference in that it can be searched concisely based on relationships.

Here, modeling the preprocessed data into a knowledge graph means structuring it to represent complex interrelationships between various entities and their properties. This process transforms structured data into network

entities and relationships, providing a visual and semantic framework to help data exploration, retrieval, and reasoning.

3.3.1 Grammar of the Graph Database

Among several graph database platform, Neo4j (Neo4j 2023) was selected as the software for implementing the graph database. It stands as the most renowned graph database management system (Angles 2012), recognized for its efficiency, scalability, and intuitive graph modeling capabilities. The language as known as Cypher supports the querying graph data. Rooted in pattern matching principles, Cypher is designed to express complex graph traversals with clarity and conciseness. Nodes are typically encapsulated within parentheses (node) while relationships are denoted with arrows (node)—[rel]-> (node). Attributes or properties can be filtered using curly braces {} (see Figure 3.5). Cypher supports various clauses, including ‘MATCH,’ ‘RETURN,’ ‘CREATE,’ and ‘DELETE,’ among others, to conduct operations ranging from data retrieval to manipulation.

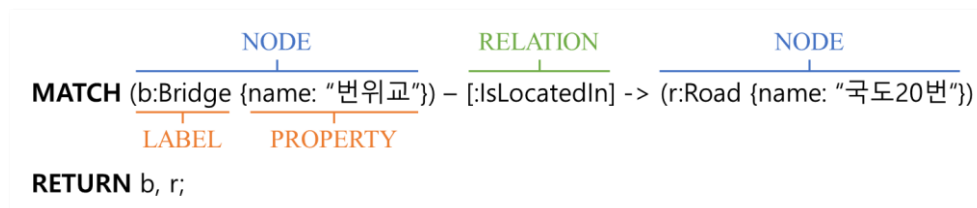


Figure 3.10 Basic grammar of Cypher query

If the query was executed as shown in Figure 3.10, the graph for the ‘변위교’ Bridge located on National Highway 20 could be plotted in the browser. In order to create an environment of graph database generation, the community edition of Neo4J was downloaded to a local Ubuntu server, and the network was opened to check the results through a browser. Additionally, scripting was possible using the "py2neo" and "neo4j" packages, which allow data input and manipulation to be performed based on Python.

3.3.2 Modeling of the Bridge Maintenance Knowledge

To capture the knowledge of bridge maintenance, semantic modeling of the data was performed in graph format. Semantic modeling of data means capturing the entities, relationships, and constraints within a particular domain in a way that machines can understand (Hull and King 1987). The goal of modeling is to represent data in a form that is close to the logical structure of an individual or organization's perspective on the world. Ontology creation, graph data design, and knowledge representation fall into these categories. In the process of semantic modeling of the bridge maintenance graph database, the author referred to existing research. Unlike previous research, this dissertation mainly used BMS data and unstructured

documents related to maintenance guidelines, so there were some areas where additional approaches were needed as shown in previous chapters.

First, the physical characteristics of the bridge were modeled. A general bridge consists of a total of 10 elements, including deck, girder, crossbeam, abutment/pier, foundation, bearing, expansion joint, pavement, drainage, and railing/curb. Each element must be modeled as independent entities because it is the unit where maintenance is performed, and data is collected. As shown in Table 3.10, entities for a total of 8,365 bridges were created based on the primary key, and entities were created for the 10 major elements included in each bridge. Depending on the size and type of the bridge, some elements may not be included.

Additionally, bridge elements are divided into super-structure and sub-structure. The super-structure composed of several spans including the deck, girder, and crossbeams. The sub-structure mainly composed of each point of bridge including abutments/piers and foundations. Other than that, bearing, expansion joints, pavement, drainage facilities, and railing/curbs are classified as secondary members. In this dissertation, data on the deck of the super-structure and the abutment/pier range of the sub-structure were entered into a graph database in detailed units of component. Depending on the bridge, it has three to five components of spans. The inspection history and

maintenance history processed in the previous chapter also created entities for the deck and pier/abutment members.

Table 3.10 Number of BMS data input in Neo4j

Table name		Number of entities	Primary key (index)
Bridge basic specification		8,365	BRDG_SEQ
Bridge element	Deck	8,194	BRDG_ELMT
	Girder	5,108	
	Secondary element	4,999	
	Pier/Abutment	8,194	
	Foundation	8,194	
	Bearing	6,232	
	Expansion joint	6,862	
	Pavement	8,194	
	Drainage	8,194	
	Railing curb	8,194	
Bridge component	Deck	32,720	BRDG_ELMT_GG
	Pier/Abutment	36,783	BRDG_ELMT_JJ
Bridge inspection	Deck	11,120	BRDG_ELMT_PJT
	Pier/Abutment	12,519	BRDG_ELMT_PJT
Bridge Repair	Deck	8,189	BRDG_ELMT_GG_PJT
	Pier/Abutment	9,846	BRDG_ELMT_GG_PJT
Bridge repair project		3,896	RPRPJT_SEQ
Roadway		186	ROADLANE_CD
Bridge Management Agency		13	SUP_MNG_AGENCY
Sub-Agency		40	SUB_MNG_AGENCY

Table 3.11 shows the entity information of the ‘Special Act’ on bridge maintenance summarized in the previous chapter. After generating the entity, the relations between them were modeled as shown in Table 3.12. Here, the name of relations is an important clue to generate the query for QA, so it was selected by considering words that users might use in their questions. In

addition, as a result of a series of experiments, it was confirmed that the directionality of the start node and end node also affects QA performance, so the entities required for complex queries were intentionally oriented so that they could be connected in one direction.

Table 3.11 Number of unstructured data input in Neo4j (Special Act)

Data	Special act	Ordinance	Regulation
Title	1	1	1
Chapter	7	8	6
Article	57	43	37
Paragraph	223	128	91
Subparagraph	164	181	62
Item	17	7	2

Table 3.12 Relationships between bridge maintenance entities

Category	Relationship	Start node	End node
Unstructured Data	IsPartOf	Article	Chapter
	Constitutes	Paragraph	Article
	IsContainedIn	Subparagraphs	Paragraph
	BelongsTo	Item	Subparagraph
	DelegatesTo	Special Act	Ordinance
		Ordinance	Regulation
System Data	LinksTo	Special Act	Article
	Has	Bridge	Element
	IsInspectedBy	Element	Inspection History
	IsRepairedBy	Element	Repair History
	Onthe	Bridge	Roadway
	Composed	Component	Element
	IsManagedBy	Bridge	Sub-agency
	Under	Sub-agency	Agency
	IsRepairedTogether	Repair project	Repair History
	IsFollowedBy	Inspection History	Repair History

```

# Node generation
For _, row in df.iterrows():
    node_properties = {k: float(v) if isinstance(v, np.float64) else v for k, v in row.items()}
    node = Node("BrdgComponent_substructure", **node_properties)

    # Save node on Neo4j
    graph.create(node)

# Relation generation
for _, row in df.iterrows():
    BRDG_ELMT_JJ = row['BRDG_ELMT_JJ']
    BRDG_ELMT = row['BRDG_ELMT']
    brdg_elmt = graph.nodes.match("BrdgElmt", BRDG_ELMT=BRDG_ELMT).first()
    brdg_elmt_jj = graph.nodes.match("Brdg_substructure", BRDG_ELMT_JJ=BRDG_ELMT_JJ).first()

    if brdg_elmt_jj is not None and brdg_elmt is not None:
        # generate relationship
        relationship = Relationship(brdg_elmt_jj, "Composed", brdg_elmt)

        # Save relationship on Neo4j
        graph.create(relationship)

```

Figure 3.11 Python code for generating node and relationships in Neo4j

In the process of adding entities and relations to the graph database, it was confirmed that input time could be reduced by creating a unique index for entities. The main relations between modeled entities from BMS data can be schematized as in Figure 3.12. It shows the management office that manages the bridge, the bridge on which it is located, the bridge's structure, and its maintenance history.

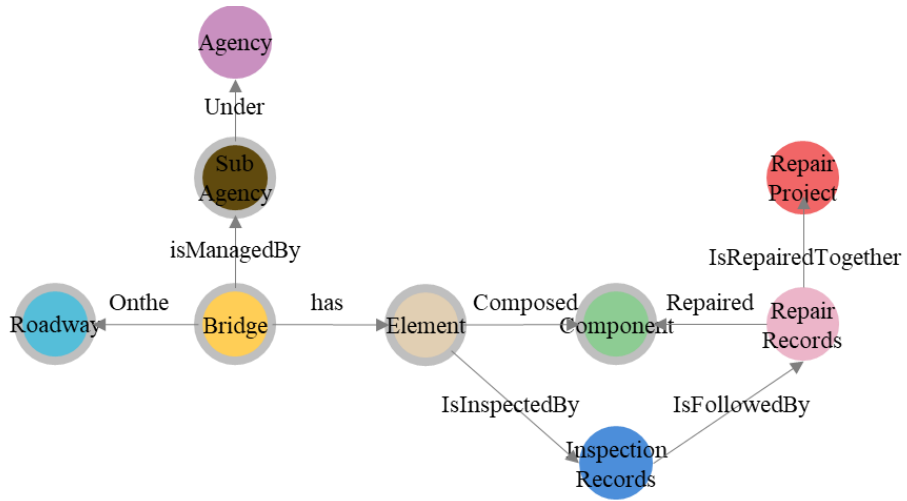


Figure 3.12 Relationship of bridge maintenance entities

The author considered that the time relationship between inspection and repair history could be important information for QA and modeled the time relationship as shown in Figure 3.13. Of course, each inspection and maintenance history have information on the time of completion as a property, but by modeling the repair history following the inspection, it was possible to express relationships between data that could not be expressed in the existing relational DB.

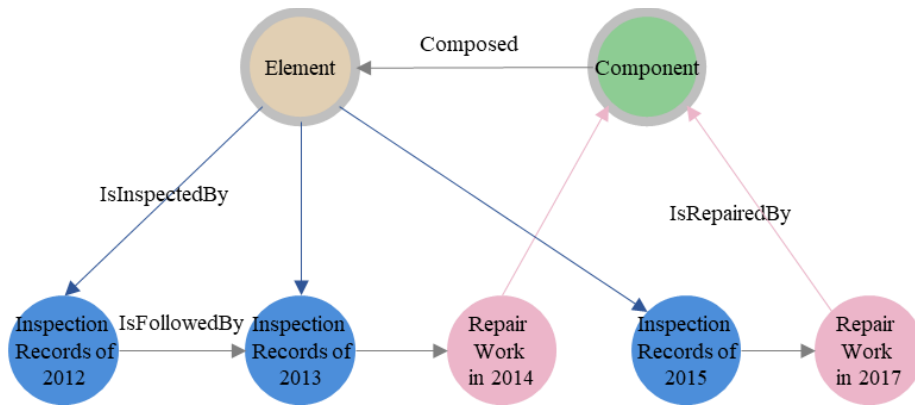


Figure 3.13 Relationship modeling between inspection and repair data

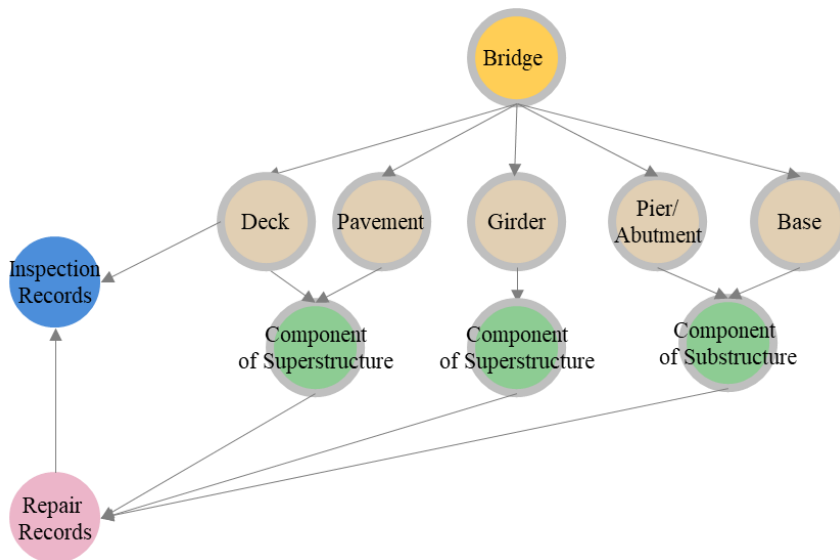


Figure 3.8 Hierarchy modeling of bridge structure

The key consideration when modeling the relation is that the bridge's inspection history is entered at the element level, and the bridge's repair history is entered at the span level. In the existing relational database (i.e., BMS), this was a factor that greatly slowed down data analysis. Since the

level information of the two separate tables was different, information loss occurred when the span-unit values were combined through an additional process of summing them element level, or when representative values were merged. However, in the graph database, the modeling is a structure that allows for additional inference without loss of information. The schema of the modeled knowledge base can be loaded in the neo4j browser as follows (Figure 3.15).

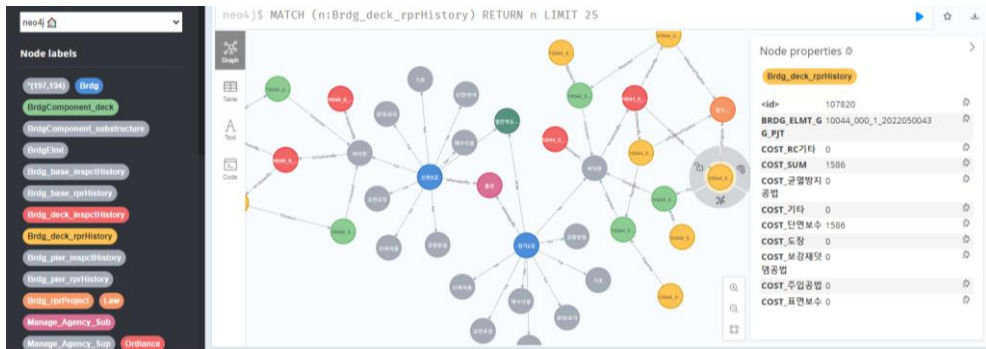


Figure 3.15 Schema of bride maintenance knowledge base

3.4 Verification of Bridge Maintenance Knowledge Base

The usability of the constructed knowledge base can be evaluated by examining whether it can answer competency questions. In this chapter, the author describes which competency questions can be answered with the proposed knowledge base. Afterwards, the author discusses additional information that can be obtained through the created knowledge base.

3.4.1 Evaluation using Competency Questions

A competency question (CQ) shows a pattern for a type of question that an ontology must be able to answer (Uschold and Gruninger 1996). CQs serve as functional requirements for the ontology, which guide both the development and evaluation of the ontology. Answering competency questions and updating the knowledge models is an interactive process that can help knowledge engineers continually improve the quality of ontology.

To verify the performance of the developed knowledge base, questions related to unstructured data and questions related to structured data were created based on raw data. Sample CQ can be found in Table 3.14. According to Table 3.14, you can see that the graph database creates a cypher query appropriate for the competency question and returns an answer.

Table 3.14 Competency questions (sample)

Category	Question	Answer	Cypher Query Generation
Bridge characteristics	일반국도39호선과 연결된 교량 10개만 알려줘	일반국도39호선과 연결된 교량 10개는 다음과 같습니다: 중추교. 미당교. 청흥교. 입동교. 유구교. 추계교. 제1거산교. 방미교. 현곡교. 풍무교'	MATCH (b:Brdg)-[:Onthe]->(r:RoadLane) WHERE r.ROADLANE_NAME = '일반국도39호선' RETURN b.BRDG_NAME LIMIT 10
	길천교는 일반국도42호선위에 있어?	네, 길천교는 일반국도42호선 위에 있습니다.	MATCH (b:Brdg)-[:Onthe]->(r:RoadLane) WHERE b.BRDG_NAME = '길천교' AND r.ROADLANE_NAME = '일반국도42호선' RETURN b, r
	부사원3교는 어느 국도에 있어?	부사원3교는 일반국도5호선에 있습니다.	MATCH (b:Brdg {BRDG_NAME: '부사원3교'})-[:Onthe]->(r:RoadLane) RETURN r.ROADLANE_NAME
	A 교량의 부재는 무엇이 있어?	교량의 부재에는 난간/연석, 배수시설, 교면포장, 신축이음, 교량받침, 기초, 교대/교각, 2차부재, 주형, 바닥판이 있습니다.	MATCH (b:Brdg)-[:has]->(be:BrdgElmt) RETURN b.BRDG_NAME, be.ELMT_NAME
Inspection	번천2교의 바닥판과 연결된 점검이력 몇개야?	번천2교의 바닥판과 연결된 점검이력은 4개입니다.'	MATCH (b:Brdg {BRDG_NAME: '번천2교'})-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(bd:Brdg_deck_inspctHistory) RETURN COUNT(bd)
	번천2교의 바닥판과 연결된 점검이력 날짜는?	번천2교의 바닥판과 연결된 점검이력의 날짜는 2015년 8월 17일, 2017년 6월 23일, 2019년 12월 17일, 2021년 6월 9일입니다.'	MATCH (b:Brdg {BRDG_NAME: '번천2교'})-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(bd:Brdg_deck_inspctHistory) RETURN bd.INSPECT_DATE
	번천2교 바닥판은 78_0이다. 이때 모든 점검이력의 누수및백태 양의 합은?	번천2교 바닥판의 누수및백태 양은 147.49입니다. 이때 모든 점검이력의 누수및백태 양의 합은 147.49입니다.'	MATCH (b:Brdg)-[:has]->(be:BrdgElmt {BRDG_ELMT: '78_0'}) MATCH (be)-[:IsInspectedBy]->(i:Brdg_deck_inspctHistory) RETURN SUM(i.`누수및백태_QTY`) AS TotalLeakageAndBackfill

3.4.2 Insights Derived from Graph Database

- **Comparison of the Retrieval Performance between Vector DB and Graph DB**

Data retrieval is a fundamental function of any QA system, and it is important to understand which database is right for the QA task. The author tried to compare the performance of vector database and graph database for the specific task of data retrieval.

A vector database is a system specifically optimized for storing and querying data in the form of vectors. With the recent development of large language model, there is increasing research on implementing the vector database, which is the base for search, in vector format (Lappalainen and Narayanan 2023). This is a structure that chunks unstructured data, vectorizes it, and then generates an answer based on the similarity of the vectors. The author briefly created vector database with the unstructured PDF file, same as the unstructured data used in graph database generation. Chroma, an AI-based open-source vector database focused on developer productivity, was selected as vector database (see Figure 3.16). The parameter settings were performed as follows in Table 3.15, and the PDF files shown in Table 3.6 were placed in the file directory and loaded into the development environment through the PDF read process. Since multiple PDFs are read, gpt-3.5-turbo-

16k was selected as the model considering the maximum number of tokens that can read the schema.

Table 3.15 Parameter setting of the vector database

Category	Description
Database	Chroma DB
Chunk size	1,000
Chunk_overlap	100
Embedding	OpenAIEmbedding
Model	gpt-3.5-turbo-16k
Max tokens	16,384

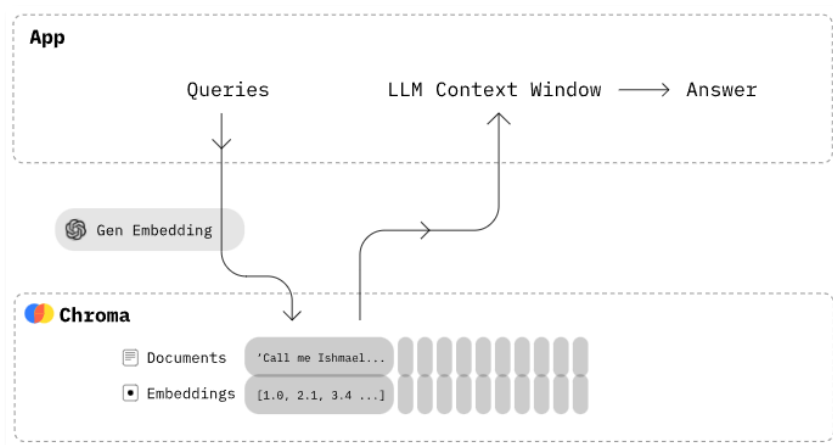


Figure 3.16 Principle of Chroma DB

As a result of entering the competency questions (See Figure 3.17), it was confirmed that there was a clear difference in answer generation in vector database and graph database. The vector database finds and returns the location of the chunk containing the natural language most similar to the question. This can quickly find the location of the desired answer and has

excellent performance for simple searches. However, usability is limited in the following cases: First, (1) when asked about the structure of the document, the legal documents used in this study are documents with hierarchy, and most well-written documents are as well. In graph DB, it is possible to generate answers to structural questions because it has been modeled structurally, but this is not possible in vector DB because there is no clue to understand the structure of the entire document. This suggests that a complex DB is needed to increase usability in QA.

Vector Database RetrievalQAChain	Execution Time: 2.5961 seconds <pre>{ 'query': '시설물의 안전 및 유지관리에 관한 특별법은 몇개의 장으로 구성되어 있지?', 'result': '시설물의 안전 및 유지관리에 관한 특별법은 총 3장으로 구성되어 있습니다.' }</pre> source_documents: [Document(page_content='시설물의안전및유지관리에관한특별법시행...법제처 -2/8-한 국가법령정보센터제5조(제3종시설물의지정·해제요청)①제1종시설물및제2종시설물의시설물의관리주체가영제5조제2항한각호외의부분에따라해당시설물을제3종시설물로지정해줄것을요청하는경우에는별지제4호서식의제3종시설물지정신청서에다음각호의서류를첨부하여같은항제1호또는제2호의자(이하이조에서“지정신청자”라한다)에게제출하여야한다.한1. 해당시설물에대한점검결과보고서(구조적안전성및결함의정도를증명할수있는사건을첨부하여야한다)한2. 시설물관리대상자본한③제3종시설물의관리주체가영제5조제3항에따라제3종시설물의지정을해제해줄것을요청하는경우에는별지제5호서식의제3종시설물지정해제신청서에다음각호의서류중해당서류를첨부하여지정권자에게제출하여야한다.한1. 해당시설물에대한법제17조제1항에따른안전점검결과보고서한2. 용도변경이있는경우건축물대장사본한3. 그밖에재난예방하기위하여계속적으로관리할필요성이없다는사실을증명하는서류한제6조(설계도서등의제출)①제1종시설물및제2종시설물건설·공공하는사업주체, 제3종시설물의관리주체한및보수·보강을실시한시설물의관리주체(이하“서류제출의무자”라한다)가법제9조제1항·제2항및제4항에따라설계도서등서류를제출하여야하는시기는별표1과같다.한②서류제출의무자가 제1항에따른서류의내용을시설물통합정보관리체계에 입력한경우에는법제9조제1항·제2항및제4항에따라서류를제출한것으로본다.한제7조(구조상요부분)영제7조제3호에서“국토교통부령으로 정하는주요부분”이라한것은하의하한해당하는부분을말한다.<개정2021.8.27.>한1. 교량발침한2. 터널의복공(覆工)부위한3. 하천시설의수문의문짝한4. 담의문체, 시공이름부및여수로(餘水路: 물이일정량을넘을때여분의물을배내기위하여만든물길)한5. 조립식건축물의연결부위한6. 상수도관로이름부한7. 환만시설중갑문의문짝중시설과계류시설, 발파제, 파제제(源障堤)및호안(護岸)의구조제한제8조(중공도)는사용승인종보법제9조제8항에따른시설물의중공도는사용승인의종보는별지제6호서식에한다.<개정2020.2.21.>한제9조(관련서류의열람방법·절차및방법)①법제10조제1항및제2항에따라열람할수있는관련서류의한법위는영별표2의서류와같다.한②제1항에따른관련서류를열람하려는자는열람의목적과범위를명시하여국토교통부장관에게 열람을신청하여야한다.한③제2항에따른관련서류의열람신청및열람신청에따른서류의열람은시설물통합정보관리체계를 통하여열람할수있다.한④제1항부터제3항까지에서규정한사항외에관련서류의열람방법·절차및방법등의세부사항은국토교통부장관이정하여고시한다.한제3장시설물의안전관리 한제1절안전점검등 한제10조(책임기술자의교육훈련등)①영제9조에따른책임기술자(이하“책임기술자”라한다)가되려는사람및한책임기술자의감독하에정밀안전진단또는성능평가를실시하려는사람은영제9조제1항및제3항에따라다음한각호외의하한해당하는교육기관에서실시하는신규교육및신규교육이수후5년마다실시하는보수교육을이수해야한다.<개정2020.12.11.>한1. 「건설기술진흥법시행령」제43조제2항에따른건설안전분야의교육기관한2. 국가또는지방자치단체소속공무원교육원한3. 「국토안전관리원법」제43조제2항에따른국토안전관리원(이하“국토안전관리원”이라한다)한4. 「공공주택관리법시행규칙」제13조각호외의하한해당하는교육기관(「공공주택관리법시행령」제115호)(20220916).ttl.pdf', 'page': 1)}}]
Graph database GraphCypherQAChain	> Entering new GraphCypherQAChain chain... Generated Cypher: <pre>MATCH (s:SpecialAct_Chapter) RETURN COUNT(s) AS NumberOfChapters Full Context: [[{'NumberOfChapters': 7}]]</pre> > Finished chain. Execution Time: 2.5972 seconds

Figure 3.17 Comparison of the response from Vector DB to Graph DB

- **Data analysis functions available through graph DB**

This chapter briefly introduces ways to improve existing data analysis results by utilizing the structure of graph database. In the graph database, tasks that could not be performed in the existing relational database (i.e., BMS) structure can be achieved through the changed structure. Graph Data Science (GDS) library built in Neo4j, provides the aggregation of algorithm for graph analysis and machine learning. One of them is embedding, a fundamental concept in machine learning and natural language processing that involves converting individual entities, such as words or sentences, into continuous vector representations. These vectors capture the semantic essence of the input data in a concise form. Graph embedding of typological information from a graph database is possible, converting the nodes of the graph into low-dimensional vectors (Grover and Leskovec, 2016). The author chose the GraphSAGE algorithm, which can preserve the typology, connectivity, and attributes of the neighboring nodes and relations as vectors. It is an inductive representation learning algorithm by sampling and aggregating features from a node's local neighborhood (Ahmed et al., 2017). The nodes of the bridges, their properties, and their relationships were projected onto an in-memory graph using the GDS library to use GraphSAGE. After projection, the GraphSAGE algorithm trained the feature properties and returned the embedded vectors. The embedded vector representing the bridges was then

imported to Python as a form of the data frame to be fed into the K-means++ clustering algorithm. The author defined these embedded values as a dataset for the proposed model. Figure 3.18 below shows a framework that compares the proposed bridge clustering methodology through graph embedding with the existing bridge clustering methods.

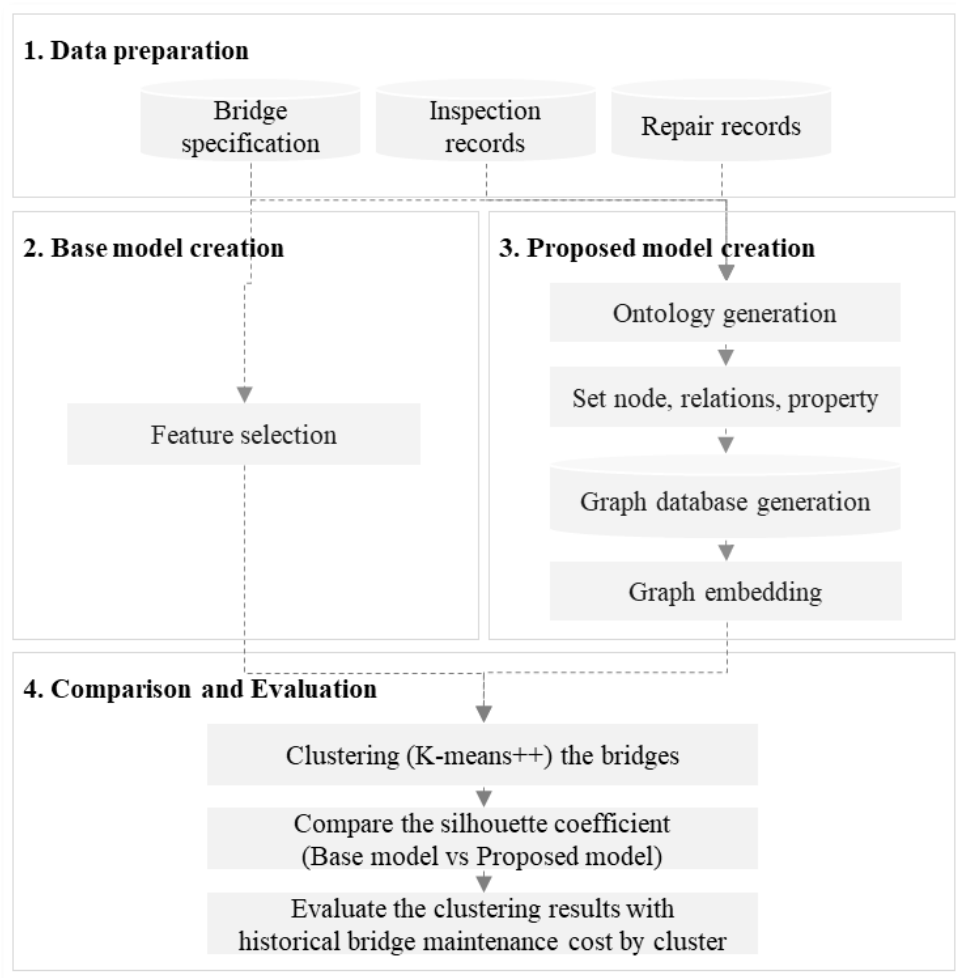


Figure 3.18 Framework for comparison of the clustering results of relational DB to graph DB

The clustering performance was hard to be quantified because it is an unsupervised learning algorithm. The silhouette coefficient of the base model and the proposed model from clustering results can be used to compare the tightness and separation of the cluster (Rousseeuw, 1987). A silhouette score presents a measure of the proximity of each point in a cluster to points in neighbouring clusters, ranging from -1 to 1. The score calculates the cohesion of a cluster by averaging the distances between one random data point and all other data points within the same cluster, as shown in the equation below. $a^{(i)}$ is the mean intra-cluster distance, and $b^{(i)}$ is the mean nearest-cluster distance for each data point.

$$\text{Silhouette coefficient}^{(i)} = \frac{b^{(i)} - a^{(i)}}{\max \{b^{(i)}, a^{(i)}\}} \quad (\text{eq.1})$$

The average silhouette coefficient supported the superiority of the proposed graph-based model with a score of 0.802 compared to the base model with a score of 0.534. The author interpreted that the reason for the good clustering results of the proposed model was that the bridge information in the graph data was linked to the inspection and maintenance project information.

Table 3.13 Comparison of the repair cost by condition grade of bridge element

Condition grade	Not clustered	Base model			Graph database model		
		Cluster 0	Cluster 1	Cluster 2	Cluster 0	Cluster 1	Cluster 2
A	15.77 (11)	0.74 (4)	-	24.35 (7)	-	20.18 (6)	10.47 (5)
B	18.39 (200)	18.12 (93)	12.29 (2)	18.75 (105)	12.68 (37)	21.38 (110)	16.16 (53)
C	43.84 (151)	41.12 (67)	134.75 (5)	40.40 (79)	44.18 (24)	31.15 (91)	75.71 (36)
D	67.65 (21)	37.53 (7)	-	82.70 (14)	-	50.69 (12)	90.26 (9)
E	93.37 (1)	93.37 (1)	-	-	-	93.37 (1)	-

In conclusion, the clustering results of the proposed model were better for the following reasons. First, in all clusters of the proposed model, a gradually increasing trend of unit cost from ‘A (Good)’ to ‘E (Worst)’ condition grades was shown. However, the base model did not show a clear trend by condition grades. Second, the proposed model showed the differences in the unit costs of maintenance works between different clusters. This meant that unnecessary or insufficient cost distribution can be reduced by distributing appropriate maintenance costs according to the characteristics of each bridge. In other words, it was possible to allocate maintenance costs more appropriate to the bridge when the representative value by group was used as shown in (c) than when the average value without clustering was used as the representative value as shown in (a) in Figure 3.19.

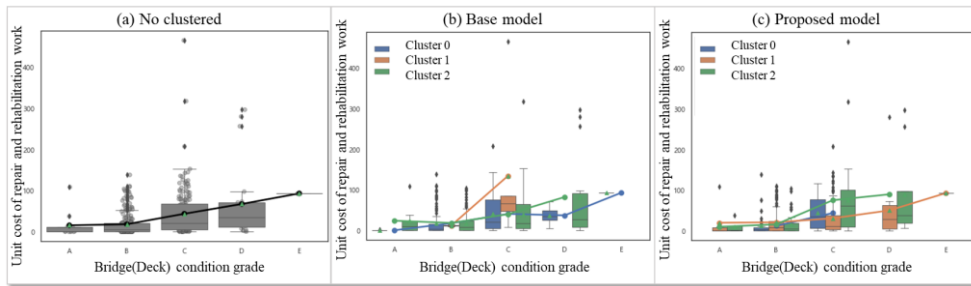


Figure 3.19 Results comparison with unit cost of the deck by clusters

Chapter 4. Implementation of the Question-Answering System using a Large Language Model

This chapter covers the entire system proposed in this dissertation to input the user's question and output the return of the answer. As described in Section 2.3. In "Research Gaps for Future Development," the author attempted to design a structure that allows users to find the answers they want from the bridge maintenance knowledge base by using a large language model with excellent natural language processing capabilities.

4.1 Classification of the Question-Answering

4.1.1 Identifying User Requirements and Expectations

Designing an optimal question-answering system involves setting the scope of questions that the system should perform. It is necessary to clarify the scope to supplement the database and develop the functions according to answer it. The scope of QA in this dissertation was set, as shown in Table 4.1. The expected main users are bridge maintenance officers, managers, and decision-makers. To clarify their expected questions, the author reviewed the Frequently Asked Question (FAQ) pages in the FMS and BMS platforms (see Figure 4.1), which the expected users predominantly use at their work. In

addition, various questions and answers were collected from the education and training platform (KALIS 2023b) for bridge inspectors deployed in the field and summarized in Appendix B. Additionally, the proposal requests of research projects commissioned by higher-level decision-makers, utilizing BMS data, were examined to gather requirements.

The author summarized the requirements of the expected user as follows: (1) the decision-makers who manage the maintenance budget mainly wanted to know the results of analysis based on data, (2) those who directly perform inspections and repairs wanted to know history information, and (3) local managers at each regional management agency wanted to know laws and guidelines related to maintenance execution. Therefore, the functions to be performed in QA could be organized as (1) retrieving regulations, (2) querying history records easily, and (3) returning prediction results by analyzing data instantly.

Table 4.1 Scope of QA

Attribute	Description
Domain	Bridge maintenance, Bridge Management, Inspection, Repair and Rehabilitation
Purpose of QA	Find the information for decision with ease of access, Analyze the stored data constantly, Predict the values not stored in database
Expected users	Bridge owners, Bridge maintenance decision makers, Inspectors, Repair workers, Managers in Agency
Data source	System data from BMS Unstructured documents from FMS

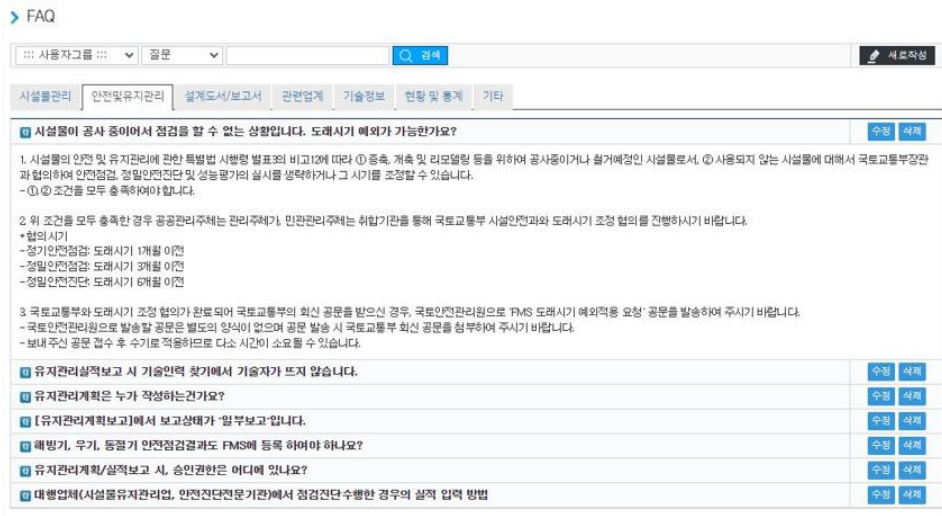


Figure 4.1. Frequently asked question in webpage of FMS



Figure 4.2 Research reports for bridge maintenance work on BMS

4.1.2 Designing Features for Query Response

Depending on the scope of QA, types of questions were categorized into the following three categories: (1) Simple retrieval, (2) Composite query, (3) Predictive reasoning. Table 4.2 shows examples of representative questions and answers for each category. As a result of organizing the types of questions, it was confirmed that the data sources and processes for finding answers for each category were different. This is related to agent design that will be introduced in the next chapter.

Table 4.2 Categorization of QA (sample)

Category	Sample Question	Desired Answer
Simple retrieval	Who draw the maintenance strategy?	Return the special act or guidelines related to bridge maintenance
	When is inspection and diagnosis performed?	
	What are the major flaws of the bridge?	
Composite query	Average of the repair cost of the bridges under Gangwon agency	Average cost
	The company that repaired the bridge A	Company name
	Number of bridges under condition grade D in 2022	Number of bridges
Predictive reasoning	Required maintenance cost when repairing bridge 10245	Predicted repair cost
	Expected condition grade of bridge ‘20235’ in next inspection	Predicted condition grade

4.2 Integration of Large Language Model in the QA System

4.2.1 Architecture of the QA System

To develop a system capable of addressing the anticipated inquiries outlined in the preceding chapter, a specific system architecture is essential. Initially, an interactive user interface is required, allowing users to input their queries. Subsequently, these inquiries are processed by a backend system, hereinafter referred to as the Question Answering (QA) System. This QA System interacts with a bridge maintenance knowledge base, accessing requisite information to formulate responses.

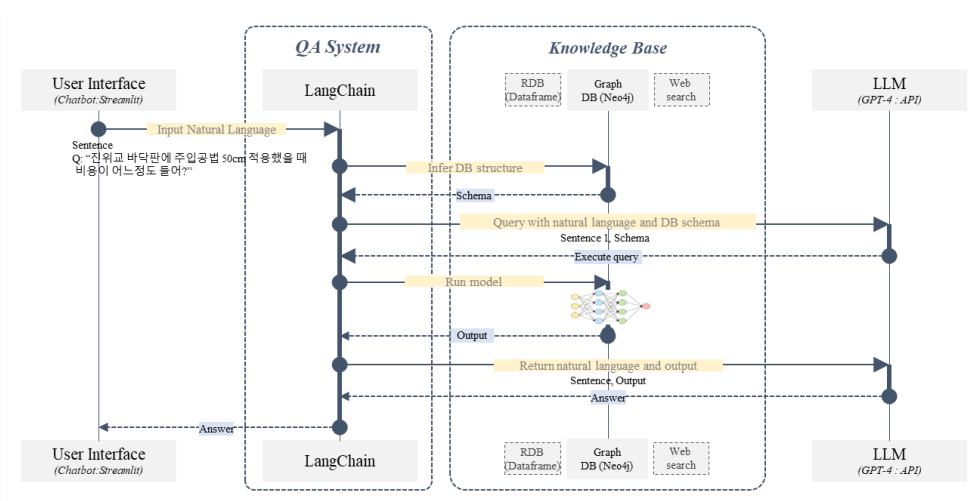


Figure 4.3 Sequence diagram of LangChain applied QA system

The key to this process is retrieving relevant data and synthesizing it into a coherent answer. To enhance the efficacy of this operation, the

implementation of an advanced natural language processing framework, powered by a large-scale language model, is proposed. This approach aims to refine the system's interpretative and generative capabilities, ensuring accurate and contextually relevant responses to user queries.

The framework that can be used to build such a QA system is the LangChain framework (Chase 2022). Agents supported by LangChain are artificial entities that can sense their environment, make decisions, and take actions. The author drew a sequence diagram as shown in Figure 4.3 to clarify how the QA system is implemented. A sequence diagram is a behavior diagram that expresses how a specific action interacts with an object in what order and has the advantage of showing what scenario the system is moving through. When a user asks a question, the LangChain framework calls the Large Language Model (LLM) and uses it to first look at the schema of the knowledge base. And based on the schema and the user's question, a Cypher query using LLM. The returned query is passed back to the knowledge base to retrieve the necessary data. Afterwards, the query and returned data are thrown back to LLM to generate the final answer. Finally, the answer in natural language form is returned to the user.

4.2.2 Selection of the Large Language Model

This chapter describes the selection process for applying LLM, an innovative technology that significantly improves natural language processing performance in the previously described QA system. Recently, the LLMs, which is known as the pre-trained foundation model, have been released competitively (Zhou et al. 2023). It refers to language models that learn massive amounts of data and have hundreds of billions of parameters or more. By learning from large-scale text data, it is possible to acquire various linguistic phenomena and knowledge, thereby improving understanding of natural language and creating new sentences. According to the research by Liu et al. (2023b), ChatGPT serviced by OpenAI, which known to have the best performance, is capable of performing the following tasks: (1) Question and Answering, (2) Text Classification, (3) Text Generation, (4) Code Generation, (5) Inference, (6) Data or information extraction, transformation, enhancement, processing, (7) Human-ChatGPT Collaboration, (8) ChatGPT Integration, and (9) Medical Application.

The reasons for adopting Large Language Model are as follows. Existing QA systems was limited to keyword matching or queries in structured databases because it is difficult to understand the difference in context and wording of questions. However, LLM can easily handle linguistic ambiguity through extensive language learning. In addition, it has the advantage of being

able to easily adapt to new types of questions without additional learning, so incorporating LLM is becoming a basic technology stack not only in current but also in future research.

There are several ways to apply LLM, such as Fine-tuning, Few-shot, One-shot, and Zero-shot learning (Brown et al. 2020). These methods offer different advantages and are suitable for different use cases, depending on the specific goals and available sources. Since the proposed QA system is based on the Korean natural language, the performance of the LLM fine-tuned in Korean was reviewed for implementation. According to the preliminary survey on the Korean LLM conducted by Joo et al., (2023), LLM specialized in Korean exists as shown in Table 4.3.

Table 4.3 Local LLM specialized in Korean (2023. 09)

List	Foundation Model	Training Dataset	Reference
Polyglot-ko	Polyglot	863 GB of Korean language data (1.2TB before processing) curated by TUNiB	(Park 2023)
KoAlpaca	Polyglot-ko, LLaMA	Stanford Alpaca	(Lee 2023a)
KoVicuna	Vicuna	Translated data using DeepL of 62k dialogues from ShaerGPT	(Lee 2023b)
KuLLM	Polyglot-ko	Translated data using DeepL of GPT-4-LLM, Vicuna, Dolly	(NIP&AI lab and Research 2023)

Amon them, the most recently developed Korean LLM, KuLLM, was downloaded on a local server to perform basic conversations without fine-

tuning and prompting. The author has conducted a preliminary test to see if local LLM can achieve the performance desired in this dissertation, and obtained results as shown in Figure 4.4. As a result, local LLM had the problem of not understanding the Cypher queries needed to call and analyze data from the knowledge base. Through this, it was possible to conclude that local LLM, which has a small scale of pre-trained data, does not meet the performance desired in this dissertation.

Figure 4.4 QA test with local LLM (KuLLM)

The author considered applying gpt-4 using API, which has the best performance recently, rather than local LLM. Looking at Table 4.4, you can see that it performs significantly better than the state-of-the-art language model.

Table 4.4 Performance of GPT-4 compared to LM SOTA (2023. 09)

Benchmark	GPT-4 / Evaluated few- shot	GPT-3.5 / Evaluated few-shot	LM SOTA / Evaluated few- shot
MMLU : Multiple-choice questions in 57 subjects	86.4% (5-shot)	70.0% (5-shot)	70.7% (5-shot U-PaLM)
HellaSwag : Commonsense reasoning around everyday events	95.3% (10-shot)	85.5% (10-shot)	84.2% LLAMA (validation set)
ARC : Grade-school multiple choice science questions. Challenge-set.	96.3% (25-shot)	85.2% (25-shot)	84.2% 8-shot PaLM
WinoGrande : Commonsense reasoning around pronoun resolution	87.5% (5-shot)	81.6% (5-shot)	84.2% 5-shot PALM
HumanEval : Python coding tasks	67.0% (0-shot)	48.1% (0-shot)	26.2% 0-shot PaLM
DROP (f1 score) : Reading comprehension & arithmetic.	80.9 (3-shot)	64.1 (3-shot)	70.8 1-shot PaLM

The author suggests the leveraged natural language processing capabilities of QA system by applying the LLM compared to existing chatbot systems, as shown in Figure 4.5. To generate answers to questions in existing chatbot systems, the following steps were required: First, the user's question was embedded to understand the user's intention using high-dimensional semantic vector values, and the necessary entities were extracted. Then, the system needed to obtain additional information through slot-filling, and finally, the text could be changed into a query by considering the context. The query retrieved an answer from the database, converted it back into natural language, and returned the answer to the users. That of several steps were performed sequentially, which typically resulted in a large amount of data loss. As a result, the performance of the previous chatbot declined and its usability decreased.

However, by applying the LLM, the process of identifying intent from natural language questions, extracting key entities, converting them into queries, and calling them from the database can be improved. Large-scale multilingual LLM (GPT-4) was considered as an API for application, and OpenAI (OpenAI 2023) provides it in the form of application programming interface (API) allows developers to use Generative Pretrained Transformer (GPT) models in their applications. An encrypted API key by could be obtained by creating an OpenAI account, which can be saved to the Operating System (OS) and used to load functions when scripting.

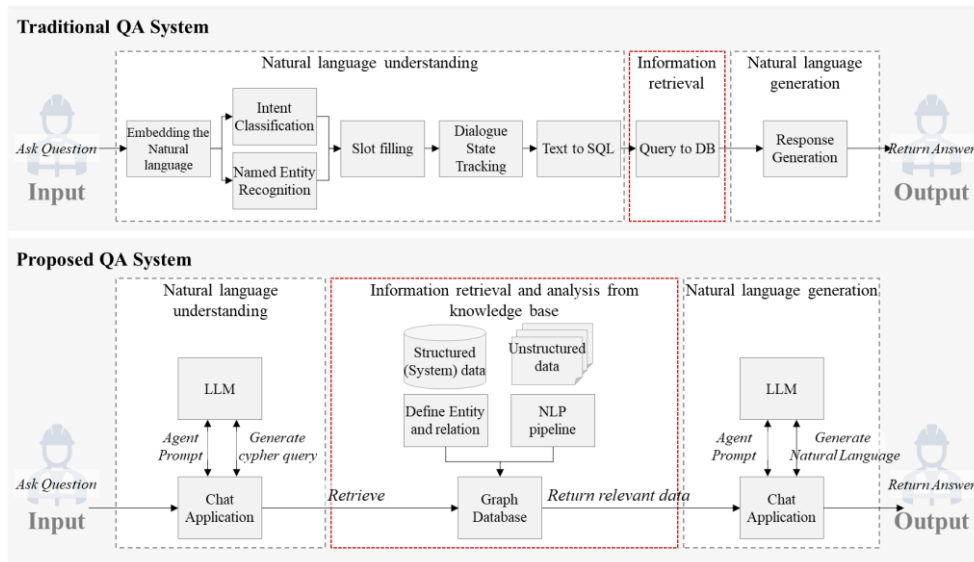


Figure 4.5 Comparison of proposed QA system with traditional system

OpenAI operates on a tiered pricing model that essentially charges users based on the number of tokens processed. In the GPT model, a token can

contain spaces, punctuation marks, parts of words, etc., and one token can be considered to consist of 3 to 4 English letters on average. As such, the proposed QA are charged each time the user asks a question as the API provided by OpenAI is used. However, it can be considered economical considering the cost incurred when installing a Local LLM and fine-tuning it to suit the desired performance.

As shown in Table 4.5, There is a prominent difference between fine-tuning and prompting techniques, and the expected performance varies depending on the task, considerate review is required when applying it. In this dissertation, the prompting technique suitable for QA is selected for development.

Table 4.5. Comparison between fine-tuning and prompting technique for LLM

Category	Finetuning	Prompting
Usage	Updating the model parameter with additional data	Providing the model with a context or instruction in the form of a prompt or query
Task	Specific tasks (e.g., translation, summarization, sentiment analysis...)	Tasks need to interact with the model to generate text or answers to question
Required data	Substantial dataset (Hundreds to thousands of examples)	Does not need specialized datasets or few example (from one to hundreds)
Resource for development	Intensive computing power require for training	No training time, immediate results

4.2.3 Techniques in Prompt Engineering

To use LLM for the desired purpose, prompt engineering must be applied. A prompt is a sequence of tokens used as the input message for the LLM. Clear and concise prompts can help the model understand the intent of the user's question and ensure the desired output (Liu et al. 2023a).

The author developed the prompt for the QA by setting the context, constraints, instructions, and examples of the prompt template. To support the prompt engineering, LangChain, which is launched in October 2022 by Harrison Chase on GitHub (Chase 2022), framework is used. It provides following modules for standard, extendable interfaces, and external integration (Figure 4.6).

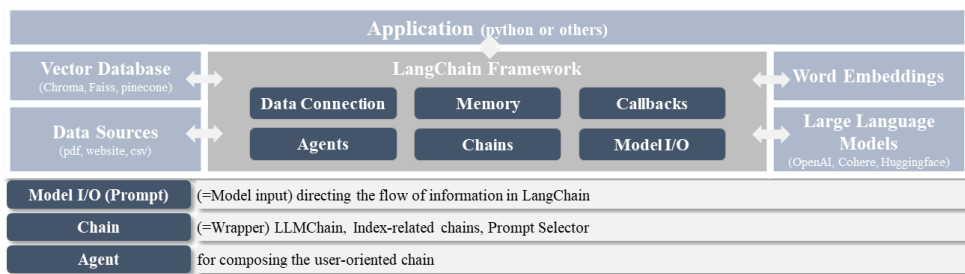


Figure 4.6 LangChain framework

First, the prompt template corresponding to model input and output should be explained. Because the prompt template used for prompting is a

natural language-based technology that avoids ambiguous language, it is very important to configure it elaborately so that the LLM can understand it well.

- Model I/O (Input/Output)

The pseudo-code (see Figure 4.7) represents the codes to call the basic template of the prompt including instructions and the few shot examples. A set of few shot examples to help the language model generate a better response. This kinds of technique called Chain-of-Thought (CoT) prompting, demonstrated its reasoning abilities in arithmetic, commonsense, and symbolic tasks surpassing even fine-tuned GPT-3 model (Wei et al. 2022). CoT ask the LLM to think step by step for reasoning the answer. This feature shows the LLM several QA cases in the same way as Few-shot learning, allowing it to better understand the context and set the direction for the answer.

```
from langchain.prompts.few_shot import FewShotPromptTemplate
from langchain.prompts.prompt import PromptTemplate

def ask_question_with_template(question):
    # Instruction과 few-shot 예시를 함수 내에서 정의
    instruction = """Chain 실행 순서: 먼저 'Intent_recognition' tool을 사용하고, 교량의 유지보수 비용 추정관련이면 'RepairCost_Prediction',
    또는 'RepairCost_Prediction_Elmt'를 선택한다. 그 외의 질문은 'Neo4j_search'를 사용하여 답변하고, 답이 없는 경우 'Serp_search'로 넘어간다.
    현재 시점은 2023년 12월 13일이므로 시간 또는 기간과 관련된 질문은 현재를 기준으로 계산해서 대답한다.
    답변 언어: 한국어"""

    few_shot_examples = """
    예시 1: [질문] '시설물의 안전 및 유지관리에 관한 특별법 제7장의 내용은?'
    [답변] """MATCH (a:SpecialAct_Chapter) RETURN a.장_내용을 " Neo4j_Search"에서 실행한다.'
    예시 2: [질문] '교량의 사고사례에 대해서 알려줘'
    [답변] """serp_search"를 실행해 교량이름을 검색하고 검색된 내용을 요약하여 반환한다.'
    예시 3: [질문] '균열이 생겼을 때 어떤 공법으로 보수해?'
    [답변] '주입공법, 표면보수, 단면보수 등을 활용할 수 있습니다'
    """

    modified_question = f'Instruction: {instruction}\n{few_shot_examples}\nYour Question: {question}'
```

Figure 4.7 Example of prompt template

In order to improve the performance of the task to be performed in this study, instructions to generate answers in Korean and priorities for tools that generate answers were entered as instructions. In addition, by entering several questions and examples of answers, the user was encouraged to provide answers close to the desired answer.

- Chains

A chain can call a single LLM and involve sequences of calls (whether to an LLM or a different utility). It works as a tool or functions in general systems (see Figure 4.8).

```
from langchain.llms import OpenAI
from langchain.chains import LLMChain

# Example of chain
llm_chain = LLMChain(prompt=prompt, llm=OpenAI(temperature=0), verbose=True)

# Defined Chain for QA system
chain_intent = LLMChain(llm=ChatOpenAI(openai_api_key = api_key), prompt=chat_prompt)
chain_neo4j = GraphCypherQAChain.from_llm(ChatOpenAI(temperature=0), graph=graph, verbose=True, return_direct=True)
chain_extraction = create_extraction_chain(schema, ChatOpenAI(temperature=0))
chain_extraction_elmt = create_extraction_chain(schema_elmt, ChatOpenAI(temperature=0))
repair_cost_chain = RepairCostPredictionChain(model_paths=model_paths, df=df)
repair_cost_chain_elmt = RepairCostPredictionChain(model_paths=model_paths, df=size)
```

Figure 4.8 Example of chain execution

```

# Define the tools for chain execution
tools = [
    Tool(
        name="Intent_recognition",
        func=chain_intent.run,
        description="이 tool은 question의 의도를 파악하여 답변생성에 필요한 tool로 연결시켜주는 tool입니다.  
교량의 보수비용과 관련된 질문일 때, 사용자의 질문에서 교량이름, 보수공법, 보수물량이 입력되면  
'RepairCost_Prediction'을 실행하고, 만약 교량이름, 교량등급이 입력되면 'RepairCost_Prediction_elmt' tool을 실행합니다.  
보수비용과 관련된 질문이 아니면 'Neo4j_search'를 실행하세요.",
    ),
    Tool(
        name="Neo4j_search",
        func=chain_neo4j.run,
        description="교량의 사양이나 유지관리 이력에 대한 쿼리가 필요할 때 활용 가능하며,  
시설물 안전 및 유지관리에 관한 특별법에 대한 답변도 가능합니다. 한국어로 답변해주세요.",
    ),
    Tool(
        name="RepairCost_Prediction",
        func=repair_cost_chain,
        description="정의되어 있는 비용추정 class를 실행하세요.",
    ),
    Tool(
        name="RepairCost_Prediction_elmt",
        func=calculate_weighted_value,
        description="교량의 부재크기와 부재의 상태등급별 가중치를 곱해서 보수보강비용을 추정하는 기능입니다.  
부재의 상태등급에는 A,B,C,D,E 등급이 있습니다. 답변은 (천원)단위로 반환하세요.",
    ),
    Tool(
        name="Serp_search",
        func=serpai.run,
        description="사용자가 요청한 답변을 다른 도구로는 찾을 수 없거나 명시적으로 웹 검색을 요청할 때  
이 도구를 사용하세요. 입력은 전체 맥락을 포함하는 질문 형식이어야 합니다..",
    )
]

```

Figure 4.9 Example of defined tools for answering

The proposed QA system defined several chains and used them as tools for generating answers. The tools defined in Figure 4.9 are used to generate answers. The first chain is a tool for classifying the intent of the question. It will determine which of the three main answer generation functions (i.e.,

retrieval, composite query, and predictive reasoning) should be used to proper answering. The “neo4j search” tool generates answers to questions based on the generated bridge maintenance knowledge base. This tool is responsible for generating answers to ‘Composite query’ and some of ‘Simple retrieval’. Next, there is a “Repair_cost_prediction” tool, which is one of the tools for utilizing machine learning models for answering the related questions. This is a chain for loading and executing the pre-trained pickle model and the variables needed for prediction. A function for maintenance cost estimation only using the bridge name and condition grade is also defined as “Repair_cost_prediction_elmt”. Lastly, in order to response for questions that require answers to data that are not in the knowledge base, the author have included an Internet search function as a “Serp API” tool. These functions include a description of the function to be called and the situation in which the tool should be used, and LLM recognizes this and generates an answer.

- **Agent**

The agent leverages an LLM to determine its actions of executing the chains and their order. As shown in Figure 4.10, the “custom agent” is designed to determine whether the previously defined chains can be activated according to the question.

```
# Execution of the Agent  
memory = ConversationBufferMemory(memory_key="chat_history", return_messages=True)  
custom_agent = ConversationalChatAgent.from_llm_and_tools(llm=ChatOpenAI(model_name='gpt-4', temperature=0), tools=tools)  
agent_executor = AgentExecutor.from_agent_and_tools(agent=custom_agent, tools=tools, memory=memory, verbose=True)
```

Figure 4.10 Example of agent execution

- **Short-term Memory**

In LangChain, memory has long-term and short-term. Since this dissertation is responsible for long-term memory in the knowledge base part, the author only discusses short-term memory needed in the QA context. Short-term memory persists state between calls of chain and agent. The function of memory is pre-requisite in this dissertation because the QA involves finding the answer through several steps like the CoT described above. As shown in Figure 4.11, memory works for memorizing the information from start to end of thought.

```

> Entering new AgentExecutor chain...
```json
{
 "action": "Intent_recognition",
 "action_input": "9번 교량에 바닥판 주입공법쓰면 보수 비용 얼마나 필요해?" }
```
Observation: 교량번호: 9번
보수공법: 바닥판 주입공법
물량: 없음
Thought:```json
{
  "action": "Final Answer",
  "action_input": "9번 교량에 바닥판 주입공법을 사용하여 보수를 진행하려면, 보수에 필요한 물량에 대한 정보가 필요합니다. 이 정보 없이는 정확한 비용을 추정할 수 없습니다. 보수에 필요한 물량을 알려주실 수 있나요?" }
```
➤ Finished chain.

Out[00]:
'9번 교량에 바닥판 주입공법을 사용하여 보수를 진행하려면, 보수에 필요한 물량에 대한 정보가 필요합니다. 이 정보 없이는 정확한 비용을 추정할 수 없습니다. 보수에 필요한 물량을 알려주실 수 있나요?'

```

Figure 4.11 Example of short-term memory in LangChain

### **4.3 Development of Answer Generation Modules**

According to the expected question types defined previous chapter, they are divided into information retrieval, composite query, and predictive reasoning. This chapter explains in detail the process by which the modules required for each question are executed.

First, in chapter 4.3.1, a data query module is described that allows LLM to find answers in the knowledge base. Second, for predictive reasoning, the author created the machine learning model needed for prediction. Although various prediction models are needed, the prediction model provided in this dissertation is the maintenance cost estimation model. Two models were built: the first is a model that returns the cost required for the repair and reinforcement method, and the second is a model that returns how much the future repair cost will be according to the condition grade of the element. Lastly, the modules that can generate the answers based on browsing or simple data retrievals are described in chapter 4.3.3.

#### **4.3.1 Data Retrieval Modules for Graph Database Interaction**

To utilize the graph-based QA function supported by LangChain, it is necessary to import packages as shown in Figure 4.12. Additionally, the authentication information and its environment settings of Neo4j for calling the graph database is necessary. When a connection is made between the

database and the QA system network, graph database-based QA becomes possible.

```
from langchain.utilities import SerpAPIWrapper
from langchain.graphs import Neo4jGraph

Prepare API keys
Os.environ["OPENAI_API_KEY"] = "L-****"
Os.environ["SERPAPI_API_KEY"] = "45ba*****"

Initialize Neo4jGraph
Graph = Neo4jGraph(url="bolt://147.00.000.00:7687", username="neo4j", password="let-me-in")

new_working_directory = "/workspace/db"
os.chdir(new_working_directory)
```

Figure 4.12 Required environment for GraphCypherQACChain operation

The constructed knowledge base can be accessed through neo4j browser by entering the ‘username’ and ‘password’ with the network address. If the environment is set up without error as shown in the Figure 4.12, the code in Figures 4.8 and 4.9 can be executed, and “GraphCypherQACChain” generates an answer to the question from the knowledge base.



Figure 4.13 Results from GraphCypherQAChain

Figure 4.13 shows an example of the response returned when specific questions related to special law provisions are asked. As the chain is executed, the correct response is inquired from the knowledge graph, and the answer is returned to users as natural language by generated through LLM.

### **4.3.2 Development of Predictive Models for Bridge Maintenance**

As discussed in Chapter 4.1, decision makers frequently make decisions about bridge maintenance based on cost. Thus, the development of a highly reliable cost estimation model is very important. The author developed a model to estimate bridge repair costs and would like to explain the process. Two models were developed, one is a prediction model that can estimate the cost required when implementing each repair method, and the other is a model that estimates the cost according to the condition grade of the element.

- **Bridge maintenance cost prediction model by repair method**

The Korean BMS has defined a total of 72 repair methods for maintenance. Among them, the repair methods written in abstract terms such as “Miscellaneous,” “RC or other” were excluded in the scope of model generation because they were regarded as bundles of ambiguous repair methods. Additionally, when the number of data did not exceed 70, those methods were also excluded. As a result, among the ten main elements of bridges, the “Foundation” element had insufficient data for developing a model because repair works for this element were rarely performed that were excluded from the research scope. For repair methods that meet the standards written above, a machine learning-based cost estimation model was created as shown in Figure 4.14.

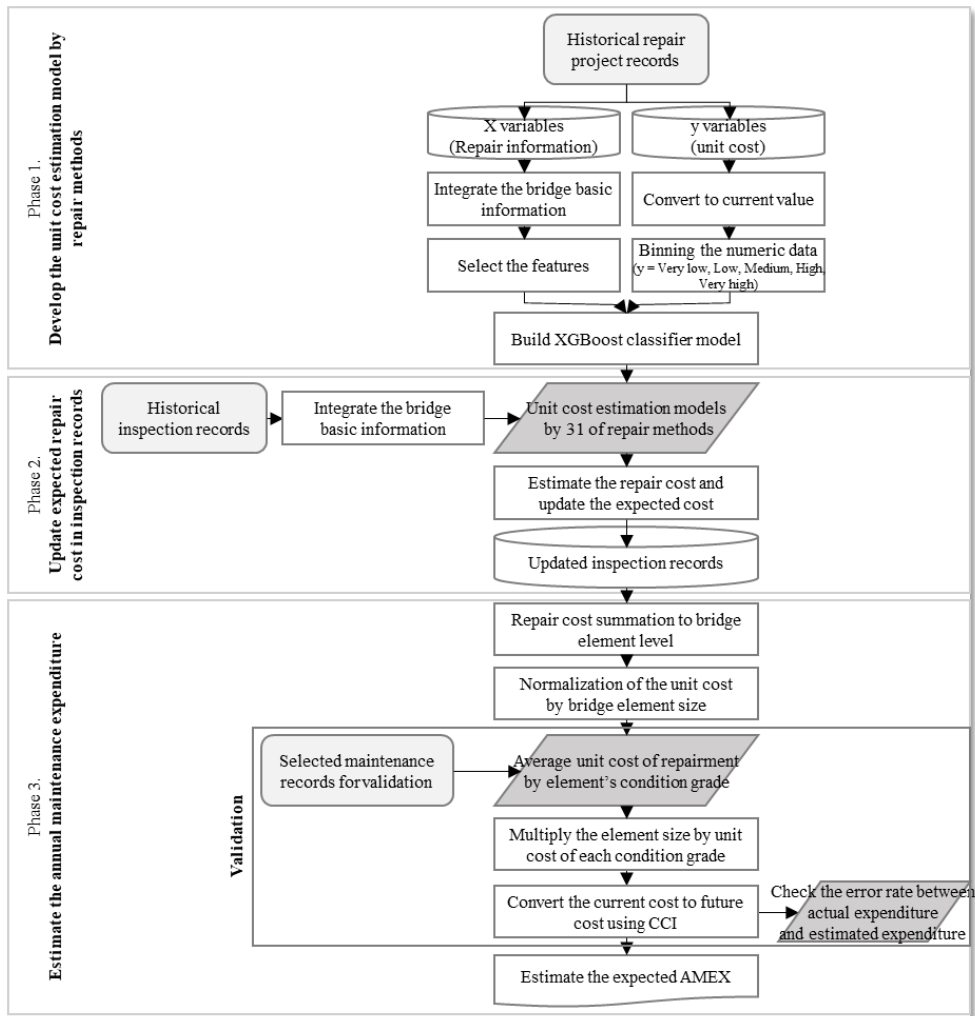


Figure 4.14 Process of generating the repair cost estimation model

In phase 1, explanatory variables and dependent variables was set using the basic specification of the bridge and repair history data. The variables below is the example of variables used for estimating the cost of repair methods of deck.

- Explanatory variables (24): “Detail type of component”, “Repair project completion date”, “Facility construction completion date”, “Bridge age”, “Quantity of the repair methods”, “Facility type”, “Road lane”, “Total length”, “Total width”, “Height”, “Depth of underwater”, “Main structure type”, “Design live load”, “Seismic design application”, “Competent authority”, “Region”, “Maximum span length”, “Average daily traffic (ADT)”, “Average daily truck traffic (ADTT)”, “Deck material”, “Deck thickness”, “Deck strength”, “Pavement type”, “Pavement thickness”.
- Dependent variable (1): “Class of the unit cost” (“Very low”, “Low”, “Medium”, “High”, “Very high”)

The refined dataset was input into the XGBoost classifier model, which is a scalable gradient tree-boosting algorithm (Chen and Guestrin 2016). The performance of the XGBoost algorithm is known to be superior to other tree-based models, such as decision trees and random forests, and less sensitive to outliers than lightGBM (light Gradient Boosting Model) when the amount of data is small (Ke et al. 2017). Many previous studies in this field have demonstrated the excellence of the XGBoost algorithm (Lim and Chi 2019; Thai 2022). Thus, the XGBoost algorithm was implemented under scikit-learn 1.0.2 version and XGBoost 1.6.2 version. When applying the algorithm, the main parameters (e.g., learning rate, n\_estimators, max\_depth, gamma, and colsample\_bytree) were tuned using a grid-search function. The five classes of target data representing unit cost were not imbalanced. Therefore,

the micro-f1 score was found to be appropriate as a performance indicator (Grandini et al. 2020). Consequently, unit cost estimation models for a total of 31 repair methods were generated. The performance of the unit cost estimation model for each repair methods was shown in Table 4.6. The average f1-score for the 31 models was 0.82.

```
Repair Cost Estimation Logic
class RepairCostPredictionChain():
 def __init__(self, model_paths, df):
 super().__init__()
 self.models = {}
 for method, path in model_paths.items():
 with open(path, 'rb') as file:
 self.models[method] = pickle.load(file)
 self.df = df
 self.class_to_amount = {
 '주입공법': (0: 63000, 1: 77000, 2: 93000, 3: 111000, 4: 198000),
 '표면보수': (0: 91000, 1: 122000, 2: 142000, 3: 167000, 4: 271000),
 '단면보수': (0: 310000, 1: 481000, 2: 588000, 3: 729000, 4: 1640000)
 }

 def __call__(self, question):
 extracted_info_list = chain_extraction.run(question)
 loaded_model = self.models[repair_method]
 class_to_amount = self.class_to_amount[repair_method]
 bridge_name = extracted_info['Bridge_name']
 repair_quantity = extracted_info['RepairQuantity']
 filtered_df = self.df[self.df['BRDG_NAME'] == bridge_name]
 filtered_df['보수물량'] = repair_quantity

 columns_ordered = ['BRDG_NAME', '보수일자', '보수물량', '바닥판종류', '바닥판두께', '바닥판강도', '바닥판포장유형',
 '준공일자', '종구분', '도로노선', '교장', '종폭', '교고', '대표상부구조', '내진설계적용여부', '위임구분',
 '관리기관2', '관리기관3', 'days']

 filtered_df = filtered_df[columns_ordered]
 X = filtered_df.drop('BRDG_NAME', axis=1)
 predictions = loaded_model.predict(X)
 amounts = [f"({class_to_amount[pred]} * repair_quantity)원" for pred in predictions]
 return amounts
```

Figure 4.15 Code for execution of the repair cost estimation model

All created models to predict the repair cost were saved in pickle format, and the parameters saved in this file were implemented in the system as shown in Figure 4.15 so that they can be called and used as needed.

Table 4.6 Performance of the repair cost estimation model by repair methods

Element	Repair methods	Data count	f1-score by class					Weighted f1-score
			Very low	Low	Medium	High	Very high	
Deck	Grouting	918	0.81	0.75	0.78	0.76	0.88	0.80
	Surface repair	5,071	0.90	0.88	0.91	0.88	0.91	0.90
	Section repair	3,160	0.65	0.63	0.66	0.51	0.63	0.62
Girder	Surface repair	751	0.83	0.89	0.89	0.88	0.91	0.88
	Injection and grouting	220	0.89	0.77	0.80	0.86	0.88	0.84
	Section repair	896	0.73	0.67	0.69	0.71	0.75	0.71
	Painting	1,871	0.88	0.84	0.82	0.81	0.84	0.83
Crossbeam	Surface repair	189	0.80	0.79	0.73	0.76	0.80	0.78
	Concrete section repair	328	0.61	0.78	0.62	0.64	0.55	0.64
	Painting	253	0.96	0.80	0.88	1.00	0.97	0.92
Pier/ Abutment	Injection and grouting	2,090	0.86	0.78	0.82	0.75	0.84	0.81
	Section repair	3,833	0.67	0.66	0.62	0.59	0.57	0.62
	Surface repair	5,412	0.85	0.82	0.82	0.85	0.86	0.84
Bearing	Main bearing repair	2,516	0.93	0.95	0.91	0.92	0.92	0.93
	Painting	158	1.00	1.00	0.82	0.73	0.70	0.81
Expansion joint	Expansion joint repair	298	0.85	0.86	0.76	0.67	0.79	0.78
	Finishing concrete	192	0.70	0.62	0.67	0.71	0.61	0.66
	Expansion joint replacement	4,036	0.83	0.78	0.76	0.77	0.82	0.79
Pavement	Asphalt surface repair	432	1.00	0.91	0.89	0.83	0.87	0.91
	Asphalt patching	108	0.80	0.91	0.71	0.73	0.67	0.75
	Asphalt cutting overlay	6,937	0.87	0.91	0.94	0.93	0.92	0.91
	Surface cleaning	272	0.95	0.85	0.87	0.84	0.72	0.85
	Concrete paving replacement	327	0.94	1.00	0.90	0.89	0.96	0.94
	Asphalt paving replacement	3,979	0.97	0.98	0.97	0.97	0.96	0.97
Drainage	Drainage repair	1,017	0.85	0.81	0.79	0.64	0.78	0.79
	Drainage replacement	899	0.78	0.78	0.82	0.80	0.87	0.81
	Inlet repair	117	0.91	1.00	0.90	0.77	0.77	0.86
	Inlet replacement	73	0.86	0.00	0.80	0.89	0.89	0.78
Railing/ Curb	Railing repair	1,283	0.85	0.85	0.84	0.83	0.88	0.85
	Curb repair	671	0.95	0.95	0.94	0.80	0.91	0.92
	Railing replacement	1,481	0.91	0.91	0.93	0.95	0.96	0.93

- **Bridge maintenance cost prediction model by condition grade**

As shown in phase 3 in Figure 4.14, the representative value by condition grade of element was gained. Table 4.7 represents the weight of each element by condition grade. The repair cost of a bridge element can be calculated by multiplying the size by the weight of each element as shown in eq. 2. Each information is stored in a dataframe, and depending on the query content, the element size and condition grade weight of the element are retrieved, and the final answer is returned.

$$\text{Repair cost of Deck A} = \text{Element Size}_{deck} \times \text{Weight}_{deck} \text{ (eq.2)}$$

Table 4.7 Unit cost of repair by condition grade of bridge elements

Element	Element size measurement (Unit)	Average element size	Average unit cost by element condition grade (Unit: 1,000 KRW/element size)				
			A	B	C	D	E
Deck	Exposed section size ( $m^2$ )	1,597.3	3.1	3.6	16.6	71.0	104.7
Girder	Exposed section size ( $m^2$ )	3,845.8	0.5	7.0	26.1	145.4	-
Crossbeam	Exposed section size ( $m^2$ )	781.0	6.4	13.4	15.9	31.7	-
Pier/ Abutment	Surface area ( $m^2$ )	1,321.1	2.5	2.9	12.1	62.2	-
Bearing	Number of bearings	35.2	258.5	394.4	692.3	2,474.6	-
Expansion joint	Length of the expansion joint (m)	46.5	224.	227.0	808.1	1,447.6	-
Pavement	Pavement area ( $m^2$ )	1,413.2	30.5	46.6	120.9	225.9	-
Drainage	Number of drainages	3.1	769.1	786.8	1,543.0	4,000.9	-
Railing/ Curb	Length of the railing/curb (m)	259.6	24.1	14.8	41.3	97.9	-

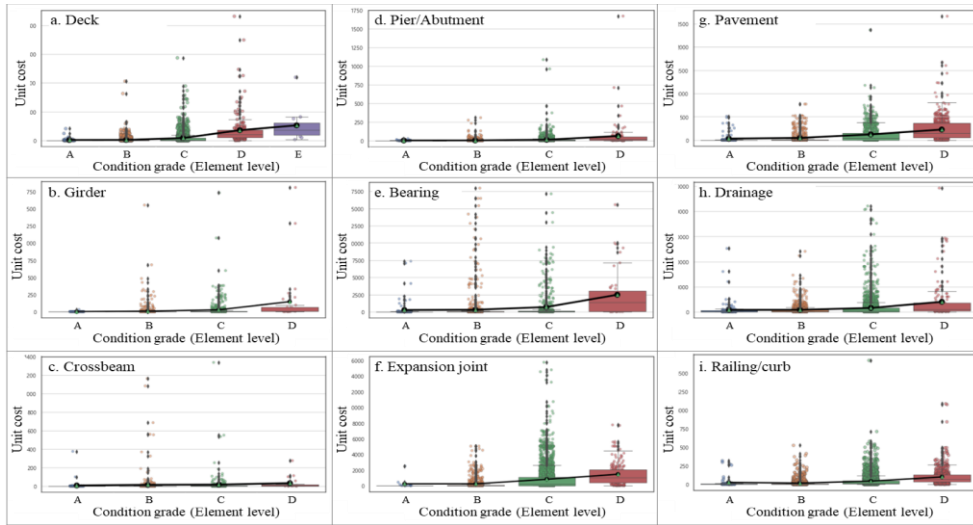


Figure 4.16 Distribution of the unit cost of repair by condition grade

#### # Schema for necessary variable

```
schema_elmt = {
 "properties": { "Bridge_name": {"type": "string"}, "ConditionGrade": {"type": "string"}},
 "required": ["Bridge_name", "ConditionGrade"]}
```

#### # weights by condition grade : Deck

```
weights = { 'A': 3.1, 'B': 3.6, 'C': 16.6, 'D': 71.0, 'E': 104.7}
```

```
def calculate_weighted_value(question):
```

```
 extracted_info_list = chain_extraction_elmt.run(question)
 extracted_info = extracted_info_list[0]
 bridge_number = extracted_info['Bridge_name']
 grade = extracted_info['ConditionGrade']
 element_size = size[size['BRDG_NAME'] == bridge_number]['Elmt_size'].iloc[0]
 weight = weights[grade]
 return element_size * weight
```

Figure 4.17 Code for execution of the repair cost estimation by condition grade of element in bridge

To estimate the expected repair cost according to condition grade of element, the size of the bridge must be retrieved based on the bridge name. Also, the weight value for each grade must be retrieved based on the grade information in the question entered by the user. Figure 4.17 shows the code implemented so that the entity to be extracted from the user's question can be specified in advance as a schema, and the corresponding value can be extracted and entered as a variable needed to generate an answer. When the function 'calculate\_weighted\_value' is called, the bridge size from the database according to the bridge name in the question can be extracted, and the weight value based on the grade in the question can be taken to return the estimated cost by multiplying the two.

### **4.3.3 System Integration and Module Orchestration**

As mentioned in Section 4.2, it was confirmed that an agent was needed to select the tools and to execute the activation of the required chain. Analyzing the intent of user input is the first function to be performed. This aspect of the system is referred to as the "Intent Recognition Agent," as illustrated in Figure 4.18. Within this intent recognition chain, the primary function involves discerning the nature of the input based on predefined prompt templates. The process entails determining whether a query necessitates the utilization of a machine learning model for generating

responses, if it can be directly answered through queries to a knowledge base, or if it requires conducting web searches due to the absence of relevant information in the knowledge base. This “Intent recognition chain” is pivotal in streamlining the QA system, ensuring that each inquiry is addressed through the most efficient and appropriate means available within the system's architecture.

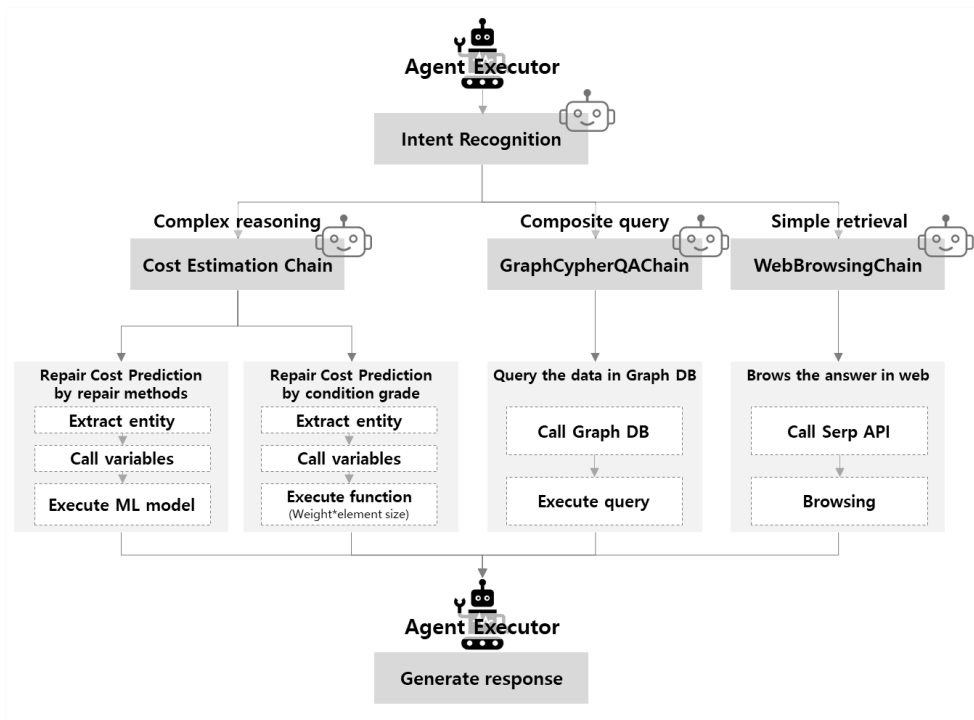


Figure 4.18 Activation flow of chains

Among the various prediction models using machine learning related to bridges, this paper proposes a function that can estimate maintenance costs

for bridges. The parameter model pre-trained based on XGBoost developed in the previous chapter is stored in the system, and variable information is inserted according to the user's question to generate an answer. In general, when repairing and reinforcing, people wonder how much it will cost when which method is applied, and when inspecting, they wonder how much it will cost depending on the condition grade. Apply a schema extraction chain that can extract the information needed to generate question answers for each case. In the case of an estimation model by repair and reinforcement method, “Bridge number”, “Repair quantity”, and “Repair method” data are required, and in the case of an estimation model by condition grade, “condition grade” and “bridge number” are required, so each is schema defined within (see Figure 4.19). If the necessary variables are not included in the questions, the necessary variables must be obtained through additional questions. The case in which additional questions are asked when all necessary variables are not collected can be confirmed through the second case in Figure 4.20.

```
Schema

schema_mthds = {"properties": { "Bridge_name": {"type": "string"}, "RepairMethod": {"type": "string"}, "RepairQuantity": {"type": "integer"},
 "required": ["Bridge_name", "RepairMethod", "RepairQuantity"]},

schema_elmt = {"properties": { "Bridge_name": {"type": "string"}, "ConditionGrade": {"type": "string"},
 "required": ["Bridge_name", "ConditionGrade"]},
```

Figure 4.19 Example of defined schema for variable extraction

Once the necessary variables were collected, the function in Figure 4.15 above could be run to generate cost estimates based on machine learning model.



Figure 4.20 Example of asking additional question to user

Generating answers based on a knowledge base is possible through ‘GraphCypherQACHain’ as shown in Figure 4.18. It connects database in Neo4j to QA system and supports the ability to create and run required queries. A lot of questions related to bridge maintenance history can be answered by

running this chain. Additionally, because answers are generated based on the connected knowledge database, there is no hallucination phenomenon, which can be considered a disadvantage of GPT-4.

Lastly, there are cases where the knowledge base and functions created cannot answer all the questions asked by users, and real-time information is needed. For this purpose, the author designed a chain that can run the Serp API function. As it is a customized QA system in for a specific domain, this function defines an instruction in the prompt to be executed when there is no response even after running through the previous chains.

## **Chapter 5. Demonstration of the QA System through User Testing**

This chapter describes a prototype of the application that implemented the bridge maintenance QA system framework proposed in this dissertation. Participants of interview tested the application, and the evaluation results were discussed in this part.

### **5.1 Prototyping of the QA System**

#### **5.1.1 Interface Development using Streamlit**

To test the usability of the proposed framework, a front-end is needed to implement the interaction between the back-end model to the user. Various resources for prototyping the web application have been developed and widely being utilized; such as Gradio (Gradio 2023), which offers an easy interface for creating demos of machine learning models, Streamlit that allows rapid development of data applications (Streamlit 2023). These tools facilitate quick iteration and testing, providing a seamless transition from concept to functional prototype. The author chose Streamlit as a framework for front-end system because it runs in a single Python script, making code management and integration easy. Since the proposed QA system has high

complexity such as database and LLM connection, it was judged that Streamlit was more advantageous. This framework is also actively used in similar previous works (Jain et al. 2022; Shukla et al. 2021), to demonstrate the their systems. For development, the Streamlit application has been installed on Ubuntu 18.04.6 LTS server. Within the virtual environment, the Python version is 3.10.13, and the Streamlit version is 1.10.1.

Please note that the code used to develop Streamlit is entered in the third column of Appendix A. ‘Codes of the Proposed Framework’ section. The user interface was designed as shown in Figure 5.1. It consists of one-page, user names (i.e., 0. Input user name) and questions could be input (i.e., 1. Ask a question about the bridge maintenance), and satisfaction scores could be received through radio buttons (i.e., 2. Evaluation of the satisfaction with the answer). A “QA history” field was created at the bottom so that users can check the history of questions entered during testing. Through this user interface, the author was able to save the user name, entered question, execution time, satisfaction score, and returned question in CSV format.

0. Input user name

Enter your name:

이규원

Save Data

## Bridge Maintenance Query Assistant

순서: 0. 사용자이름 입력 1. 질문 입력 (Ask클릭) 2. 답변 만족도 평가 (Save Data클릭)

### 1. Ask question about the bridge maintenance

Enter your question:

종계교의 바닥판에 점검이력 있어?

Ask

### 2. Evaluation of the satisfaction with the answer

Click your satisfaction score (1: Not Satisfied, 7: Very Satisfied)

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7

Save Data

### QA History

Q: 종계교의 바닥판에 점검이력 있어?

A: 종계교의 바닥판에 대한 점검 이력이 있습니다. 가장 최근의 점검은 2019년 10월 31일에 이루어졌으며, 그 때의 점검 등급은 'B'였습니다. 균열이 17.78의 수치로 기록되었으나, 다른 주요 손상은 보고되지 않았습니다. 이전에 2014년 12월 24일에도 점검이 이루어졌고, 그 때는 재료 분리가 25.12, 누수 및 백태가 221.25, 균열이 1.6, 박락이 2.16, 철근 노출/부식이 2.68로 기록되었습니다. 이 때의 점검 등급은 'D'였습니다.

Execution time: 25.41 seconds

Figure 5.1 User interface for the proposed QA system

### 5.1.2 Design of User Testing

The author decided that the system proposed in this study should be tested twice by users to confirm the results of the initial test and subsequent supplementation. The first user test was conducted to determine whether the developed system operates, evaluate its usability, and identify areas for improvement. Six construction management majors were recruited as participants. The participants had an average of 2.2 years of experience in construction management and all of them had experience in using an LLM. Two interview participants had experience analyzing BMS data. Participants

in the first user testing attended the laboratory in Building 35 of Seoul National University at the designated time and conducted individual interviews (See Figure 5.2).



Figure 5.2 User testing environment

After correcting the recommendations and errors identified in the first user test, the author conducted a second user test to evaluate the feasibility of practical use. Four construction management majors were recruited as participants. They have 5.75 years of experience in this field. Two participants had experience analyzing BMS data, and all four participants had experience using LLM (Table 5.1).

Table 5.1 Personal information of experiment participants

Test	No	Work experience (unit=year)	Final degree	Specialty (Major)	Experience for analyzing BMS data	Experience of using LLM
First	1	7	Master	Construction Management	O	O
	2	1	Bachelor	Construction Management	X	O
	3	1	Bachelor	Construction Management	X	O
	4	1	Bachelor	Construction Management	X	O
	5	1	Bachelor	Construction Management	X	O
	6	2	Bachelor	Construction Management	O	O
Second	7	5	Master	Construction Management	X	O
	8	3	Master	Construction Management	X	O
	9	7	Ph.D	Construction Management	O	O
	10	8	Ph.D	Construction Management	O	O

User testing was conducted for approximately 40 minutes per participant. First, (1) the author provided a brief introduction to the purpose and scope of the system. At this time, the high-level concepts of the bridge maintenance ontology were explained. (2) Users were asked to enter 10 competency questions that represented basic templates for answerable questions, and within each template, users can select and enter detailed information such as bridge name, repair method, and quantity according to the user's preference. Those were provided as examples to help interview participants understand the functions that can be performed in the system. (3) Users were allowed to ask 10 open-ended questions related to bridge maintenance work and receive

answers. (4) Improvements and inconveniences were summarized by participants based on the interview results. Each step took about 10 minutes.

Interview for the usability evaluation of the Bridge maintenance QA system

Date: \_\_\_\_\_

Time: \_\_\_\_\_

[Personal Information: 개인 정보]

Name	Age	Final degree	Major

Affiliation	Position	Work Experience
		(year)

[Check list: 사전 질문]

1) Do you have experience analyzing data from Bridge Management System?

If so, for what purpose?

\_\_\_\_\_

2) Have you ever checked the special laws on safety and maintenance of facilities?

If so, for what purpose?

\_\_\_\_\_

3) Do you have any experience using Large Language Models (ex. ChatGPT)?

If so, to what extent?

\_\_\_\_\_

[교량 이름 선택]

보사원2교	정령교	중마교	고덕교	가산교
통암교	덕소교	정령대교	안학교	화진교
당지교	학암교	서호교	사천교	여운교
포산교	신리교	유구교	안변교	묘탄교
수산교	단양교	봉황교	소월교	월곡교
서하교	청룡교	단촌교	가북교	중월교

[보수보강 공법명: 비선택]

수입공법	표면보수	단면보수
------	------	------

[교량의 상태등급: A ~ E등급]

[점검이력 손상 목록: 비선택]

균열	누수/침하	도랑손상	도랑탈락
형상균열	벽면	벽면	변형
부식	도막	재료분해(콘크리트, 강재)	경각구 손상
철근노출/부식	콘크리트박락	파손	피복유막 부착

[시스템 개선사항] 자유롭게 작성해주시요.

ex. 주기가 불리할 기능, 불편 속도.

Figure 5.3 Check list for the user test participants

The 10 competency questions used at this time are shown in Table 5.2. For each competency question, a sample name was provided so that the bridge name, damage, repair method, condition grade, etc. could be optionally entered (see right side of Figure 5.3). As a result, about 20 question and answer pairs were created and collected for each experiment participant, and the QA pairs collected here were used to calculate accuracy.

Table 5.2. Competency questions for user testing

Class	Question	Expected Answer
Simple retrieval	시설물의 안전 및 유지관리에 관한 특별법은 어떻게 구성되어 있는가?	시설물안전법은 “~”
	시설물의 안전 및 유지관리에 관한 특별법은 몇개의 조와 연결되어 있지?	시설물의 안전 및 유지관리에 관한 특별법은 7개의 조와 연결되어 있습니다.
	시설물의 안전 및 유지관리에 관한 특별법 제7장의 내용은?	제7장의 내용은 “~”
Composite query	{BridgeName} 바닥판과 연결된 점검이력 몇개야?	{BridgeName}의 바닥판과 연결된 점검 이력은 총 {Value}개입니다.
	{BridgeName} 바닥판의 점검이력 시점 알려줘	{BridgeName}의 바닥판 점검 이력에 따르면, {Date}에 점검이 이루어졌습니다
	{BridgeName}는 어느 국도에 있어?	{RoadName}에 위치해 있습니다.
	{BridgeName}의 부재는 무엇이 있어?	{BridgeName}의 부재는 다음과 같습니다:
	{BridgeName}의 바닥판의 {DefectName} 양의 합은?	총 {DefectName}의 합은 {Value}입니다.
Predictive reasoning	{BridgeName}에 {RepairMethod}으로 {보수물량}만큼 보수했을 때 비용 얼마나 들어?	필요한 보수비용은 약 {Value}원입니다
	{BridgeName}에 {RepairMethod}을 보수했을 때 비용 얼마가 필요해?	ValueError:

An additional monitor was installed so that users could check how the entered questions were being processed. The back-end process that users can check is displayed as shown in Figure 5.4.

```

> Entering new AgentExecutor chain...
```json
{
  "action": "Neo4j_search",
  "action_input": "트 산 교 는 어 느 국 도 에 있 어 ?"
}
...

> Entering new GraphCypherQAChain chain...
Generated Cypher:
MATCH (b:Brdg {BRDG_NAME: '트 산 교 '})-[:0nthe]->(r:RoadLane)
RETURN r.ROADLANE_NAME

> Finished chain.

Observation: [{r.ROADLANE_NAME: '일반국도 35호선'}]
Thought:```json
{
  "action": "Final Answer",
  "action_input": "트 산 교 는 일반국도 35호선에 위치해 있습니다 ."
}
...

> Finished chain.

```

Figure 5.4 Given information for participants about the back end

The evaluation of each question was conducted by checking the results and then selecting the satisfaction score. The satisfaction score used a Likert scale ranging from 1 to 7 points. This 7-point Likert scale has been used since 1932, and it provides two moderate opinions along with two extremes, two intermediate, and one neutral opinion to the respondent as shown in Figure 5.5. The interview results were automatically saved as shown in Figure 5.6.

1 2 3 4 5 6 7

○ ○ ○ ○ ○ ○ ○

Not satisfied ----- Neutral ----- Veri satisfied

Figure 5.5 Likert scale

User Name	Input	Output	Execution	Satisfaction
Participant_10	시설물의 안전 및 유지	시설물의 안전 및 유지관	42.8359	3
Participant_10	시설물의 안전 및 유지	시설물의 안전 및 유지관	16.81836	6
Participant_10	시설물의 안전 및 유지	시설물의 안전 및 유지관	14.34765	4
Participant_10	장평교의 바닥판과 연	장평교의 바닥판과 연결된	21.14938	7
Participant_10	장평교의 바닥판의 점	장평교의 바닥판 점검이	12.95539	7
Participant_10	장평교는 어느 국도에	장평교는 지방도325호선	18.16238	7
Participant_10	장평교의 부재는 무엇	죄송합니다, 장평교의 부	7.615284	
Participant_10	중마교의 부재는 무엇	중마교의 부재로는 바닥	14.15483	7
Participant_10	중마교의 바닥판의 균	중마교의 바닥판의 균열	13.91339	5
Participant_10	중마교에 표면보수로	중마교에 표면보수로 10	14.1114	6
Participant_10	고덕교에 C등급일 때	고덕교의 보수보강비용은	11.33196	7

Figure 5.6 Saved data from user testing

5.2 Analysis of User Testing Results

5.2.1 Descriptive Statistics

- **First user testing**

The results of the first user testing can be summarized as follows. The satisfaction scores for the generated answer according to each question and the execution time are organized as shown in Figure 5.7. An average satisfaction score of 4.90 was obtained on the 7-point Likert scale, a score close to satisfied. However, there were also results that gave low satisfaction scores in the range of 1 to 3, which will be discussed in detail at next chapter (Chapter 5.2.2). The average execution time was 13.03 seconds, ranging from a minimum of 5.19 seconds in the short case to a maximum of 44.60 seconds in the long case.

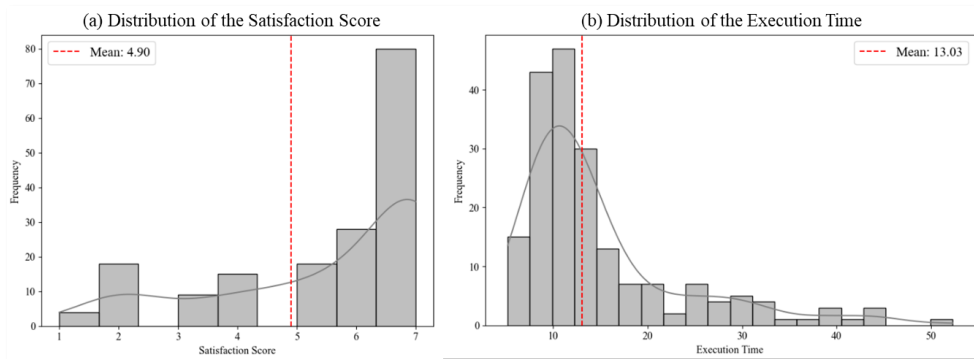


Figure 5.7 Distribution of data from user (testing 1)

Additional statistics were checked to see differences in satisfaction and execution time by question type. There is a difference between the results when the competency questions provided by the researcher were input and when the open-end questions were asked, as shown in Table 5.3. It was confirmed that the satisfaction score when asking open-end questions was 1.58 lower than that of competency questions, and the execution time also increased by about 1.14 seconds. The higher satisfaction score from competency questions indicates that the system proposed in this dissertation is not capable of answering all questions; thus, it implies that it must be used for a specific purpose because its scope is clearly limited.

Table 5.3. Average satisfaction score and execution time by category of question (testing 1)

Category	Satisfaction Score	Execution Time (Sec.)
Competency Question	5.76	12.39
Open-ended question	4.18	13.53

Depending on the method of generating the answer, there was a difference in satisfaction score and execution time as shown in Table 5.4. In the case of ‘Simple retrieval’, it was found that it took longer to execute than the other two methods because it included a step of passing over to internet browsing if the answer was not found. The satisfaction score was highest for ‘predictive reasoning’, followed by ‘composite query’ and ‘simple retrieval’.

Table 5.4 Average satisfaction score and execution time by response generation module (testing 1)

Method of generating answer	Satisfaction score	Execution Time (Sec.)
Simple retrieval	4.56	18.57
Composite query	4.98	10.51
Predictive reasoning	5.36	11.02

To check the comprehensive data distribution, a scatter plot (Figure 5.8) was drawn. The color of data point represents the answer generation method, and the shape of data point represents the classification of the question. There were differences for each question-answering generation method, and this system can reach the conclusion that satisfaction is high in competency questions generated through predictive reasoning.

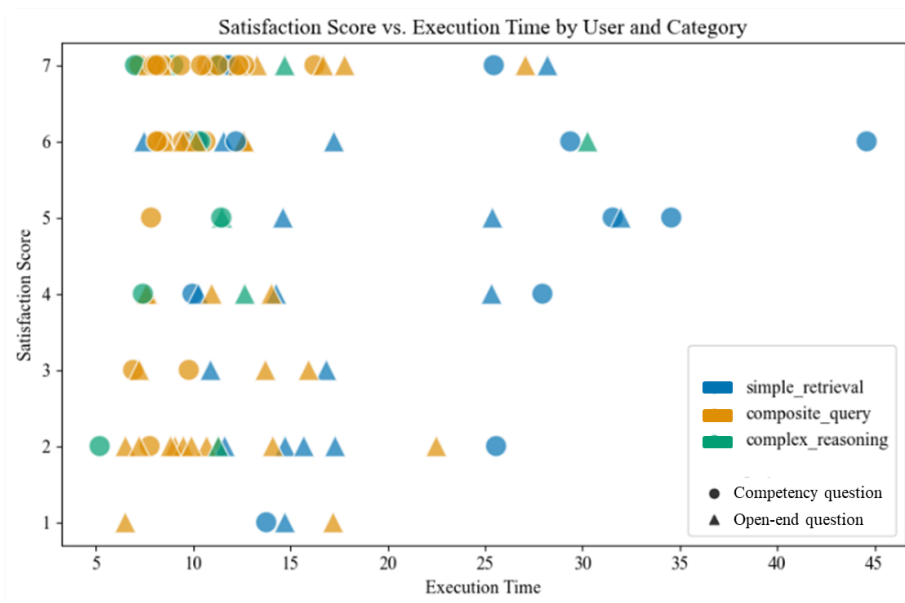


Figure 5.8 Scattered plot of data from user (testing 1)

- **Improvements made through User Feedback**

Common improvement requests obtained through the first test can be summarized as follows. First, there was an opinion that it would be better if the response speed was a little faster. This is a common opinion among people who have previously used ChatGPT and is something that needs to be improved in the future.

Second, there was several confusions about the language of the answers. When a question is asked in Korean, it is correct to answer in Korean, but some answers were given in English. This was mostly a case of ‘simple retrieval’ and is an error that occurs while searching the Internet and can be corrected by adjusting the prompt template.

Third, there was a case that errors occur when users ask questions that require querying multiple entities simultaneously. This problem occurs due to the limitation of the number of tokens when data from too many entities is searched exceed the limits. It can be improved by showing a few-shot example of creating a query that can query the answer by parts, rather than a method of searching the entire data.

And there was the problem of short-term memory in system that remember the variables from previous questions. Even though the user did not enter variables in the question, there were cases where variables

remembered from previous questions were used. This can be improved by deleting the existing context and initializing the session.

Table 5.5 Requirements for improvements from user testing 1

Category		Requirements
System improvement		▪ Reduction of the execution time ▪ Confusion of the response generation language (Kor/Eng) ▪ Retention of preceding variables due to short-term memory
	Front-end	▪ Display of the answer generation process (selected tool in agent)
	Back-end	▪ Prediction of expected defect, cause of defect, required repair method for defect ▪ Establishment of sequential relationship of time ▪ Information of the similar bridges as a reference
Additional features		
Limitations		▪ Simultaneous inquiry of multiple entities ▪ Answers exceeding the number of tokens

The following requests were made for implementation of additional features. From the user's perspective, it is difficult to clearly understand how the knowledge base and QA system were structured. Therefore, there was an opinion that a function was needed to provide guidance on how to modify the question when a user's question comes in and it is difficult to answer. In addition, this QA system only includes a 'predictive reasoning' function for maintenance cost estimation, but additional functions such as requesting an estimate of whether defects have occurred, inferring the cause of the defects, and suggesting repair and reinforcement methods to solve the defects were

requested. This is possible through creating and adding machine learning models. Additionally, there were users who wanted to know how the answer was created in the user interface, whether the answer was found in a knowledge base or web browsing. These improvement requests were partially improved by adding instructions and a few shots example before the second user testing.

- **Second user testing**

The results of the second user testing can be summarized as follows. The satisfaction scores for the generated answer according to each question and the execution time are organized as shown in Figure 5.9. An average satisfaction score of 6.34 was obtained on the 7-point Likert scale, a score close to highly satisfied. Additionally, the number of scores from 1 to 3 indicating dissatisfaction has decreased significantly. The average execution time was 18.27 seconds, ranging from a minimum of 6.81 seconds in the short case to a maximum of 52.26 seconds in the long case.

Similar with the first testing, the competency questions provided by the researcher had high satisfaction score than open-ended questions. The satisfaction score when asking open-end questions was 0.27 lower than that of competency questions, and the execution time also increased by about 3.63 seconds. This can be interpreted as an increase in execution time due to more

chain reviews by adding a few more examples to the prompt to generate the answer, but it also brings us closer to giving the user the answer they want.

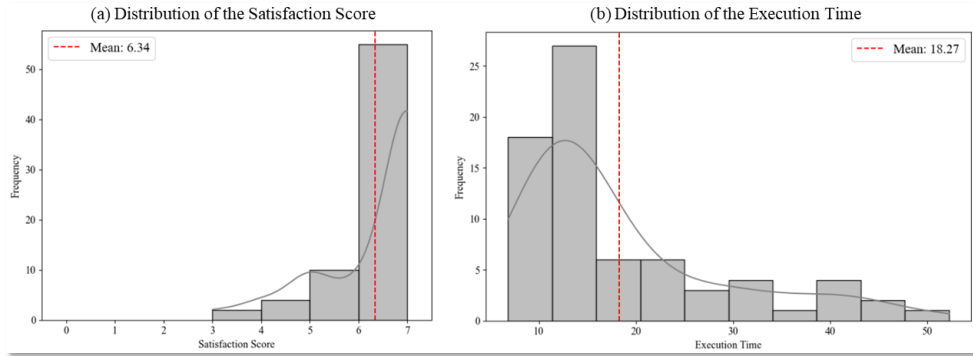


Figure 5.9 Distribution of data from user (testing 2)

Table 5.6. Average satisfaction score and execution time by category of question (testing 2)

Category	Satisfaction Score	Execution Time (Sec.)
Competency Question	6.46	16.61
Open-ended question	6.19	20.24

According to Table 5.7, it can be seen that the satisfaction level of ‘simple retrieval’ has greatly improved compared to the first user testing. It can be interpreted that the added example in prompt template improve the performance of data retrieval in the knowledge base or web browsing. However, as a result, the execution time required to generate an answer also increased.

Table 5.7 Average satisfaction score and execution time by response generation module (testing 2)

Method of generating answer	Satisfaction score	Execution Time (Sec.)
Simple retrieval	6.50	24.46
Composite query	6.26	12.75
Predictive reasoning	6.36	26.07

To check the comprehensive data distribution, a scatter plot (Figure 5.10) was drawn. Although the overall satisfaction score has increased, the execution time for each answer generation function shows a similar trend, which is interpreted to be shorter overall in the case of ‘composite queries’.

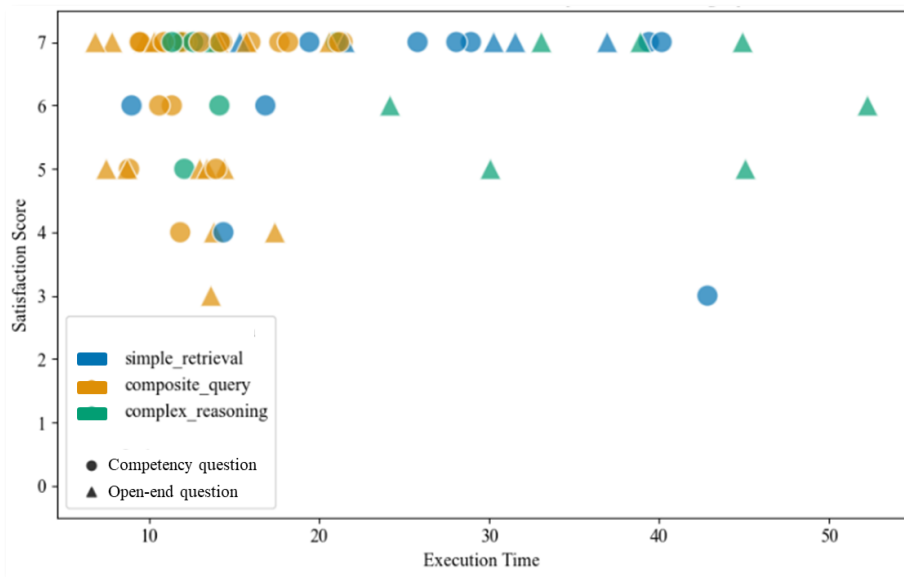


Figure 5.10 Scattered plot of data from user (testing 2)

When the results of the first and second tests are comprehensively reviewed, the results can be obtained as shown in the following tables 5.8 and 5.9. The average satisfaction score was 5.49 points, which can be interpreted as 'generally satisfied' on a scale of 1 to 7. The average response execution speed was 14.97 seconds.

Table 5.8 Average satisfaction score and execution time by category of question (total)

Category	Satisfaction	Execution Time (Sec.)
Closed-ended Question	5.76	12.39
Open-ended question	4.14	13.29

Table 5.9 Average satisfaction score and execution time by response generation module (total)

Category	Satisfaction	Execution Time (Sec.)
Simple retrieval	4.56	18.57
Composite query	4.93	10.40
Predictive reasoning	5.36	11.02

5.2.2 Discussion and Implications

The author comprehensively organized the data collected through two rounds of user testing as follows. A total of 194 QA pairs were created from users, 122 in the first test and 72 in the second test. The author manually checked these answers to see whether the correct answer was returned. As a result, it was possible to derive the answer generation accuracy for each first and second user testing as shown in Figure 5.11. The average accuracy of the first test was low at 61.4%, but it improved to 85.7% in the second test.

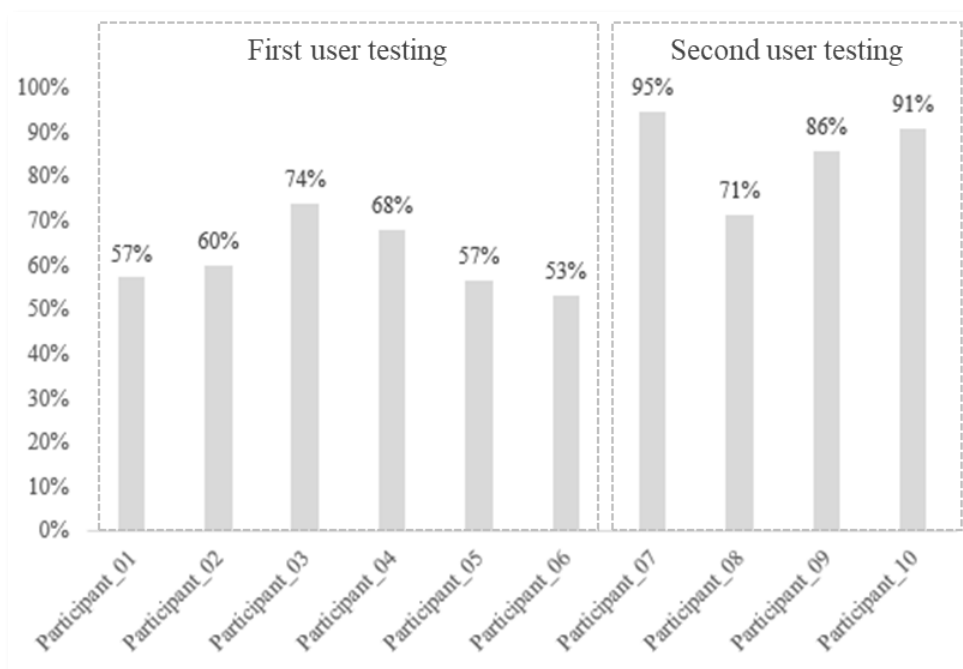


Figure 5.11 Accuracy of answers by participants

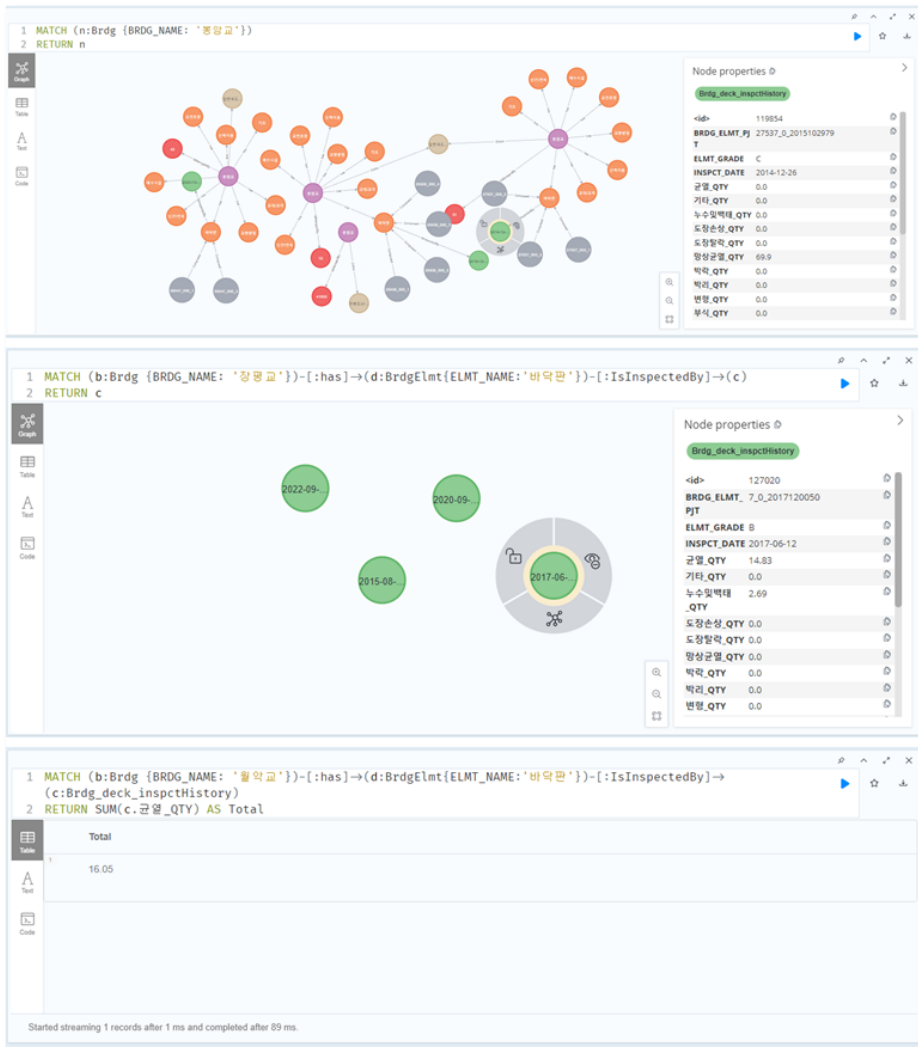


Figure 5.12. Manual checking of returned answers

The process of checking the answer is to input the generated query directly into neo4j and compare it with the returned results (see Figure 5.12). Several implications were obtained through this process, the first of which was that multi-hop was possible to find an answer by crossing multiple relations. When asking a question based on a bridge, it is possible to look up

the property values in the ‘inspection history’ entity through the ‘element’ entity, and it is possible to sum up several values of these properties or reorganize them in descending order. To obtain an answer with a general Cypher query, the MATCH, RETURN, SUM, and DESC functions must be used together, so although it is a fairly complex function, it suggests that it can be implemented relatively easily.

Second, if the user asks a question without linguistic confusion, the answer returned through query creation rarely contains errors. In the question 'What elements does bridge ‘A’ compose?', which was included in the competency question, the Korean word ‘부재’ has the meaning of 'element', but also has the meaning of 'absence'. Because of this, there were some cases in which the LLM model failed to return answers due to linguistic confusion. In cases where other answers were generated well, no incorrect answers were returned.

Third, there were cases where multiple values were returned even though the question required a single answer. In this case, the label in the knowledge base was duplicated. In the case of bridges in Korea, there are often multiple bridges with the same name. In this case, multiple answers are returned because all bridges with same name are queried. However, since it is difficult for the user’s aspect to understand this internal processing, it may be perceived as returning an incorrect answer. Therefore, the author suggest a

method to avoid duplicate labels when building a knowledge base or to specify that the value was generated by searching multiple values when generating an answer.

Chapter 6. Conclusions

This chapter summarizes and discusses the objectives, achievements, and contributions of the research. Current limitations and recommendations for future research are also provided.

6.1 Summary of Research Objectives and Achievements.

Bridges, one of the major social infrastructures, are widely distributed throughout the country. Although a variety of information related to bridge maintenance is being collected in systems such as BMS, there are a lack of users who can make data-based decisions based on this information. In this context, it is difficult to properly maintain the bridge, which should be managed objectively based on data. Therefore, the primary goal of this dissertation was to support a way for bridge maintenance operators to access and analyze the knowledge of bridge maintenance easily. To achieve this goal, the author set the objective to develop a QA system for bridge maintenance workers by integrating heterogeneous data as a graph-based knowledge base for easy queries and generating answers.

To accomplish this objective, the specific objectives were: (1) Integration of the heterogeneous data from different sources with the form of

knowledge base to easily curate the index and path for answering; (2) Development of an LLM-based QA agent to understand user questions and process them with knowledge base and orchestrate the modules to generate answers in appropriate forms required for the QA agent's decision-making; (3) Demonstration of the system usability through prototyping of it.

By addressing the research objectives, this research achieved the results and findings as follows:

- (1) The author developed a knowledge base of bridge maintenance with unstructured data and structured data. A graph database was selected as the optimal format for storing and managing data and answering complex questions, and its completeness was verified through competency questions. Through the creation of a graph-based knowledge base, the author was able to obtain additional insight that improved the performance of retrieval and clustering analysis of similar bridges using relationships between connected nodes.
- (2) The author completed a pipeline of using LLM for QA system by applying GPT-4, which has high natural language processing performance. Prompt engineering was applied based on the LangChain framework. Ultimately, the author developed a system that modularizes several functions that can generate answers such as bridge maintenance-related knowledge retrieval, composite queries

related to past maintenance history, and machine learning-based maintenance cost predictive reasoning.

- (3) A user interface was developed using Streamlit for demonstration of the QA system. It is designed to evaluate the user's satisfaction by asking questions and checking the answers. User satisfaction was evaluated on the Likert-scale, and a score of 5.49 (generally satisfied) was obtained on a scale between 1 and 7, and it was confirmed that it took an average of 14.97 seconds to generate a final answer. Using 194 QA-pairs generated through ten of participants in user test, it was confirmed that the QA system proposed in this dissertation could achieve an accuracy of 85.7% in QA task.

6.2 Research Contributions

The proposed framework can provide the following contributions to the academic field: First, the author expanded the bridge maintenance knowledge base. This is an expanded concept than the existing ontology creation and has academic value in that it combines structured data and unstructured data. Also, in this process, an efficient database format and graph generation method suitable for the bridge maintenance knowledge base was provided by devising and organizing various data modeling methods. By building a QA system based on a domain-specific database, it is possible to provide accurate and in-

depth answers to specific areas. This means that it can provide more reliable information than typical open domain QA systems.

Second, the author proposes an advanced system that can perform data analysis, which previously had to be performed through complex codes, through questions in the form of natural language. In other words, it has improved a structure that can answer within minutes from a task that would take hours to days depending on the level of understanding of the data structure. Through this, practitioners who had difficulty analyzing data can easily derive the desired results, which can support improved bridge maintenance. This can not only improve work efficiency but also contribute to innovation. Also, this provides a new framework model applicable to a variety of fields.

6.3 Recommendations for Future Research

To this end, the author discovered the following limitations in the process of creating a knowledge base and linking the LLM and recommend that future researchers consider the following points.

First, the author proposes an expansion of the knowledge base. In this dissertation, the data used at the current stage of bridge management were integrated into a knowledge base. However, there is a lot of additional heterogeneous data generated from the digitalization of the construction field.

For example, data such as sensor data, 3D models, or images generated from digital twins can also be connected to the database. This will enable new knowledge creation and inference.

Second, the utilization of LLM proposed in this framework can be improved. Instead of using GPT-4 as an API, the future researcher will be able to create a dedicated LLM that can be saved local server. This can be a fundamental way to solve current security problems. Additionally, by reducing unnecessary parameters in model, it will be possible to design a highly efficient but economical LLM structure. Of course, it is expected that various attempts and experiments will be needed to surpass the current level of performance, but it is an area well worth researching.

Third, because the current test aims to verify the initial framework, it was conducted by recruiting about 10 people majoring in the field. However, in order to obtain more realistic results, it is necessary to collect and test with field experts in the future. This is designated as a research goal to be carried out in the near future.

Finally, although the system proposed in this study is primarily aimed at being used to help field staff who have difficulty retrieving and analyzing appropriate data due to frequent rotation or lack of manpower, the following recommendations are made to expand the user experience: For example, in the case of experts, this system can increase the reliability and explanatory

power of the answer by showing not only the answer value provided as the answer but also the raw data used to generate the answer. This function can be easily implemented by adding a command to the answer generation module according to the results of comparing the answer generation time and user's satisfaction.

Appendix A. Codes of the Proposed Framework

- **Codes for Knowledge Base Generation**

```
from neo4j import GraphDatabase
from py2neo import Graph, Node, Relationship
import pandas as pd
import numpy as np
host = "bolt://147.47.35.***:7687"
user = "neo4j"
password = "letmein"
gds = GraphDatabase.driver(host, auth=(user, password))

for idx, row in df.iterrows():
    node_properties = {k: float(v) if isinstance(v, np.float64) else v for k, v in row.items()}
    node_properties["index"] = idx
    node = Node("SpecialAct_Chapter", **node_properties)
    graph.create(node)
for _, row in df_1_3.iterrows():
    CHAPTER = row['chapter']
    ARTICLE = row['new_index']
    chapter = graph.nodes.match("SpecialAct_Chapter", new_index=CHAPTER).first()
    article = graph.nodes.match("SpecialAct_Article", new_index=ARTICLE).first()
    if chapter is not None and article is not None:
        relationship = Relationship(article, "IsPartOf", chapter)
        graph.create(relationship)
```

- **Codes for table extraction from the unstructured data**

```
import camelot
import pandas as pd
pd.set_option('display.max_colwidth', -1)
file_path = "02_시설물의 안전 및 유지관리에 관한 특별법 시행령(대통령령).pdf"
tables = camelot.read_pdf(file_path, pages='all', flavor='lattice')
dataframes = [table.df for table in tables]
number_of_tables = len(dataframes)
df2 = dataframes[15]
df2 = df2.replace("\n", "", regex=True)
df2.columns = df2.iloc[0]
df2 = df2.drop(0).reset_index(drop=True)
```

- **Codes for execute the Streamlit application**

```
# Initialization of session state
if 'data' not in st.session_state:
    st.session_state['data'] = pd.DataFrame(columns=['User Name', 'Input', 'Output', 'Execution Time', 'Satisfaction'])

# Define Streamlit app function
def main():

    # Initiate if the history is empty
    if 'history' not in st.session_state:
        st.session_state['history'] = []

    st.title('Bridge Maintenance Query Assistant')
    st.text('순서: 0.사용자이름 입력 1.질문 입력(Ask클릭) 2.답변 만족도 평가(Save Data클릭)')
    st.sidebar.title('0. Input user name')
    user_name = st.sidebar.text_input('Enter your name:')

    # Define a text input
    st.subheader('1. Ask question about the bridge maintenance ')
    question = st.text_input('Enter your question:')

    # Define a button and its action
    if st.button('Ask'):
        start_time = time.time()
        answer = ask_question_with_template(question)
        execution_time = time.time() - start_time

        st.write(answer)
        st.write(f"실행 시간: {execution_time:.2f}초")

        new_data = pd.DataFrame({'User Name': user_name, 'Input': question, 'Output': answer,
                                'Execution Time': execution_time})
        st.session_state['data'] = pd.concat([st.session_state['data'], new_data], ignore_index=True)
        # Append data in history
        st.session_state['history'].append((question, answer, execution_time))

    st.subheader('2. Evaluation of the satisfaction with the answer')
    satisfaction = st.radio('Click your satisfaction score (1: Not Satisfied, 7: Very Satisfied)', range(1, 8))
```

```

# Save satisfaction score in the session_state
st.write('<style>div.row-widget.stRadio > div{flex-direction:row;}</style>',
unsafe_allow_html=True)
st.session_state['satisfaction'] = satisfaction

if st.button('Save Data'):
    current_data = st.session_state['data']
    if 'satisfaction' in st.session_state:
        current_data.loc[current_data.index[-1], 'Satisfaction'] = st.session_state['satisfaction']

    filename = "user_queries.csv" if not user_name else f"{user_name}_queries.csv"
    current_data.to_csv(filename, index=False, encoding='cp949')
    st.write('Data saved to', filename)

st.subheader('QA History')

# Display the previous question, answer, execution time
for q, a, t in reversed(st.session_state['history']):
    st.write("Q:", q)
    st.write("A:", a)
    st.write(f"Execution time: {t:.2f} seconds")
    st.write("---")

# Run the main function in Streamlit.
if __name__ == '__main__':
    main()

```

Appendix B. Frequently Asked Questions in Existing System

• Question and answers from the education system on KALIS (Sample)

국토안전관리원 국토안전교육원 (<https://www.kalisedu.or.kr/user/info/InfoBoardUserList.do>)

Question	Answer
3종 시설물 대상에 해당된다면 무조건 정기 안전점검을 실시해야 하나요?	아닙니다. 3종 시설물은 중앙행정기관의 장 또는 지방자치단체의 장이 「시설물안전법」 시행령 제5조(제3종시설물의 지정·해제 등)에 따라 지정한 시설물만 해당됩니다.
성능평가와 내진성능평가는 같은 말인가요?	아닙니다. 「시설물안전법」 제1장 제2조(정의)에 따르면 "내진성능평가"란 지진으로부터 시설물의 안전성을 확보하고 기능을 유지하기 위하여 「지진·화산재해대책법」 제14조제1항에 따라 시설물별로 정하는 내진설계기준에 따라 시설물이 지진에 견딜 수 있는 능력을 평가하는 것을 말하며, "성능평가"란 시설물의 기능을 유지하기 위하여 요구되는 시설물의 구조적 안전성, 내구성, 사용성 등의 성능을 종합적으로 평가하는 것을 말한다. 내진설계 대상 시설물 중 내진성능평가를 받지 않은 시설물에 대하여 정밀안전진단을 실시하는 경우에는 해당 시설물에 대한 내진성능평가를 포함하여 실시하여야 한다. (시설물안전법 제12조③항)
성능평가과정의 대상시설물은 어떤 것이 있나요?	「시설물안전법」 시행령 별표 13을 참고하시기 바랍니다.

시설물통합정보관리시스템 (<https://www.fms.or.kr/com/mainFrame.do>)

Question	Answer
시설물 정기안전점검 시기 관련하여 문의드립니다. 시설물안전법 시행령 별표3 비고9번, '정밀안전진단의 실시 완료일이 속한 반기에 실시하여야 하는 정기안전점검은 생략할 수 있다' 관련한 질의입니다. 정밀안전점검 용역이며 과업기간 2023년 06월 26일 ~ 2023년 07월 25 (계약일: 2023-06-14)입니다 7월에 준공되어 하반기 정기안전점검용역 생략가능한지 문의드립니다.	정기안전점검의 상위 점검인 정밀안전점검의 종료일이 하반기(7월)인 경우 하반기의 정기안전점검은 생략이 가능함을 알려드립니다.
제3종시설물 지정관련 해서 문의드립니다. 국가안전대진단(집중안전점검)기간을 통해 제3종시설물로 지정되어야 하는 게 있었습니다. 관리주체(제3종시설물 지정요청) -> 취합기관 승인까지 했습니다. 이후에 절차에 대해 궁금합니다. 지정 시 행정절차와 시스템상의 절차가 궁금합니다. 참고로, 시스템상의 절차는 매뉴얼을 찾아봤지만 이해가 안됩니다.	제3종시설물 지정요청에 대한 승인 이후 절차는 취합기관 또는 지정기관의 아이디로 3종시설물 지정등록을 실시하여야 합니다. 지정등록 하는 방법은 FMS 총괄 매뉴얼(p.116) 제3종시설물 지정등록을 참고해주시기 바랍니다.

1) 1차 개발행위 하여 옹벽 연장 100m 이상에 높이 4.9m 준공 2) 2차 개발행위 하여 1차 개발행위 한 곳에 옹벽 2단으로 4.9m 시공 중일 때 1차 개발행위 할 때는 시특법 2종 시설물이 아니고 2차 개발행위가 준공된다면 시특법 2종 시설물이라고 알고 습니다. 그러면 2차 옹벽 준공 시에 안전점검종합보고서를 만들고 FMS에 등록하여야 하나요? 아니면 2차 개발행위 시작할 때 FMS 등록 먼저 해야 하나요?

지방에 있는 연장 300m 터널인데 차량이나 기차는 안 다니는 '사람'만 다니는 터널입니다. 이 터널을 중 시설물로 등록하고 관리를 해야 하는지 궁금합니다. 관리를 하게 된다면 몇 중시설물에 등록해야 하는지도 궁금합니다.

유지관리계획은 누가 작성하는 건가요?

1. 여름철 정기안전점검 기간에 철거공사가 시작될 경우에 [시설물관리]-[시설물철거]에서 철거 정보 등록할 경우 정기안전점검들 안해도 되는지??

재난위험시설(D급)으로 지정된 제3종 시설물의 경우, 관리주체가 [시설물관리]-[시설물철거]후 취합기관에서 승인하는 절차로 가능한지? 아니면 관리주체가[시설물관리]-[제3종시설물지정요청] 후 취합기관의 승인 및 해제고시를 절차로 해야 되는지? 3. 또한, 해당 절차는 철거공사가 완료된 경우 해야되는지, 그 전에 해도 되는지 확인 바랍니다.

기준에 따라 내진성능평가를 받은 시설물의 같은 경우 최초 정밀안전진단을 실시하였을 경우 다시 내진성능평가를 하여야 하는지 궁금합니다. 아니면 최초에 받은 내진성능평가를 참고하여 2회차 정밀안전진단시에도 내진성능평가를 안해도 되는지 궁금합니다. 아니면 정밀안전진단시 항시 내진성능평가를 받아야 하는지 궁금합니다.

2차 옹벽 준공 시 시설물 안전법의 2종시설물 범위에 포함되신다면 준공 전에 시설물안전법 제9조에 의거하여 건설, 공급하는 사업주체는 설계도서, 시설물관리대장을 제출하여야 함을 알려드립니다.

터널시설물의 경우 시설물 안전법 시행령 별표1에 의거해 도로터널 및 철도터널로 구분하여 1,2종시설물로 지정되어 있습니다. 문의하신 내용 중 연장 300m 터널의 경우 도로터널 2종시설물에 해당되어 보이나, 도로터널의 경우 도로법 제10조에 해당하는 도로인지 판단이 필요해 보입니다. 해당하실 경우에는 2종시설물로 등록해주시기 바랍니다.

유지관리계획은 「시설물의 안전 및 유지관리에 관한 특별법」 제6조제1항에 따라

'관리주체'가 작성하여야 합니다. 다만, 제7조에 따른 제3종시설물 중

「공동주택관리법」 제2조제2호에 따른 의무관리대상 공동주택이 아닌 공동주택 등 민간관리주체 소관 시설물 중 대통령령으로 정하는 시설물의 경우에는 특별자치시장, 특별자치도지사, 시장, 군수 또는 구청장이 작성하여야 합니다.

「시설물의 안전 및 유지관리에 관한 특별법」 시행령 별표3의 비고12호에 따라 ①증축, 개축 및 리모델링 등을 위하여 공사 중이거나 철거예정인 시설물로서, ②사용되지 않는 시설물에 대해서는 국토교통부장관과 협의하여 안전점검, 정밀안전진단 및 성능평가의 실시를 생략하거나 그 시기를 조정할 수 있습니다. 정기안전점검의 실시를 생략하거나 그 시기를 조정하려는 경우 ①,②의 조건을 모두 충족하여야 합니다. 위 내용은

국토교통부(시설안전과)와 협의하시어,국토안전관리원으로 FMS 수정 요청 공문을 국토교통부 회신 공문과 함께 송부하여 주시면 처리가 가능합니다. *정기안전점검 : 도래시기 1개월 이전 협의완료

FMS에서 3종시설물의 철거요청은 3종시설물 해제(고시) 완료 후, 가능합니다. 관리주체가 3종시설물을 해제 요청한 후, 3종지정기관에서 승인, 해제/알림, 고시 완료하여야 해제등록 절차가 완료됩니다.

「시설물의 안전 및 유지관리에 관한 특별법」 제12조제3항에 따라 관리주체는 「지진·화산해대책법」 제14조제1항에 따른 내진설계 대상 시설물 중 내진성능평가를 받지 않은 시설물에 대하여 정밀안전진단을 실시하는 경우에는 해당 시설물에 대한 내진성능평가를 포함하여 실시하여야 합니다. 따라서, 기존에 내진성능평가를 실시한 시설물의 경우 정밀안전진단 시 내진성능평가를 포함하여 실시할 의무가 없음을 알려드립니다.

Appendix C. Competency Question List for Knowledge Base

• Simple Retrieval

No	Question	Answer	Process
1	시설물의 안전 및 유지관리에 관한 특별법은 어떻게 구성되어 있는가?	시설물 안전법은 총 8장으로 구성되어 있습니다. 제1장은 총칙을 다루고, 제2장은 기본계획 등을 다룹니다. 제3장은 시설물의 안전관리에 대해 다루고, 제4장은 안전점검등의 대행에 관한 내용을 포함합니다. 제5장은 시설물의 유지관리에 대해 다루고, 제6장은 시설물의 보수에 관한 내용을 포함합니다. 제7장은 보칙을 다루고, 마지막으로 제8장은 벌칙에 대한 내용을 다룹니다.	>Entering new GraphCypherQACChain chain... Generated Cypher: MATCH (law:Law)-[:LINKS_TO]->(chapter:SpecialAct_Chapter) RETURN law.name, chapter.장, chapter.장_내용 Full Context: [{ 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제2장', 'chapter.장_내용': '기본계획 등'}, { 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제3장', 'chapter.장_내용': '시설물의 안전관리'}, { 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제4장', 'chapter.장_내용': '안전 및 유지관리에 관한 특별법', 'chapter.장_내용': '안전 및 유지관리에 관한 특별법', 'chapter.장': '제8장', 'chapter.장_내용': '벌칙'}, { 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제1장', 'chapter.장_내용': '총칙'}, { 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제7장', 'chapter.장_내용': '보칙'}, { 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제5장', 'chapter.장_내용': '시설물의 유지관리'}, { 'law.name': '시설물의 안전 및 유지관리에 관한 특별법', 'chapter.장': '제4장', 'chapter.장_내용': '안전점검등의 대행'}] > Entering new GraphCypherQACChain chain... Generated Cypher: MATCH (law:Law)-[:LINKS_TO]->(chapter:SpecialAct_Chapter) RETURN COUNT(chapter) AS num_of_chapters Full Context: [{ 'num_of_chapters': 7}]
2	시설물의 안전 및 유지관리에 관한 특별법은 몇개의 장을 포함하고 있지?	시설물의 안전 및 유지관리에 관한 특별법은 7개의 장을 포함하고 있습니다.	> Entering new GraphCypherQACChain chain... Generated Cypher: MATCH (s:SpecialAct_Article)-[:IsPartOf]->(s:SpecialAct_Chapter) RETURN COUNT(DISTINCT s) Full Context: [{ 'COUNT(DISTINCT s)': 57}]
3	시설물의 안전 및 유지관리에 관한 특별법은 몇개의 조와 연결되어 있지?	시설물의 안전 및 유지관리에 관한 특별법은 57개의 조와 연결되어 있습니다	> Entering new GraphCypherQACChain chain... Generated Cypher: MATCH (a:SpecialAct_Chapter {장: '제7장'}) RETURN a.장_내용 Full Context: [{ 'a.장_내용': '보칙'}]
4	시설물의 안전 및 유지관리에 관한 특별법 제7장의 내용은?	제7장은 '보칙'에 관한 내용입니다	

No	Question	Answer	Generated Cypher
1	일반국도39호선과 연결된 교량 10개만 알려줘	일반국도39호선과 연결된 교량 10개는 다음과 같습니다:1. 중추교2. 미당교 3. 청흥교 4. 입동교 5. 유구교 6. 추계교 7. 제1거산교 8. 방미교 9. 현곡교 10. 풍무교'	MATCH (b:Brdg)-[:Onthe]->(r:RoadLane) WHERE r.ROADLANE_NAME = '일반국도39호선' RETURN b.BRDG_숫자 LIMIT 10
2	길천교는 일반국도42호선위에 있어?	네, 길천교는 일반국도42호선 위에 있습니다.'	MATCH (b:Brdg)-[:Onthe]->(r:RoadLane) WHERE b.BRDG_NAME = '길천교' AND r.ROADLANE_NAME = '일반국도42호선' RETURN b, r
3	부사원3교는 어느 국도에 있어?	부사원3교는 일반국도5호선에 있습니다.'	MATCH (b:Brdg {BRDG_NAME: '부사원3교'})-[:Onthe]->(r:RoadLane) RETURN r.ROADLANE_NAME
4	변천2교의 바닥판과 연결된 점검이력 몇개야?	변천2교의 바닥판과 연결된 점검이력은 4개입니다.'	MATCH (b:Brdg {BRDG_NAME: '변천2교'})-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(bd:Brdg_deck_inspctHistory) RETURN COUNT(bd)
5	변천2교의 바닥판과 연결된 점검이력 날짜는?	변천2교의 바닥판과 연결된 점검이력의 날짜는 2015년 8월 17일, 2017년 6월 23일, 2019년 12월 17일, 2021년 6월 9일입니다.'	MATCH (b:Brdg {BRDG_NAME: '변천2교'})-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(bd:Brdg_deck_inspctHistory) RETURN bd.INSPECT_DATE
6	변천2교의 난간종류는?	변천2교의 난간 종류는 콘크리트 난간입니다.'	MATCH (b:Brdg {BRDG_NAME: '변천2교'})-[:has]->(be:BrdgElmt) RETURN be.ELMT_DETAIL
7	변천2교 바닥판은 78_0이다. 이때 모든 점검이력의 누수및백태 양의 합은? 어느 점검이력에 얼마나 있는지도 보여줘	변천2교 바닥판의 모든 점검 이력의 누수및백태 양의 합은 147.49입니다. 아래는 각 점검 이력과 해당 양입니다 2021년 6월 9일에 점검한 이력: 누수및백태 양은 0.26입니다.2019년 12월 17일에 점검한 이력: 누수및백태 양은 5.54입니다. 2017년 6월 23일에 점검한 이력: 누수및백태 양은 17.08입니다. 2015년 8월 17일에 점검한 이력: 누수및백태 양은 124.61입니다.	MATCH (b:Brdg)-[:has]->(be:BrdgElmt {BRDG_ELMT: '78_0'}) MATCH (be)-[:IsInspectedBy]->(i:Brdg_deck_inspctHistory) RETURN SUM(i.누수및백태_QTY) AS Total_Leakage, i.INSPECT_DATE AS Inspection_Date

Appendix D. User Testing Sheet

- Participant No.1

Interview for the usability evaluation of the Bridge maintenance QA system

Date: 2023-11-24

Time: 13:09

[Personal Information: 개인 정보]

Name	Age	Final degree	Major
정 세 환	32	M. Sc.	토목공학 (건설관리)
Affiliation		Position	Work Experience
서울대학교 건설정보공학부		박사과정생	7 (year)

[Check list: 사전 질문]

1) Do you have experience analyzing data from Bridge Management System? Yes.

If so, for what purpose?

무시각적 손상 분석에 관련된 자료 수집을 위한 자료의 분석

2) Have you ever checked the special laws on safety and maintenance of facilities? Yes.

If so, for what purpose?

교량시설물의 유지관리 및 점검 체계/절차 파악

3) Do you have any experience using Large Language Models (ex. ChatGPT)? Yes.

If so, to what extent?

ChatGPT, fine-tuning of an existing LLM (e.g. BERT)

[교량 이름 샘플]

부사원3교	장평교	중마교	고덕교	가산교
봉암교	덕소교	청평대교	연하교	화진교
양지교	학담교	서호교	사천교	여운교
토산교	신리교	유구교	만년교	모란교
수산교	단양교	봉황교	소월교	울악교
서하교	청룡교	단촌교	가부교	증들교

와

[보수보강 공법명: 바닥판]

주입공법	표면보수	단면보수
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[점검이력 손상 목록: 바닥판]

균열	누수밀백택	도장손상	도장탈락
함상균열	박락	박리	변형
부식	오염	재료분리(공동, 공극)	정착구 부식
철근노출/부식	콘크리트박락	파손	피복두께 부족

[시스템 개선사항] 자유롭게 작성해주세요.

ex: 추가가 필요한 기능, 답변 속도.

답변 속도 측면

- 간단한 질문이 피터링 장으로 해결될 것 같은 질문이 예상보다 시간이 오래 걸림.

(예: 00 사용설명서 관련하는 교량 10개 알려주)

- 시스템 인터페이스에 보조선 때, 해당이 어디에서 문제가 발생했는지 알려 주려함 → 해당이 내 질문의 의도를 파악해 못했을 때, 질문을 더 자세히 바꿔야 하도록 알려 주려함.

- 정정사항을 요약해서 알려주는 것은 매우 좋음.

- 관련4무호로 관련 마당 교량을 알려달라고 하는 것은 기본적인 질문 끝으로 답이 끝나지 않아 당황스러움.

- 기존 질문 content를 삭제하고 새로운 취재할 수 있다면 좋음 것 같음.

추가되면 좋음 기능. 문장

- 손상 받아 여백 측정/예측 기능.

- 손상이 알려진 후로 관련 지시 → 이를 해결하기 위한 보수방법 관련 지시

- 주로 관련 대상 교량 식별 기능 (예: 손상문자로 가장 많은 10개 교량). - 교량 미리 등급 예측

- 손상문자로 보수금액을 수리비 잔디 지시.

- Participant No.10

Interview for the usability evaluation of the Bridge maintenance QA system

Date: 2023.12.06.

Time: 20:15

[Personal Information: 개인 정보]

Name	Age	Final degree	Major
장태연	32	박사	건설관리
Affiliation		Position	Work Experience
삼성전자		Staff Engineer	7(year)

[Check list: 사전 질문]

1) Do you have experience analyzing data from Bridge Management System?

If so, for what purpose?

있음. 교량의 부재 상태, 손상 크기, 보수 비용을 추정하고자 함.

2) Have you ever checked the special laws on safety and maintenance of facilities?

If so, for what purpose?

있음. BMS 데이터 분석을 위한 교량 유지관리 기본 배경지식 습득 목적.

3) Do you have any experience using Large Language Models (ex. ChatGPT)?

If so, to what extent?

있음. 영문번역 / Python 코드 수정 / 사회이슈 검색

[교량 이름 샘플]

부사원3교	장평교	중마교	고덕교	가산교
봉암교	덕소교	청평대교	연하교	화진교
양지교	학담교	서호교	사천교	여운교
토산교	신리교	유구교	만년교	모란교
수산교	단양교	봉황교	소월교	월악교
서하교	청룡교	단촌교	가부교	중흥교

[보수보강 공법명: 바닥판]

주입공법	표면보수	단면보수
------	------	------

[교량의 상태등급: A - E등급]

[점검이력 손상 목록: 바닥판]

균열	누수및백태	도장손상	도장탈락
망상균열	박락	박리	변형
부식	오염	재료분리(공동, 공극)	경각구 부식
철근노출/부식	콘크리트백태	파손	피복두께 부족

[시스템 개선사항] 자유롭게 작성해주세요.

단순검색

- 실시간 필요한 정보는 아니지만, 단순검색 시 사용자의 기대 답변 속도(구글링, ChatGPT는 매우 빠름..)을 생각했을 때 답변 속도가 느린 편임.
- 기본개념을 질의하는 경우에는 개인적인 생각으로는 구글검색 결과가 조금 더 유용할 것 같음.
- 단순 내용에 대한 질문이라면 더 구체적인(많은 정보가 담긴) 답변을 주어도 좋을 것 같음(예. 시 설물안전법 7장 내용을 모두 기입, 법 내용을 볼 수 있도록 하이퍼링크 연결)

복합쿼리/복합 추론

- 손상량에 대한 답변 시 단위가 없음. 데이터 상 다양한 단위의 손상량이 있는데, 단위별 정보를 제공하는 것도 필요할 수 있음.
- 또한 사용자는 손상량에 대한 단위를 붙여서 질문할 것으로 예상되는데, 이러한 쿼리에 대한 답변이 준비되어 있는지 확인이 필요함.

복합추론

- 추론의 방식(예. 어떤 이력데이터를 활용했는지, 활용한 알고리즘/모델은 무엇인지, 정확도는 얼마나 되는지)을 사용자가 알 수 있도록 간단한 설명을 덧붙이는 것도 좋을 것 같음.

기타

- 의미는 같으나 테스트한 문장이 아닌 다른 문장으로 질의했을 때도 동일한 답변을 줄지 검증하는 것도 필요함.

Appendix E. Back-end Process of User Testing

- Agent execution process of Back-end

```
> Entering new AgentExecutor chain...
```json
{
 "action": "Neo4j_search",
 "action_input": "시설물의 안전 및 유지관리에 관한 특별법의 구성"
}
...
> Entering new GraphCypherQAChain chain...
Generated Cypher:
MATCH (s:SpecialAct_Chapter)-[:IsPartOf]->(a:SpecialAct_Article)-[:Constitutes]-
>(p:SpecialAct_Paragraph)-[:IsContainedIn]->(sp:SpecialAct_SubParagraph)-[:BelongsTo]-
>(i:SpecialAct_Item)
WHERE s.장_내용 CONTAINS '시설물의 안전 및 유지관리에 관한 특별법'
RETURN s, a, p, sp, i
> Finished chain.
Observation: []
Thought:```json
{
 "action": "Serp_search",
 "action_input": "시설물의 안전 및 유지관리에 관한 특별법의 구성은 어떻게 되어 있는가?"
}
Observation:
[http://www.law.go.kr/%EB%B2%95%EB%A0%B9/%EC%8B%9C%EC%84%A4%EB%AC%BC%EC%9D
%98%20%EC%95%88%EC%A0%84%20%EB%B0%8F%20%EC%9C%A0%EC%A7%80%EA%B4%80%
EB%A6%AC%EC%97%90%20%EA%B4%80%ED%95%9C%20%ED%8A%B9%EB%B3%84%EB%B2%9
5, '제1조(목적) 이 법은 시설물의 안전점검과 적절한 유지관리를 통하여 재해와 재난을 예방하고
시설물의 효용을 증진시킴으로써 공. 중(公衆)의 안전을 확보하고 나아가 ..., '본 「시설물의 안전 및
유지관리의 실시 세부지침(성능평가 편)(이하 「세부지침」 이라 한다)은 「시설 물의 안전 및
유지관리에 관한 특별법」 (이하 「법」 이라 한다) 제43조 및 같은 「법」 시행령(이하 「령」 이라
한다) 제30조 및 같은 「법」 시행규칙(이하 「규칙」 이라 한다) 제32조에 따 ..., '서울기술사학원
토목시공기술사 시설물의 안전 및 유지관리에 관한 특별법 핵심설명.', '이 법률의 목적은 시설물의
안전점검과 적절한 유지관리를 통하여 재해와 재난을 예방하고 시설물의 효용을 증진시킴으로써
공중(公衆)의 안전을 확보하고 ..., '③ 전기설비, 기계설비 또는 계측시설을 포함하는 복합된
시설물(건축물은 제외한다)의 구조안전에 관련되는 전기설비, 기계설비 또는 계측시설에 대한
정밀안전진단을 ..., '시설물의 안전 및 유지관리에 관한 특별법 96. 2. 건축법
..... 133. 3. 재난 및 안전관리 기본법 151. 4.
학교안전사고 예방 및 보상에 관한 ..., '국토안전관리원 #소규모 #취약시설 #안전점검 #시설물
#사회복지시설 #안전점검 #시설물안전법 #안전사고 22년도 사회복지시설 안전 및 유지관리에 ..., '...
유지하여야 한다고 규정. 하고 있음. 단, 교량 및 터널은 「시설물의 안전 및 유지관리에 관한
특별법(이하. 시설물안전법)」 에 따라 안전점검을 실시하여야 함. 구분.』
Thought:```json
{
 "action": "Final Answer",
 "action_input": "시설물의 안전 및 유지관리에 관한 특별법은 시설물의 안전점검과 적절한
유지관리를 통하여 재해와 재난을 예방하고 시설물의 효용을 증진시킴으로써 공중의 안전을 확보하는
것을 목적으로 합니다. 이 법은 복합된 시설물의 구조안전에 관련되는 전기설비, 기계설비 또는
계측시설에 대한 정밀안전진단을 포함하고 있습니다. 또한, 교량 및 터널은 이 법에 따라 안전점검을
실시하여야 합니다. 자세한 내용은 법률 텍스트를 참조하시는 것이 좋습니다."
}
> Finished chain.
> Entering new AgentExecutor chain...
```json
{
  "action": "Neo4j_search",
  "action_input": "시설물의 안전 및 유지관리에 관한 특별법은 몇개의 조와 연결되어 있지?"
}
> Entering new GraphCypherQAChain chain...
Generated Cypher:
```

MATCH (s:SpecialAct_Article)-[:IsPartOf]->(c:SpecialAct_Chapter) RETURN COUNT(DISTINCT c) AS num_of_chapters > Finished chain. Observation: [{ 'num_of_chapters': 7}] Thought: ``json { "action": "Final Answer", "action_input": "시설물의 안전 및 유지관리에 관한 특별법은 총 7개의 조와 연결되어 있습니다."} ``
> Entering new AgentExecutor chain... ``json { "action": "Neo4j_search", "action_input": "모란교는 어느 국도에 있어?"} `` > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg)-[:Onthe]->(r:RoadLane) WHERE b.BRDG_NAME = '모란교' RETURN r.ROADLANE_NAME > Finished chain. Observation: [{ 'r.ROADLANE_NAME': '일반국도23호선'}] Thought: ``json { "action": "Final Answer", "action_input": "모란교는 일반국도 23호선에 위치해 있습니다."} ``
> Entering new AgentExecutor chain... ``json { "action": "Neo4j_search", "action_input": "중마교의 손상 이력 중 가장 많은 것은 무엇인가?"} `` > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg)-[:IsManagedBy]->(m:Manage_Agency_Sub)-[:Under]->(s:Manage_Agency_Sup) WHERE b.BRDG_NAME = '중마교' MATCH (b)-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(i:Brdg_deck_inspetHistory) WITH i, be ORDER BY i.부식_QTY DESC LIMIT 1 RETURN i, be > Finished chain. Observation: [{ 'i': { 'INSPCT_DATE': '2014-12-24', '박리_QTY': 0.0, '콘크리트백태_QTY': 0.0, '재료분리(공동,공극)_QTY': 0.0, '망상균열_QTY': 144.00, '누수및백태_QTY': 0.1, '도장탈락_QTY': 0.0, '부식_QTY': 0.0, '오염_QTY': 0.0, '정착구 부식_QTY': 0.0, 'BRDG_ELMT_PJT': '25015_0_2015102370', 'QTY': 0.0, '변형_QTY': 0.0, '철근노출/부식_QTY': 0.0}, 'be': { 'ELMT_SIZE': 940.5, 'ELMT_DETAIL': '콘크리트 슬래브', 'SIZE_UNIT': '콘크리트 슬래브', 'ELMT_NAME': '바닥판'}}] Thought: ``json { "action": "Final Answer", "action_input": "중마교에서 가장 많이 발견된 손상은 '망상균열'이며, 그 양은 144.0입니다. 또한, '균열'도 발견되었으며 그 양은 5.6입니다."} ``
> Entering new AgentExecutor chain... ``json { "action": "Neo4j_search", "action_input": "과손이 가장 많이 발생한 교량"} `` > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg)-[:IsManagedBy]->(m:Manage_Agency_Sub) RETURN b.BRDG_NAME, count(*) AS damage_count ORDER BY damage_count DESC LIMIT 1 > Finished chain. Observation: [{ 'b.BRDG_NAME': '신기교', 'damage_count': 9}] Thought: ``json { "action": "Final Answer", "action_input": "과손이 가장 많이 발생한 교량은 '신기교'이며, 총 9회의 과손이 발생했습니다."} ``
> Entering new AgentExecutor chain... ``json

<pre> { "action": "Neo4j_search", "action_input": "도장손상이 가장 많이 발생한 교량"}''' > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg)-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(bdih:Brdg_deck_inspctHistory) RETURN b.BRDG_NAME, SUM(bdih.도장손상_QTY) AS Total_Damage ORDER BY Total_Damage DESC LIMIT 1 > Finished chain. Observation: [{ 'b.BRDG_NAME': '오금교', 'Total_Damage': 3102.23}] Thought:'''json { "action": "Final Answer", "action_input": "도장손상이 가장 많이 발생한 교량은 '오금교'이며, 총 손상량은 3102.23입니다."}''' </pre>
<pre> > Entering new AgentExecutor chain... '''json { "action": "Neo4j_search", "action_input": "만년교의 건설년도"}''' > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg {BRDG_NAME: '만년교'}) RETURN b.CONST_DATE > Finished chain. Observation: [{ 'b.CONST_DATE': '1999-12-01' }] Thought:'''json { "action": "Final Answer", "action_input": "만년교는 1999년에 건설되었습니다."}''' </pre>
<pre> > Entering new AgentExecutor chain... '''json { "action": "Serp_search", "action_input": "망상 균열이 무엇인가요?"}''' Observation: ['콘크리트에서의 망상균열이란 콘크리트 또는 모르타르 표면층의 수축에 의하여 발생하는 · 그물 모양의 미세한 불규칙 균열이 생성되는 것을 발생된느 균열을 칭합니다. 1) 철근균열에 따른 균열. 철근이 거푸집 속에서 편재되어 콘크리트의 피복 ...!', '응결상태에 있는 콘크리트가 진동이나 충격울기, 발파, 다짐, 시동장비의 부주의한 사용 ...!', '콘크리트 타설 후 수십일 이후에 방사형의 망상균열이 통상 불규칙하게 발생한다. (2) 콘크리트의 침하 및트 구조물에 발생하는 균열의 원인 및 보수 보강방법에 대하여 기술하시오. 1.5K views · 2 years ago ...more ...!', '... 망상균열 및 백태 (a) 주철근방생하는 균열은 미세균열, 선상균열, 거북등균열 등이 있다.', '경부고속철도 자갈래도에 부설된 프리스트레스트 콘크리트(PSC) 침목 중 다수의 침목에서 종상당수의 ...!', '콘크리트 표면에 나타난 망상균열은 양생이 부적절한 경우 또는 높은 단위수량이나 과다한 흙손질 등으로 인해 굵은 골재가 가라앉고 시멘트취급건물에서는 망상균열이. 관찰되고 ... 추정되며, 2 호기 좌측 하단부 등에서 관찰되는 망상균열은 미세한 폭을 가진 잔균열로서.'] Thought:'''json { "action": "Final Answer", "action_input": "망상균열은 콘크리트 또는 모르타르 표면층의 수축에 의해 발생하는 그물 모양의 미세한 불규칙 균열을 말합니다. 이는 콘크리트 타설 망상균열은 양생이 부적절한 경우 또는 높은 단위수량이나 과다한 흙손질 등으로 인해 굵은 골재가 가라앉고 시멘트 및 잔 골재가 상승할 때 나타날 수 있다}''' </pre>
<pre> '''> Entering new AgentExecutor chain... '''json { "action": "RepairCost_Prediction_elmt", "action_input": "신리교, C"}''' Observation: 40039.200000000004 Thought:'''json { "action": "Final Answer", "action_input": "신리교의 C등급일 때 보수보강비용은 약 40,039.2 천원이 필요합니다."}''' </pre>
<pre> > Entering new AgentExecutor chain... '''json { "action": "RepairCost_Prediction", "action_input": "단양교 표면보수 5"}''' </pre>

<pre> /data/leege/CustomAgent_20231203.py:183: SettingWithCopyWarning: A value is trying to be set on a copy of a slice from a DataFrame. Try using .loc[row_indexer,col_indexer] = value instead filtered_df['보수물량'] = repair_quantity Observation: ['455000원'] Thought:```json { "action": "Final Answer", "action_input": "단양교에 표면보수로 5만㎤ 보수했을 때 비용은 약 455,000원이 드는 것으로 예상됩니다."}``` </pre>
<pre> > Entering new AgentExecutor chain... ```json { "action": "Neo4j_search", "action_input": "월악교의 바닥판의 균열 양의 합"}``` > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg {BRDG_NAME: '월악교'})-[:has]->(be:BrdgElmt)-[:IsInspectedBy]- >(bdih:Brdg_deck_inspctHistory) RETURN SUM(bdih.균열_QTY) AS Total_Crack_Quantity > Finished chain. Observation: [{Total_Crack_Quantity: 16.05}] Thought:```json { "action": "Final Answer", "action_input": "월악교의 바닥판의 균열 양의 합은 16.05입니다."}``` </pre>
<pre> > Entering new AgentExecutor chain... ```json{ "action": "Neo4j_search", "action_input": "유구교, 만년교, 모란교 중에서 누수 및 백태가 가장 많은 교량"}``` > Entering new GraphCypherQAChain chain... Generated Cypher: MATCH (b:Brdg)-[:IsManagedBy]->(:Manage_Agency_Sub)-[:Under]->(:Manage_Agency_Sup) WHERE b.BRDG_NAME IN ['유구교', '만년교', '모란교'] MATCH (b)-[:has]->(be:BrdgElmt)-[:IsInspectedBy]->(bdih:Brdg_deck_inspctHistory) RETURN b.BRDG_NAME, SUM(bdih.누수및백태_QTY) AS LeakageAndSpalling ORDER BY LeakageAndSpalling DESC LIMIT 1 > Finished chain. Observation: [{b.BRDG_NAME: '모란교', 'LeakageAndSpalling': 4.89}] Thought:```json { "action": "Final Answer", "action_input": "유구교, 만년교, 모란교 중에서 누수 및 백태가 가장 많은 교량은 모란교입니다."}``` > Finished chain. </pre>

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국문 초록

교량 유지관리 지식베이스와 거대언어모델을 활용한 전문 질의응답 시스템 개발

이규은

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대표적인 사회기반시설물인 교량의 안전한 이용을 위해 주기적인 점검진단과 보수보강 조치와 같은 유지관리 업무는 매우 중요하다. 많은 국가에서 시설물 유지관리 규정 및 기준들을 법제화 및 매뉴얼화하고 있으며 교량관리시스템 (BMS)과 같은 플랫폼을 개발하여 교량의 제원 정보와 유지관리 이력 정보를 수집 및 관리하고 있다. 그러나 매년 증가하는 교량의 개수에 비해 이를 효율적으로 관리할 전문가의 수는 부족하며, 데이터 기반 의사결정에 활용되어야 할 많은 양의 유지관리 이력 정보와 매뉴얼들은 구조화되지 않은 이질적인 형태로 분산되어 있어 적극적인 활용이 어렵다. 많은 선행 연구자들이 데이터에 기반한 의사결정을 위해 데이터베이스를

구축하고 머신러닝 기반 예측 모델들을 개발하고 있지만, 다양한 연구 성과에도 불구하고, 데이터를 활용해야 할 실질적인 사용자가 필요한 정보를 찾는 데는 어려움이 있다.

이를 해결하기 위해, 본 연구는 교량 유지관리 업무 담당자들이 데이터의 검색 및 분석된 결과를 기반으로 의사결정을 할 수 있도록 지원할 수 있는 질의응답 시스템을 개발하고자 한다. 이를 위해 먼저, BMS 및 관계법령 센터에 저장되어 있는 다양한 교량 유지관리 관련 이중 데이터를 수집하고 구조화하여 지식베이스를 생성한다. 둘째로, 지식베이스를 기반으로 교량 유지관리와 관련된 사용자의 질의에 대한 답변을 생성할 수 있는 모듈을 생성한다. 셋째로 거대언어모델의 자연어 추론 능력을 활용해 사용자의 자연어 질의를 쿼리로 변경하고, 쿼리를 이용해 지식베이스에서 정보를 찾아, 다시 답변을 반환할 수 있는 시스템을 설계한다. 마지막으로 사용자 테스트를 통해 개발된 시스템이 사용자의 질의 목적에 맞는 답변을 반환했는지 검증한다.

지식베이스 구축 결과와 질의응답 시스템을 검증하기 위해 저자가 직접 역량질문 데이터셋을 구성하여 역량질문에 대한 답변을 지식베이스 기반으로 생성할 수 있는지 검증하였다. 역량질문을 활용한 검증결과 구축된 지식베이스가 필요한 답변을 생성하기 적합한 환경임을 확인하였다. 또한, 질의응답 시스템을 웹기반 어플리케이션으로 개발하여, 사용자가 직접 테스트를 통해 질의 응답을 할 수 있는 환경을 구축했다. 10명의 참가자를 모집하여 사용자 테스트를 실시하였으며, 질의에 따라 생성된 답변과 응답속도 등에 따라 평가된 사용자 만족도를 분석했다. 분석된 결과 평균 이상의

만족도와 14.97초의 평균 응답 생성 시간을 통계적으로 확인할 수 있었다. 생성된 질의-응답 세트를 활용하여 정확도를 검토한 결과 85.7%의 정확도를 가지는 것을 확인할 수 있었다.

결론적으로, 본 논문은 교량 유지관리 분야의 정보 혁신 및 담당자의 업무 지원을 위해, 이중데이터를 결합한 교량유지관리 지식 베이스를 구축하였으며 거대언어모델을 활용한 질의응답 시스템을 개발하여 사용자의 데이터 검색 및 분석에 대한 접근성을 확장시켰다. 이는 교량 유지관리 실무에서 데이터에 기반한 의사결정을 지원하는데 활용될 수 있으며, 궁극적으로 안전한 시설물 이용에 기여할 수 있다.

주요어: 시설물 유지관리; 지식베이스; 거대언어모델; 질의응답 시스템

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